Anytime-Valid Certification of Robustness Margins for Safe Control via E-Values

Supplementary Ablation Report

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Overview

Assumption and theorem numbers refer to the arXiv paper [1]. Section 1 gives the computations made after the simulation: the exceedance risk (Section 1.1), the metrics (Section 1.2), the parameters and constants (Section 1.3) and the assumption checks (Section 1.4). Section 2 gives the ablation results, and Section 3 collects all tables and figures in the order of the text. For the results, read this Overview and the findings in the opening text of Sections 2.3–2.9.

Setup

As in [1, Sec. II], the system is $x_{t+1} = f(x_t, u_t, w_t)$, with the state x_t, the input u_t and the disturbance w_t, and $\hat{f}$ is the nominal one-step predictor. The decision-time filtration $\mathcal{F}_t$ contains the information available after u_t is selected and before x_{t+1} is observed. The barrier function h defines the safe set $\mathcal{C} := \{x : h(x) \geq 0\}$.

A robustness margin $c_t \geq 0$ enters the tightened nominal barrier condition, and the CBF safety filter applies $u_t := u_t(c_t)$,

$$h\big(\hat{f}(x_t, u_t)\big) - c_t \geq (1-\gamma)h(x_t), \quad \gamma \in (0,1], \qquad u_t(c) \in \operatorname*{arg\,min}_{u \in \mathcal{U}(x_t,c)} \|u - u_t^{\mathrm{nom}}\|^2, \tag{1}$$

where $\mathcal{U}(x_t, c)$ is the set of inputs that satisfy the tightened condition in (1) with $c_t = c$, and u_t^{nom} is the nominal input.

The barrier loss Y_t measures the model mismatch in the barrier. The exceedance risk $p_t(c)$ is the probability that it exceeds a margin c, and the exceedance risk limit bounds this risk at every decision time t,

$$Y_t := \big[h\big(\hat{f}(x_t, u_t)\big) - h(x_{t+1})\big]_+, \qquad p_t(c) := \mathbb{P}(Y_t > c \mid \mathcal{F}_t), \qquad p_t(c_t) \leq \alpha_0, \tag{2}$$

with the CBF-violation probability α_0. If $Y_t \leq c_t$, the applied input satisfies the true barrier condition $h(x_{t+1}) \geq (1-\gamma)h(x_t)$. The exceedance risk limit in (2) thus bounds the conditional probability of a CBF violation at t by α_0.

By Assumption 1, $\mathcal{U}(x_t, c_{\max})$ is nonempty, so the filter stays feasible, and applying $u_t(c_{\max})$ ensures the barrier-loss bound $Y_t \leq c_{\max}$ almost surely. The candidate margins form the finite grid $\Gamma \subseteq [0, c_{\max})$, and the deployed margin c_t is selected from $\Gamma \cup \{c_{\max}\}$.

Certification and Deployment

For every candidate $c \in \Gamma$, e-value tests collect evidence (Sections IV and V of [1]) against the null hypothesis $H_0(c)$ that c is α-insufficient, $p_t(c) \geq \alpha$ at all t, with the margin insufficiency $\alpha \in (0, \alpha_0]$. A certified candidate is deployable while the current risk-increase bound of one of its certificates is at most $\alpha_0 - \alpha$. The method proceeds in six steps.

1. Evidence. Each test uses the exceedance indicators $Z_{t-1}(c) = \mathbf{1}_{\{Y_{t-1} > c\}}$.
2. Tests. Test $r = 1, 2, \ldots$ starts at $\sigma_r = (r-1)K$, where K is the test period. From $E_{\sigma_r}^{\lambda,r}(c) = 1$, it updates one process per gain $\lambda \in \Lambda$ and averages them,

$$E_t^{\lambda,r}(c) = E_{t-1}^{\lambda,r}(c)\big[1 + \lambda(\alpha - Z_{t-1}(c))\big], \qquad E_t^r(c) = \frac{1}{|\Lambda|}\sum_{\lambda \in \Lambda} E_t^{\lambda,r}(c), \qquad t > \sigma_r. \tag{3}$$

3. Certification. Test r certifies c once its evidence reaches $1/\delta_r$,

$$\tau_r(c) := \inf\left\{t > \sigma_r : E_t^r(c) \geq \frac{1}{\delta_r}\right\}, \qquad \delta_r = \frac{\delta}{b(r+1)\ln^2(r+1)}, \qquad \sum_{r\geq 1}\delta_r = \frac{\delta\kappa}{b} \leq \delta, \tag{4}$$

with the error confidence δ, the budget normalizer $b \geq \kappa$ and $\kappa := \sum_{n\geq 2} 1/(n\ln^2 n) \approx 2.11$. Certification stores the certificate (c, r) with the monitoring period $[\sigma_r, \tau_r(c))$.

4. Deployment. The current risk-increase bound scales the certificate bound $d_t^{\mathrm{V}}(c;\sigma,\tau)$ of variant $\mathrm{V} \in \{\mathrm{U}, \Delta, \mathrm{L}, \mathrm{C}\}$, the largest risk-increase term over $\sigma \leq s < \tau$ (next subsection), by the bound factor $\phi \in [0, 1]$ of [1, App. F-B]. With the set $\mathcal{S}_t$ of candidates certified by time t, the deployable set and the deployment rule are

$$d_t^{(r)}(c) = \phi\, d_t^{\mathrm{V}}\big(c;\sigma_r,\tau_r(c)\big), \qquad \mathcal{D}_t := \Big\{c \in \mathcal{S}_t : \min_{r:\,\tau_r(c)\leq t} d_t^{(r)}(c) \leq \alpha_0 - \alpha\Big\},$$
$$c_t := \begin{cases} \min \mathcal{D}_t, & \mathcal{D}_t \neq \emptyset, \\ c_{\max}, & \mathcal{D}_t = \emptyset. \end{cases} \tag{5}$$

5. Supporting test. For a deployed margin $c_t \in \mathcal{D}_t$, the supporting test index is

$$r_t := \max \arg\min_{r:\tau_r(c_t)\leq t} d_t^{(r)}(c_t), \tag{6}$$

the most recently started test with the smallest current risk-increase bound among those that certified c_t. If $\mathcal{D}_t = \emptyset$, then $r_t := 0$.

6. Guarantees. Assumption 2 requires, for times $\sigma < \tau \leq t$ and every $s \in \{\sigma, \ldots, \tau - 1\}$,

$$p_t(c) \leq p_s(c) + d_t(c;\sigma,\tau), \tag{7}$$

and each variant uses $d_t(c;\sigma,\tau) = d_t^{\mathrm{V}}(c;\sigma,\tau)$. This gives the exceedance risk limit in (2). Let $\phi = 1$, and let a certificate (c, r) with $d_t^{(r)}(c) \leq \alpha_0 - \alpha$ have $p_s(c) < \alpha$ at some s of its monitoring period. Then

$$p_t(c) \leq p_s(c) + d_t^{(r)}(c) < \alpha + (\alpha_0 - \alpha) = \alpha_0.$$

A certificate without such an s is a false certification. Under Assumptions 1 and 2, the following guarantees hold.

- For finite K, Theorem 3 gives $p_t(c_t) \leq \alpha_0$ at all decision times with probability at least $1 - \delta$. Its proof also bounds the false-certification probability by δ.
- For $K = \infty$, only test 1 starts, at $\sigma_1 = 0$. Theorems 1 and 2 give the same two bounds with the error level $\delta_1 = \delta/(2b\ln^2 2)$ in place of δ.
- For $\phi < 1$, the current risk-increase bound need not satisfy Assumption 2, and no guarantee applies.

Variants and Baselines

The four variants of [1, App. D-B] differ only in the risk-increase term inside the certificate bound $d_t^{\mathrm{V}}(c;\sigma,\tau)$.

Variant	Risk-increase term at s, $\sigma \leq s < \tau$	Check at (t, s)
V-U	$L_x^{\mathrm{geom}}\|x_t - x_s\| + L_u^{\mathrm{geom}}\|u_t(c) - u_s\| + \Delta_t^{\mathrm{geom}}$	uniform
V-Δ	$L_x^{\mathrm{geom}}\|x_t - x_s\| + L_u^{\mathrm{geom}}\|u_t(c) - u_s\| + \Delta_t^{\mathrm{geom}}(c)$	for c
V-L	$\min\{\bar{P}(c), G_{t,s}(c)\} + \Delta_t^{\mathrm{geom}}$	uniform
V-C	$(1 - \Delta_{t,s})B_{t,s}(c) + \Delta_{t,s}\bar{P}(c)$	for c

The terms use the drift term $\Delta_{t,s}$, the s-dependent form of Δ_t, and

$$\begin{aligned} \Delta_{t,s} &= \min\{1, L_\mu(t-s)\}, \qquad B_{t,s}(c) = \min\{\bar{P}(c), G_{t,s}(c) + \eta_{t,s}(c)\}, \\ \Delta_t^{\mathrm{geom}} &= \bar{\eta}_{t,s} + \Delta_{t,s}, \qquad \Delta_t^{\mathrm{geom}}(c) = \eta_{t,s}(c) + \Delta_{t,s}. \end{aligned} \tag{8}$$

Here $\eta_{t,s}(c)$ is the geometric correction, $\bar{\eta}_{t,s}$ the uniform geometric correction, $\bar{P}(c)$ the fraction bound, $G_{t,s}(c)$ the clearance term, and L_x^{geom} and L_u^{geom} the constants, all defined there. The effective drifts Δ_t^{geom} and $\Delta_t^{\mathrm{geom}}(c)$ depend on s through $\Delta_{t,s}$ and the corrections.

The geometric check for c at (t, s) ensures that one obstacle determines the clearance ψ near the predicted nose point $z_t(c)$. The uniform geometric check requires the check for every $c' \in \Gamma$. A failed required check sets the certificate bound to 1, which prevents deployment with that certificate.

The certificate bound $d_t^{\mathrm{V}}(c; \sigma, \tau)$ is the maximum of the term over $\sigma \le s < \tau$, capped at 1 for V-U, V-Δ and V-L. V-C needs no cap, since its terms are at most $\bar{P}(c) \le 1/2$. On the same history and monitoring period, the ordering holds, with the arguments $(c; \sigma, \tau)$ omitted,

$$d_t^{\mathrm{C}} \le d_t^{\mathrm{L}} \le d_t^{\mathrm{U}}, \qquad d_t^{\mathrm{C}} \le d_t^{\Delta} \le d_t^{\mathrm{U}}. \tag{9}$$

The baselines of [1, App. F-A] are unfiltered nominal control (u_t^{nom}), CBF safety filters with the fixed margins 0 and $c_{\max}$, ACI and QT. ACI and QT track the $(1-\alpha_0)$-quantile of Y_t online, clipped to $[0, c_{\max}]$, and target an average exceedance frequency α_0. Hence $p_t(c_t)$ may exceed α_0 at some decision times.

Scenarios and Settings

As in [1, App. F-A], a unicycle robot drives to a goal past cones in the Corridor and past walls and cones in the Maze. The disturbance displaces the position,

$$w_t = A_t e(\Phi_t), \qquad A_t \mid \mathcal{F}_t \sim (1-\beta_t)\,\mathrm{U}_1 + \beta_t\,\mathrm{U}_2, \qquad \beta_t := \tfrac{1}{2}[1 + \sin(2L_\mu t)], \tag{10}$$

with $e(\Phi) := (\cos\Phi, \sin\Phi)$, the direction Φ_t uniform on $[0, 2\pi)$, and U_1 and U_2 uniform on $[0, c_{\max}/2]$ and $[c_{\max}/2, c_{\max}]$. The magnitude distribution μ_t^A thus shifts between small and large magnitudes, with $|\beta_{t+1} - \beta_t| \le L_\mu$ for the distribution-drift bound L_μ. The controller knows neither μ_t^A nor β_t.

A setting $(\alpha_0, \alpha, \delta, K)$ fixes the CBF-violation probability, the margin insufficiency, the error confidence and the test period.

Name	$(\alpha_0, \alpha, \delta, K)$	Shown variant	Remark
Main grid	375 settings	all four	Table 3
Default	$(0.15, 0.10, 0.05, 50)$	V-C	in the main grid
S1	$(0.30, 0.25, 0.20, 50)$	V-C	selected setting
S2	$(0.30, 0.25, 0.15, 50)$	V-C	selected setting
S3	$(0.30, 0.20, 0.20, 50)$	V-L	selected setting
S4	$(0.30, 0.25, 0.025, 100)$	V-Δ	selected setting
S5	$(0.30, 0.25, 0.10, 10)$	V-C	selected setting
S*	$(0.25, 0.225, 0.15, 50)$	all four	Maze only, outside the main grid (Section 2.2.1)

Every setting runs all four variants. Where a figure or table shows one variant per setting (the trajectory figures, the bound-factor tables and the detailed-check tables), it shows the variant of this column. "S1–S5 with their variants" denotes these five pairs.

Unless stated otherwise, $\phi = 1$, $b = \kappa$, rounded up to 2.11, and $L_\mu = 0.0005$. The comparisons also use $L_\mu = 0.005$. Each method runs $N_{\mathrm{MC}} = 200$ trajectories per scenario and setting. All methods share the initial states and, for each L_μ, the disturbance realizations.

1 Simulation Details

This section gives the computations behind the values in the tables of Section 3, made after the simulation from the recorded trajectories. The controller never uses them. Sections 1.1–1.4 give the exceedance risk, the metrics, the parameters and constants, and the assumption checks. They extend Appendices E and F of [1].

1.1 Exceedance-Risk Evaluation

The exceedance risk $p_t(c)$ in (2) is evaluated along the recorded trajectories with the full simulated disturbance distribution, which the controller never uses. Sections 1.1.1–1.1.4 give its angular form, the quadrature rule, the closed form for one affine obstacle, and the numerical risk bounds.

1.1.1 Angular Risk

The exceedance risk $p_t(c)$ is an average over the disturbance magnitude of an angular risk, which has an exact arc-length form. At each recorded decision time t, the state, the applied input and the observed history are held fixed.

As in [1, App. D], z_t is the predicted nose point, $\psi = \min_i \psi_i$ is the clearance over the obstacles i, and $\{Y_t > c\} = \{\ell(x_t, u_t, w_t) > c\}$ for $c \geq 0$, with the shortfall

$$\ell(x_t, u_t, w) = \psi(z_t) - \psi(z_t + w).$$

In the simulation, the disturbance is $w_t = A_t e(\Phi_t)$ in (10). The magnitude A_t has the distribution μ_t^A, and, given A_t, the direction Φ_t is uniform on $[0, 2\pi)$. Averaging over the magnitude gives

$$p_t(c) = \int_0^{c_{\max}} p_t(c \mid A)\, \mu_t^A(dA), \tag{11}$$

with the angular risk

$$p_t(c \mid A) := \frac{1}{2\pi} \int_0^{2\pi} \mathbf{1}_{\{\psi(z_t + Ae(\Phi)) < \psi(z_t) - c\}}\, d\Phi. \tag{12}$$

Since ψ is the minimum of the ψ_i, the directions in (12) split over the obstacles,

$$\begin{aligned} \{\Phi : \psi(z_t + Ae(\Phi)) < \psi(z_t) - c\} &= \textstyle\bigcup_i \mathcal{A}_{t,i}(c, A), \\ \mathcal{A}_{t,i}(c, A) &:= \{\Phi : \psi_i(z_t + Ae(\Phi)) < \psi(z_t) - c\}. \end{aligned} \tag{13}$$

Since ψ_i is 1-Lipschitz, $\psi_i(z_t + Ae(\Phi)) \geq \psi_i(z_t) - A$. Hence $\mathcal{A}_{t,i}(c, A)$ is empty unless A exceeds the entry magnitude

$$a_{t,i}(c) := \psi_i(z_t) - \psi(z_t) + c. \tag{14}$$

Each set $\mathcal{A}_{t,i}(c, A)$ has an explicit form for the three obstacle shapes.

- A circular obstacle has $\psi_i(z) = \|z - o\| - R$ with the center o and the effective radius R (a cone has $R = \rho_j + \rho_{\text{body}}$). It has $\Phi \in \mathcal{A}_{t,i}(c, A)$ if and only if $\|z_t + Ae(\Phi) - o\| < R + \psi(z_t) - c$. The set is empty when $R + \psi(z_t) - c \leq 0$. For a positive right-hand side and $A > 0$, expanding the square shows

that membership is equivalent to the angle between $e(\Phi)$ and $o - z_t$ having cosine larger than

$$\frac{A^2 + \|o - z_t\|^2 - (R + \psi(z_t) - c)^2}{2A\|o - z_t\|}.$$

Hence $\mathcal{A}_{t,i}(c, A)$ is an arc centered at the direction of $o - z_t$.

- An affine obstacle with unit normal n has $\psi_i(z_t + Ae(\Phi)) = \psi_i(z_t) + An^\top e(\Phi)$. By (14), $\Phi \in \mathcal{A}_{t,i}(c, A)$ if and only if the angle between $e(\Phi)$ and $-n$ has cosine larger than $a_{t,i}(c)/A$.
- A rectangle O_k has $\psi_i(z) = \mathrm{sdist}(z, O_k) - \rho_{\mathrm{body}}$. With the signed offset $\varrho_t(c) := \rho_{\mathrm{body}} + \psi(z_t) - c$, which is the same for every rectangle, $\Phi \in \mathcal{A}_{t,i}(c, A)$ if and only if $\mathrm{sdist}(z_t + Ae(\Phi), O_k) < \varrho_t(c)$. For $\varrho_t(c) > 0$, the set $\{z : \mathrm{sdist}(z, O_k) < \varrho_t(c)\}$ is the rectangle dilated by $\varrho_t(c)$, bounded by the sides shifted outward by $\varrho_t(c)$ and by arcs of the circles of radius $\varrho_t(c)$ about the corners. For $\varrho_t(c) \le 0$, it is the rectangle eroded by $|\varrho_t(c)|$, bounded by the sides shifted inward by $|\varrho_t(c)|$.

For a rectangle, $\mathcal{A}_{t,i}(c, A)$ is thus a finite union of arcs. Their end points lie among the intersections of the circle $\{z_t + Ae(\Phi) : \Phi \in [0, 2\pi)\}$ with the side lines, the side lines shifted by $\varrho_t(c)$, and, for $\varrho_t(c) > 0$, the corner circles of radius $\varrho_t(c)$. The condition in (13) at the midpoint between two consecutive end points decides whether the arc between them belongs to $\mathcal{A}_{t,i}(c, A)$.

With $|\cdot|$ the total length of a union of arcs, computed by sorting and merging the arcs, the angular risk is

$$p_t(c \mid A) = \frac{1}{2\pi}\Big|\bigcup_{i:\, a_{t,i}(c) < A} \mathcal{A}_{t,i}(c, A)\Big|. \tag{15}$$

This evaluates (12) exactly up to floating-point rounding.

1.1.2 Quadrature Rule

The arc-length form (15) gives $p_t(c \mid A)$ exactly, and $p_t(c)$ is its integral over A. The integrand can grow like a square root at the entry magnitudes, so the quadrature rule first removes this behavior by a substitution.

By (10), μ_t^A mixes the uniform distributions U_1 on $\mathcal{I}_1$ and U_2 on $\mathcal{I}_2$ with the weights

$$\beta_{t,1} := 1 - \beta_t, \quad \beta_{t,2} := \beta_t, \qquad \mathcal{I}_1 := [0, c_{\max}/2], \quad \mathcal{I}_2 := [c_{\max}/2, c_{\max}],$$

each with the density $2/c_{\max}$ on its support. Since ψ is 1-Lipschitz, $p_t(c \mid A) = 0$ for $A \le c$. Restricting each support to its part $\mathcal{I}_j(c) := \mathcal{I}_j \cap [c, c_{\max}]$ above c specializes (11) to

$$p_t(c) = \frac{2}{c_{\max}} \sum_{j=1}^{2} \beta_{t,j} \int_{\mathcal{I}_j(c)} p_t(c \mid A)\, dA. \tag{16}$$

The entry magnitudes (14) in the interior of $\mathcal{I}_j(c)$ split it into the set $\mathcal{M}_{t,j}(c)$ of magnitude intervals. On the interior of each $\mathcal{J} = [\underline{A}, \overline{A}] \in \mathcal{M}_{t,j}(c)$, the set of obstacles in the union (15) does not change. At a positive, nondegenerate entry magnitude $\underline{A} = a_{t,i}(c)$, the cosine bound that defines $\mathcal{A}_{t,i}(c, A)$ equals one and has a nonzero derivative. Since

$$\arccos(1 - x) = \sqrt{2x} + O(x^{3/2}),$$

the length of the arc grows like $(A - \underline{A})^{1/2}$, and $p_t(c \mid A)$ can have an unbounded derivative at $\underline{A}$.

The substitution $A = \underline{A} + (\overline{A} - \underline{A})\vartheta^2$, $\vartheta \in [0, 1]$, has $dA = 2(\overline{A} - \underline{A})\vartheta\, d\vartheta$ and $(A - \underline{A})^{1/2} = (\overline{A} - \underline{A})^{1/2}\vartheta$.

It gives

$$\int_{\mathcal{J}} p_t(c \mid A)\, dA = \int_0^1 2(\overline{A} - \underline{A})\vartheta\, p_t\big(c \mid \underline{A} + (\overline{A} - \underline{A})\vartheta^2\big)\, d\vartheta, \tag{17}$$

in which a term of order $(A - \underline{A})^{1/2}$ in $p_t(c \mid A)$ becomes a term of order ϑ^2, smooth at $\vartheta = 0$.

The 24-point Gauss–Legendre rule on $[0, 1]$, with nodes ϑ_m and weights λ_m, integrates such smooth integrands accurately. Applied to the right-hand side of (17), it gives, for $\mathcal{J} \in \mathcal{M}_{t,j}(c)$ and $m = 1, \ldots, 24$, the nodes and weights

$$A_{\mathcal{J},m} := \underline{A} + (\overline{A} - \underline{A})\vartheta_m^2, \qquad \varpi_{\mathcal{J},m} := \frac{4\beta_{t,j}}{c_{\max}}(\overline{A} - \underline{A})\vartheta_m \lambda_m. \tag{18}$$

Substituting this rule into (16) yields the estimated exceedance risk of [1, App. E-A],

$$\widehat{p}_t(c) := \sum_n \varpi_{t,n}\, p_t(c \mid A_{t,n}). \tag{19}$$

Its nodes $A_{t,n}$ and weights $\varpi_{t,n}$ are the $A_{\mathcal{J},m}$ and $\varpi_{\mathcal{J},m}$ over $j \in \{1, 2\}$, $\mathcal{J} \in \mathcal{M}_{t,j}(c)$ and m. They depend on c through the intervals, and the weights are positive since $\vartheta_m, \lambda_m > 0$.

1.1.3 Affine Closed Form

When one affine obstacle determines ψ on the disk of radius $c_{\max}$ about z_t, the angular risk equals

$$P_\infty(c, A) = \frac{1}{\pi} \arccos \frac{c}{A} \quad (A > c), \qquad P_\infty(c, A) = 0 \quad \text{otherwise}, \tag{20}$$

which does not depend on the clearance $\psi(z_t)$. Then $p_t(c)$ in (16) has a closed form, and the numerical risk bounds need no bisection.

The rectangle condition of [1, App. D-B] is checked in coordinates. For a rectangle $[\underline{z}_1, \bar{z}_1] \times [\underline{z}_2, \bar{z}_2]$, the disk of radius $c_{\max}$ about z lies in the exterior face region to the right or in the corner region to the upper right if

$$\begin{aligned} \text{face region to the right:} \quad & z_1 - \bar{z}_1 \geq c_{\max}, \quad \min\{z_2 - \underline{z}_2, \bar{z}_2 - z_2\} \geq c_{\max}, \\ \text{corner region to the upper right:} \quad & z_1 - \bar{z}_1 \geq c_{\max}, \quad z_2 - \bar{z}_2 \geq c_{\max}. \end{aligned} \tag{21}$$

The other faces and corners follow by symmetry. The obstacle is affine on a face region and circular, with $R = \rho_{\text{body}}$, on a corner region.

Let the nearest obstacle i_z be the smallest index that minimizes $\psi_i(z_t)$. If $\min_{i \neq i_z} \psi_i(z_t) \geq \psi(z_t) + 2c_{\max}$ and the rectangle condition (21) hold at z_t and i_z is affine, then $\psi(z_t + w) = \psi_{i_z}(z_t + w)$ for $\|w\| \leq c_{\max}$, and $p_t(c \mid A) = P_\infty(c, A)$ in (20). For $0 < c < A$,

$$\frac{d}{dA}\Big[A \arccos \frac{c}{A} - c \operatorname{arcosh} \frac{A}{c}\Big] = \arccos \frac{c}{A} + \frac{c}{\sqrt{A^2 - c^2}} - \frac{c}{\sqrt{A^2 - c^2}} = \arccos \frac{c}{A}.$$

Hence, for a magnitude interval $[\underline{A}, \overline{A}]$ with $c \leq \underline{A}$,

$$\int_{\underline{A}}^{\overline{A}} P_\infty(c, A)\, dA = F_c(\overline{A}) - F_c(\underline{A}), \qquad F_c(A) := \frac{1}{\pi}\Big[A \arccos \frac{c}{A} - c \operatorname{arcosh} \frac{A}{c}\Big]. \tag{22}$$

At $c = 0$, the continuous extension is $F_0(A) = A/2$. Substituting (22) with $[\underline{A}, \overline{A}] = \mathcal{I}_j(c)$ into (16) gives $p_t(c)$ without quadrature. The numerical risk bounds of Section 1.1.4 use this closed form whenever it applies.

1.1.4 Numerical Risk Bounds

The estimate $\widehat{p}_t(c)$ of (19) carries no error guarantee. The trajectory events therefore use the numerical risk bounds of [1, App. E-A],

$$p_t^-(c) \le p_t(c) \le p_t^+(c),$$

which enclose $p_t(c)$ in exact arithmetic. The allowances at the end of this section address floating-point error.

The bounds use a partition $\mathcal{B}_{t,j}(c)$ of $\mathcal{I}_j(c)$ into magnitude intervals, initially $\{\mathcal{I}_j(c)\}$. An interval $\mathcal{J} \in \mathcal{B}_{t,j}(c)$ with midpoint $A_{\mathcal{J}}^{\mathrm{mid}}$ and half-width $\varepsilon_{\mathcal{J}}$ has the interval probability under μ_t^A

$$\pi_{\mathcal{J}} := 4\beta_{t,j}\varepsilon_{\mathcal{J}}/c_{\max}.$$

Since ψ is 1-Lipschitz, every direction Φ and every $A \in \mathcal{J}$ satisfy

$$\left|\psi(z_t + Ae(\Phi)) - \psi(z_t + A_{\mathcal{J}}^{\mathrm{mid}} e(\Phi))\right| \le |A - A_{\mathcal{J}}^{\mathrm{mid}}| \le \varepsilon_{\mathcal{J}}.$$

Hence a direction that satisfies the condition of (12) at $A_{\mathcal{J}}^{\mathrm{mid}}$ with $c + \varepsilon_{\mathcal{J}}$ satisfies it at A with c, and a direction that satisfies it at A with c satisfies it at $A_{\mathcal{J}}^{\mathrm{mid}}$ with $c - \varepsilon_{\mathcal{J}}$. Integrating over Φ, every $A \in \mathcal{J}$ satisfies

$$p_t(c + \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \le p_t(c \mid A) \le p_t(c - \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}). \tag{23}$$

For $c' < 0$, the evaluation sets $p_t(c' \mid A_{\mathcal{J}}^{\mathrm{mid}}) := 1$. This keeps the upper bound valid, since every probability is at most one.

Integrating (23) over each $\mathcal{J}$ and summing with (16),

$$\begin{aligned} p_t^-(c) &= \textstyle\sum_{j=1}^{2} \sum_{\mathcal{J} \in \mathcal{B}_{t,j}(c)} \pi_{\mathcal{J}}\, p_t(c + \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \le p_t(c), \\ p_t^+(c) &= \textstyle\sum_{j=1}^{2} \sum_{\mathcal{J} \in \mathcal{B}_{t,j}(c)} \pi_{\mathcal{J}}\, p_t(c - \varepsilon_{\mathcal{J}} \mid A_{\mathcal{J}}^{\mathrm{mid}}) \ge p_t(c), \end{aligned} \tag{24}$$

for every such partition, with each angular risk evaluated by (15).

The width $p_t^+(c) - p_t^-(c)$ is the sum over the intervals of $\pi_{\mathcal{J}}$ times the difference of the two angular risks in (24). Each bisection step replaces the interval with the largest such contribution by its two halves. The refinement stops at the first of the following conditions.

- The threshold of the event lies outside $[p_t^-(c), p_t^+(c)]$, with the margin insufficiency α for the false-certification event and the CBF-violation probability α_0 for the deployment event, both defined in (27).
- The width reaches the refinement limit $p_t^+(c) - p_t^-(c) \le 10^{-5}$.
- The number of magnitude intervals reaches the refinement limit 128.

A failed risk evaluation gives $[p_t^-(c), p_t^+(c)] = [0, 1]$.

Two allowances account for floating-point error.

- The length allowance $\varepsilon_{\mathrm{len}}$ is 256 machine epsilons times the largest of 1, $c_{\max}$, the absolute coordinates of z_t and the absolute clearances $\psi_i(z_t)$. In (24), each angular risk is evaluated at $c \pm (\varepsilon_{\mathcal{J}} + \varepsilon_{\mathrm{len}})$ instead of $c \pm \varepsilon_{\mathcal{J}}$. In the affine case, the bounds substitute (22) into (16) at $c \pm \varepsilon_{\mathrm{len}}$.
- Each angular risk in (24) and each of the two bounds is moved outward by 10^{-10}.

Rigorous roundoff coverage is not established, and the bounds do not use outward-rounded interval arithmetic.

1.2 Metrics

This subsection collects all metrics of the tables: the main metrics of [1, App. E], the bracketing of the events and their upper confidence bounds, the secondary metrics, the temporal variation, and how each table entry is formed.

1.2.1 Main Metrics

The following table lists the main metrics, and $^+$ marks a decision fraction.

Purpose	Symbol	Name	Definition
Performance	$\widehat{m}_c/c_{\max}$	normalized mean robustness margin	(25)
	$\widehat{T}_{\mathrm{mis}}$	mean mission time	(25)
Safety	$\widehat{q}^{+}_{\mathrm{CBF}}$	CBF-violation fraction	(26)
	$\widehat{q}^{+}_{\mathrm{unsafe}}$	safe-set-violation fraction	(26)
	$\widehat{m}_{p^{\max}}$	mean estimated peak exceedance risk	(34)
Guarantees	$\widehat{q}_{\mathrm{cert}}$	false-certification frequency	(28), event (27)
	$\widehat{q}_{\mathrm{dep}}$	deployment-event frequency	(28), event (27)
	$\overline{q}$	95% upper confidence bound	(28), Section 1.2.3
Assumption 2	$\widehat{q}_{\mathrm{A2}}$, $\widehat{q}_{\mathrm{pw}}$	fractions of trajectories with a bound excess $e_{\mathrm{A2},t}$ or $e_{\mathrm{pw},t}$ above $\varepsilon_{\mathrm{chk}}$	(46), Section 1.4
Evaluation	$\widehat{p}_t(c)$	estimated exceedance risk	(19)

Trajectory k executes $N_k \le T$ transitions, with the decision horizon T, and $I_{\mathrm{goal},k} := 1$ if it reaches the goal and 0 otherwise. For a decision indicator $I_{k,t}$ and $\mathcal{K}_+ := \{k : N_k \ge 1\}$,

$$
\begin{aligned}
T_{\mathrm{mis},k} &:= T_s[I_{\mathrm{goal},k}N_k + (1 - I_{\mathrm{goal},k})T], \\
\widehat{T}_{\mathrm{mis}} &:= \frac{1}{N_{\mathrm{MC}}} \sum_{k=1}^{N_{\mathrm{MC}}} T_{\mathrm{mis},k}, \\
m_{c,k} &:= \frac{1}{N_k} \sum_{t<N_k} c_{k,t}, \quad q_k^+ := \frac{1}{N_k} \sum_{t<N_k} I_{k,t}, \\
\widehat{m}_c &:= \frac{1}{|\mathcal{K}_+|} \sum_{k\in\mathcal{K}_+} m_{c,k}, \quad \widehat{q}^{+} := \frac{1}{|\mathcal{K}_+|} \sum_{k\in\mathcal{K}_+} q_k^+.
\end{aligned}
\tag{25}
$$

The mission time $T_{\mathrm{mis},k}$ is the time to reach the goal, or TT_s if the goal is missed. Dividing $\widehat{m}_c$ by $c_{\max}$ gives the normalized mean robustness margin, and $\widehat{q}^{+}$ is the decision fraction of I.

The CBF-violation and safe-set-violation fractions are the decision fractions of

$$
\begin{aligned}
I_{\mathrm{CBF},k,t} &:= \mathbf{1}_{\{h(x_{k,t+1})<(1-\gamma)h(x_{k,t})\}}, \\
I_{\mathrm{unsafe},k,t} &:= \mathbf{1}_{\{h(x_{k,t+1})<0\}}.
\end{aligned}
\tag{26}
$$

The false-certification and deployment events are

$$\begin{aligned}\mathcal{A}_{\mathrm{cert},k} &:= \{\exists c \in \Gamma,\ r \geq 1 :\ \tau_{k,r}(c) \leq N_k,\ \min_{\sigma_r \leq s < \tau_{k,r}(c)} p_{k,s}(c) \geq \alpha\},\\ \mathcal{A}_{\mathrm{dep},k} &:= \{\exists\, 0 \leq t < N_k : p_{k,t}(c_{k,t}) > \alpha_0\}.\end{aligned} \tag{27}$$

Over the N_{MC} trajectories, an event $\mathcal{A}$ has the trajectory frequency and the 95% upper confidence bound

$$\widehat{q}(\mathcal{A}) := \frac{1}{N_{\mathrm{MC}}} \sum_{k=1}^{N_{\mathrm{MC}}} \mathbf{1}_{\mathcal{A}_k}, \qquad \mathbb{P}\big(q(\mathcal{A}) \leq \overline{q}(\mathcal{A})\big) \geq 0.95. \tag{28}$$

For finite K, Theorem 3 and its proof bound the probabilities of both events by δ, and for $K = \infty$ Theorems 1 and 2 bound them by δ_1. An event frequency is consistent with its guarantee if its upper confidence bound is at most δ, or δ_1 for $K = \infty$.

1.2.2 Events and Counts

The false-certification and deployment events (27) are defined with the exceedance risk, which is unknown. The confirmed and possible events bracket them with the numerical risk bounds of Section 1.1.4.

Let $\mathcal{A}^{-}_{\mathrm{cert},k}$ and $\mathcal{A}^{-}_{\mathrm{dep},k}$ be the two events with every $p_{k,t}(c)$ replaced by $p^{-}_{k,t}(c)$, and let $\mathcal{A}^{+}_{\mathrm{cert},k}$ and $\mathcal{A}^{+}_{\mathrm{dep},k}$ be the events with $p^{+}_{k,t}(c)$. Both events are increasing in the exceedance risks. Since $p^{-}_{k,t}(c) \leq p_{k,t}(c) \leq p^{+}_{k,t}(c)$ by (24),

$$\mathcal{A}^{-}_{\mathrm{cert},k} \subseteq \mathcal{A}_{\mathrm{cert},k} \subseteq \mathcal{A}^{+}_{\mathrm{cert},k}, \qquad \mathcal{A}^{-}_{\mathrm{dep},k} \subseteq \mathcal{A}_{\mathrm{dep},k} \subseteq \mathcal{A}^{+}_{\mathrm{dep},k}. \tag{29}$$

The confirmed events are $\mathcal{A}^{-}_{\mathrm{cert},k}$ and $\mathcal{A}^{-}_{\mathrm{dep},k}$, and the possible events are $\mathcal{A}^{+}_{\mathrm{cert},k}$ and $\mathcal{A}^{+}_{\mathrm{dep},k}$. For trajectory k, they occur under the following conditions.

Event	Confirmed if	Possible if
Deployment	$p^{-}_{k,t}(c_{k,t}) > \alpha_0$ at some $t < N_k$	$p^{+}_{k,t}(c_{k,t}) > \alpha_0$ at some $t < N_k$
False certification	a certificate (c, r) has $p^{-}_{k,s}(c) \geq \alpha$ at every s of its monitoring period	a certificate (c, r) has $p^{+}_{k,s}(c) \geq \alpha$ at every s of its monitoring period

A trajectory that ends early because of sampled contact, filter infeasibility or a numerical error counts as a possible event of both types. Summing the indicators in (29) over the N_{MC} trajectories gives

$$\text{confirmed count} \leq \text{count of the event} \leq \text{possible count}.$$

Every upper confidence bound uses the possible count, and Section 2.1.3 states which counts the tables print.

1.2.3 Upper Confidence Bounds

The upper confidence bound $\overline{q}(\mathcal{A})$ of (28) is the one-sided 95% Clopper–Pearson bound computed from the possible count $n_{\mathcal{A}}$ of Section 1.2.2.

For a binomial count $n \sim \mathrm{Bin}(N_{\mathrm{MC}}, q)$ and $n_{\mathcal{A}} < N_{\mathrm{MC}}$,

$$\mathbb{P}(n \leq n_{\mathcal{A}}) = 1 - F_{\mathrm{Beta}(n_{\mathcal{A}}+1,\, N_{\mathrm{MC}}-n_{\mathcal{A}})}(q).$$

Hence $\overline{q}(\mathcal{A})$ is the value of q at which $n_{\mathcal{A}}$ or fewer events have probability 0.05. When $n_{\mathcal{A}} = N_{\mathrm{MC}}$, $\overline{q}(\mathcal{A})$ is one. For $n_{\mathcal{A}} = 0$, $F_{\mathrm{Beta}(1,N_{\mathrm{MC}})}(q) = 1-(1-q)^{N_{\mathrm{MC}}}$, so setting $\mathbb{P}(n \le 0) = (1-q)^{N_{\mathrm{MC}}}$ to 0.05 gives

$$\overline{q}(\mathcal{A}) = 1 - 0.05^{1/N_{\mathrm{MC}}}. \tag{30}$$

For example, for $N_{\mathrm{MC}} = 200$ this bound is 0.0149, which the tables print as 1.5%.

1.2.4 Secondary Metrics

The secondary metrics complement the main metrics of Section 1.2.1 and use their notation. Each mean averages a per-trajectory quantity X_k over its own set $\mathcal{K}_X$ of trajectories,

$$\widehat{m}_X := \frac{1}{|\mathcal{K}_X|} \sum_{k \in \mathcal{K}_X} X_k. \tag{31}$$

The following table lists the secondary metrics. The first six describe certification and deployment, and the last five describe the barrier value, the barrier loss, the filter intervention and the body clearance.

Symbol	Name	Per-trajectory quantity	Averaged over
$\widehat{q}^{+}_{\mathrm{nocert}}$	no-certificate fraction	$I_{\mathrm{nocert},k,t}$ (32)	$\mathcal{K}_+$
$\widehat{q}^{+}_{\mathrm{nodep}}$	no-deployable-margin fraction	$I_{\mathrm{nodep},k,t}$ (32)	$\mathcal{K}_+$
$\widehat{q}^{+}_{\mathrm{use}}$	smaller-margin-use fraction	$I_{\mathrm{use},k,t}$ (32)	$\mathcal{K}_+$
$\widehat{m}_{\tau^{\mathrm{use}}}$	mean first-use time (s)	τ^{use}_k (33)	$\mathcal{K}_{\mathrm{use}}$
$\widehat{m}_{p^{\max}}$	mean estimated peak exceedance risk	$\widehat{p}^{\max}_k$ (33)	$\mathcal{K}_{p^{\max}}$
$\widehat{m}_{\mathrm{cert}}$	mean certificate count	$n_{\mathrm{cert},k}$ (35)	all
$\widehat{m}_{h^{\min}}$	mean minimum barrier value	$h^{\min}_k$ (35)	all
$\widehat{q}^{+}_{Y}$	barrier-loss exceedance fraction	$I_{Y,k,t}$ (36)	$\mathcal{K}_+$
$\widehat{m}_Y/c_{\max}$	normalized mean barrier loss	$m_{Y,k}/c_{\max}$ (36)	$\mathcal{K}_+$
$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}$	control intervention, in unscaled input coordinates	$m_{\|u-u^{\mathrm{nom}}\|^2,k}$ (36)	$\mathcal{K}_+$
$\widehat{m}_{\psi^{\min}}$	mean minimum sampled body clearance	$\psi^{\min}_k$ (below)	$\mathcal{K}_+$

A metric marked $^{+}$ is the decision fraction (25) of its indicator, and $\mathcal{K}_+ := \{k : N_k \ge 1\}$. The three decision indicators use the set $\mathcal{S}_{k,t}$ of certified candidates and the deployable set $\mathcal{D}_{k,t}$ in (5),

$$\begin{aligned} I_{\mathrm{nocert},k,t} &:= \mathbf{1}_{\{\mathcal{S}_{k,t}=\emptyset\}}, \\ I_{\mathrm{nodep},k,t} &:= \mathbf{1}_{\{\mathcal{S}_{k,t}\neq\emptyset,\mathcal{D}_{k,t}=\emptyset\}}, \\ I_{\mathrm{use},k,t} &:= \mathbf{1}_{\{c_{k,t}<c_{\max}\}}. \end{aligned} \tag{32}$$

The first-use time τ^{use}_k is the first decision with a margin below $c_{\max}$, and the estimated peak exceedance risk is $\widehat{p}^{\max}_k$,

$$\begin{aligned} \tau^{\mathrm{use}}_k &:= \inf\{0 \le t < N_k : c_{k,t} < c_{\max}\}, \\ \widehat{p}^{\max}_k &:= \max_{0\le t<N_k,\ c_{k,t}<c_{\max}} \widehat{p}_{k,t}(c_{k,t}). \end{aligned} \tag{33}$$

With $\mathcal{K}_{\text{use}} := \{k : \tau_k^{\text{use}} < \infty\}$ and $\mathcal{K}_{p^{\max}} := \{k : \widehat{p}_k^{\max} \text{ is defined}\}$, the two means are

$$\begin{aligned} \widehat{m}_{\tau^{\text{use}}} &:= \frac{T_s}{|\mathcal{K}_{\text{use}}|} \sum_{k \in \mathcal{K}_{\text{use}}} \tau_k^{\text{use}}, \\ \widehat{m}_{p^{\max}} &:= \frac{1}{|\mathcal{K}_{p^{\max}}|} \sum_{k \in \mathcal{K}_{p^{\max}}} \widehat{p}_k^{\max}. \end{aligned} \tag{34}$$

The factor T_s gives the first-use time in seconds. The certificate count $n_{\text{cert},k}$ counts the certificates (c, r) issued by the end of the trajectory, and the minimum barrier value is $h_k^{\min}$,

$$\begin{aligned} n_{\text{cert},k} &:= \#\{(c, r) : \tau_{k,r}(c) \le N_k\}, \\ h_k^{\min} &:= \min_{0 \le t \le N_k} h(x_{k,t}). \end{aligned} \tag{35}$$

The loss and intervention quantities use the exceedance indicator $Z_{k,t}(c) = \mathbf{1}_{\{Y_{k,t} > c\}}$ and the nominal input u_t^{nom} in (1),

$$\begin{aligned} I_{Y,k,t} &:= Z_{k,t}(c_{k,t}), \\ m_{Y,k} &:= \frac{1}{N_k} \sum_{t < N_k} Y_{k,t}, \\ m_{\|u-u^{\text{nom}}\|^2,k} &:= \frac{1}{N_k} \sum_{t < N_k} \|u_{k,t} - u_{k,t}^{\text{nom}}\|_2^2. \end{aligned} \tag{36}$$

The minimum sampled body clearance is the smallest clearance at the decision times,

$$\psi_k^{\min} := \min_{1 \le t \le N_k} \psi(y_{k,t}),$$

with the body center $y_{k,t}$ and the clearance ψ of [1, App. D-B].

1.2.5 Temporal Variation

The temporal statistics measure how much the normalized robustness margin $c_{k,t}/c_{\max}$ and the estimated exceedance risk $\widehat{p}_{k,t}(c_{k,t})$ vary along a trajectory over its first $N_{\text{var}} := 200$ decisions. Section 2.9 reports them.

The variable $\upsilon_{k,t}$ is either of these two quantities. The set $\mathcal{K}_\upsilon$ contains the trajectories with at least N_{var} decisions and finite $\upsilon_{k,t}$ for all $t < N_{\text{var}}$. For $k \in \mathcal{K}_\upsilon$, the trajectory statistics and their means (31) are

$$\begin{aligned} \bar{\upsilon}_k &:= \frac{1}{N_{\text{var}}} \sum_{t=0}^{N_{\text{var}}-1} \upsilon_{k,t}, \qquad V_{\upsilon,k} := \frac{1}{N_{\text{var}} - 1} \sum_{t=0}^{N_{\text{var}}-1} (\upsilon_{k,t} - \bar{\upsilon}_k)^2, \\ J_{\upsilon,k} &:= \Big[\frac{1}{N_{\text{var}} - 1} \sum_{t=1}^{N_{\text{var}}-1} (\upsilon_{k,t} - \upsilon_{k,t-1})^2\Big]^{1/2}, \qquad \widehat{V}_\upsilon := \frac{1}{|\mathcal{K}_\upsilon|} \sum_{k \in \mathcal{K}_\upsilon} V_{\upsilon,k}, \quad \widehat{J}_\upsilon := \frac{1}{|\mathcal{K}_\upsilon|} \sum_{k \in \mathcal{K}_\upsilon} J_{\upsilon,k}. \end{aligned} \tag{37}$$

Here $V_{\upsilon,k}$ is the sample variance and $J_{\upsilon,k}$ is the root mean square of the $N_{\text{var}} - 1$ one-step changes of $\upsilon_{k,t}$. Their means are the temporal variance $\widehat{V}_\upsilon$ and the root-mean-square change $\widehat{J}_\upsilon$.

1.2.6 Table Entries

Each row gives one method in one scenario and setting, with the means $\pm$ standard deviations over the trajectories of the metrics of Sections 1.2.1 and 1.2.4, the counts of Section 1.2.2 and the upper confidence bounds of Section 1.2.3. Means over empty sets and standard deviations over fewer than two trajectories are undefined. Contact, the event $\psi(y_{k,t}) \le 0$ for some $1 \le t \le N_k$, is detected only at the decision times, where it is sampled. Section 2.1 gives the key to the columns.

1.3 Parameters, Nominal Input and Lipschitz Constants

This subsection gives the parameters, the nominal input and the safety filter, and it derives the Lipschitz constants K_x^z and K_u^z of the predicted nose point. Their products with L_{geom} are the constants L_x^{geom} and L_u^{geom} of V-U and V-Δ in (45). Table 1 in Section 3 gives the parameters of both scenarios and the input bounds, and Table 2 gives the obstacle coordinates.

1.3.1 Nominal Input and CBF Safety Filter

Both scenarios steer by the heading error $\tilde{\theta}$ to a target point, and the function clip saturates its argument at the corresponding bound of Table 1. The following table compares the two scenarios.

Item	Corridor	Maze
Target point	goal	one look-ahead distance beyond the point nearest to y on a fixed reference curve
Nominal input	$u_t^{\mathrm{nom}} = (v^{\mathrm{nom}}, \omega^{\mathrm{nom}})$	$u_t^{\mathrm{nom}} = (a^{\mathrm{nom}}, \nu^{\mathrm{nom}})$
Filter objective	$(v - v^{\mathrm{nom}})^2 + w_\omega(\omega - \omega^{\mathrm{nom}})^2$	$(a - a^{\mathrm{nom}})^2 + w_\nu(\nu - \nu^{\mathrm{nom}})^2$
Finite input set	41 equally spaced yaw rates and the nominal yaw rate, each with the feasible speed closest to v^{nom}, selected analytically	41×25 equally spaced grid of accelerations and yaw accelerations, augmented by the nominal components a^{nom} and ν^{nom}

In the Maze, the nearest-point search uses a nondecreasing progress index and a forward window of at most 400 sampled points. The nominal input of the Corridor, and the desired yaw rate ω^{des}, desired speed v^{des} and nominal input of the Maze, are

$$\begin{aligned}
&\text{Corridor:} && v^{\mathrm{nom}} := v_{\mathrm{c}} \max\{\cos\tilde{\theta}, 0\}, && \omega^{\mathrm{nom}} := \mathrm{clip}(k_\theta \tilde{\theta}),\\
&\text{Maze:} && v^{\mathrm{des}} := v_{\mathrm{c}} \max\{\cos\tilde{\theta}, 0\} \min\{1, d_{\mathrm{goal}}/(0.6\,\mathrm{m}) + 0.3\}, && \omega^{\mathrm{des}} := \mathrm{clip}(k_\theta \tilde{\theta}),\\
& && a^{\mathrm{nom}} := \mathrm{clip}(k_v(v^{\mathrm{des}} - v)), && \nu^{\mathrm{nom}} := \mathrm{clip}(k_\omega(\omega^{\mathrm{des}} - \omega)),
\end{aligned} \tag{38}$$

where d_{goal} is the distance to the goal. The safety filter (1) replaces the objective $\|u - u_t^{\mathrm{nom}}\|^2$ by the filter objective of the table and minimizes it over the finite input set. Table 1 also gives the feedback coefficients k_θ, k_ω, k_v, the cruise speed v_{c}, the look-ahead distance and the weights w_ω and w_ν.

1.3.2 Lipschitz Constants

This part derives the Jacobian bounds of the predicted nose point $z(x,u) = q(\hat{f}(x,u))$. With the induced norm $\|\cdot\|_2$, the target is

$$\|\partial_x z\|_2 \le K_x^z, \qquad \|\partial_u z\|_2 \le K_u^z, \tag{39}$$

if the absolute speed is at most $\bar{v}$ throughout each sampling period. The speed limits of Table 1 give $\bar{v} = 0.6\,\mathrm{m/s}$ in the Corridor and $\bar{v} = 0.8\,\mathrm{m/s}$ in the Maze, and these values give the entries K_x^z and K_u^z of

Table 1. The bounds extend to differences on every convex operating region whose state-input pairs satisfy this speed bound.

Within one sampling period, the input is constant. Fix (x, u) and let $\tilde{t} \in [0, T_s]$ be the time elapsed within the period. By the vehicle models of [1, App. F-A], the speed and the heading are

$$v(\tilde{t}) = v + a\tilde{t}, \qquad \theta(\tilde{t}) = \theta + \omega\tilde{t} + \nu\tilde{t}^2/2, \tag{40}$$

with $a = \nu = 0$ in the Corridor, where v and ω are inputs. Hence $\hat{f}(x, u)$ has the heading $\theta^+ = \theta(T_s)$ and the position $y^+ = y + \int_0^{T_s} v(\tilde{t})\, e(\theta(\tilde{t}))\, d\tilde{t}$. Since the nose point is $q(x) = y + l\, e(\theta)$, the predicted nose point is

$$z = y + \int_0^{T_s} v(\tilde{t})\, e(\theta(\tilde{t}))\, d\tilde{t} + l\, e(\theta^+). \tag{41}$$

Differentiating (41) gives the columns of the Jacobians. With $e_\perp(\theta) := (-\sin\theta, \cos\theta)$, the derivative of $e(\theta)$ is $e_\perp(\theta)$. By (40), $\theta(\tilde{t})$ has the derivatives 1, $\tilde{t}$, and $\tilde{t}^2/2$ in θ, ω, and ν, and $v(\tilde{t})$ has the derivatives 1 and $\tilde{t}$ in v and a. Neither depends on y. By the chain rule under the integral, $\partial_y z = I_2$ and

$$\begin{aligned}
&\partial_\theta z = \textstyle\int_0^{T_s} v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + l\, e_\perp(\theta^+), && \partial_v z = \textstyle\int_0^{T_s} e(\theta(\tilde{t}))\, d\tilde{t},\\
&\partial_\omega z = \textstyle\int_0^{T_s} \tilde{t}\, v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + lT_s\, e_\perp(\theta^+), && \partial_a z = \textstyle\int_0^{T_s} \tilde{t}\, e(\theta(\tilde{t}))\, d\tilde{t},\\
&\partial_\nu z = \textstyle\int_0^{T_s} \frac{\tilde{t}^2}{2}\, v(\tilde{t})\, e_\perp(\theta(\tilde{t}))\, d\tilde{t} + \frac{lT_s^2}{2}\, e_\perp(\theta^+).
\end{aligned} \tag{42}$$

The terms outside the integrals come from $l\, e(\theta^+)$ with $\theta^+ = \theta(T_s)$.

Each column is bounded through the norm of its integrand. Since the norm of an integral is at most the integral of the norm, and since $|v(\tilde{t})| \le \bar{v}$, $\|e\| = \|e_\perp\| = 1$ and $\int_0^{T_s} \tilde{t}^k\, d\tilde{t} = T_s^{k+1}/(k+1)$,

$$\begin{aligned}
&\|\partial_\theta z\| \le l + \bar{v}T_s, && \|\partial_v z\| \le T_s,\\
&\|\partial_\omega z\| \le lT_s + \bar{v}T_s^2/2, && \|\partial_a z\| \le T_s^2/2,\\
&\|\partial_\nu z\| \le lT_s^2/2 + \bar{v}T_s^3/6.
\end{aligned} \tag{43}$$

The column bounds give the matrix norms. Let J collect the columns of $\partial_x z$ other than $\partial_y z$, so that $\partial_x z = [I_2, J]$ with

$$\begin{aligned}
&\text{Corridor:} \quad J = \partial_\theta z, && \partial_u z = [\partial_v z, \partial_\omega z],\\
&\text{Maze:} \quad J = [\partial_\theta z, \partial_v z, \partial_\omega z], && \partial_u z = [\partial_a z, \partial_\nu z].
\end{aligned}$$

The matrix $[I_2, J][I_2, J]^\top = I_2 + JJ^\top$ has the largest eigenvalue $1 + \|J\|_2^2$. Since the spectral norm is at most the Frobenius norm $\|\cdot\|_F$,

$$\|\partial_x z\|_2^2 = 1 + \|J\|_2^2 \le 1 + \|J\|_F^2, \qquad \|\partial_u z\|_2^2 \le \|\partial_u z\|_F^2, \tag{44}$$

where each squared Frobenius norm is the sum of the squared column norms. Substituting (43) into (44) yields $\|\partial_x z\|_2 \le K_x^z$ and $\|\partial_u z\|_2 \le K_u^z$. The resulting constants are those of Table 1. In the Corridor, for example,

$$\|J\|_F^2 = \|\partial_\theta z\|^2 \le (l + \bar{v}T_s)^2, \qquad \|\partial_u z\|_F^2 \le T_s^2 + (lT_s + \bar{v}T_s^2/2)^2.$$

The Jacobian bounds extend to differences along segments. Let (x, u) and (x', u') lie in a convex operating region that satisfies the speed bound, and let $(x_\lambda, u_\lambda) := (x', u') + \lambda(x - x', u - u')$ for $\lambda \in [0, 1]$. By convexity, this segment lies in the region, so the Jacobian bounds hold along it. Integrating the Jacobian

of z along the segment,

$$z(x,u) - z(x',u') = \int_0^1 \left[\partial_x z(x_\lambda,u_\lambda)(x-x') + \partial_u z(x_\lambda,u_\lambda)(u-u')\right] d\lambda.$$

Bounding the norm of the integrand by $K_x^z\|x-x'\| + K_u^z\|u-u'\|$ gives the Lipschitz bound of V-U and V-Δ, $\|z(x,u) - z(x',u')\| \le K_x^z\|x-x'\| + K_u^z\|u-u'\|$.

Two properties of the Maze keep these bounds valid for the simulated predictor. By (40), $v(\tilde{t})$ is affine in $\tilde{t}$. Hence $|v(\tilde{t})| \le \bar{v}$ holds when the current speed v and the predicted speed $v + aT_s$ satisfy the limits of Table 1. The nominal one-step predictor $\hat{f}$ of the Maze evaluates the integral in (41) with the 12-point Gauss–Legendre rule on $[0, T_s]$. The derivatives of the computed nose point are therefore the same rule applied to the integrands in (42).

The rule has positive weights and integrates 1, $\tilde{t}$ and $\tilde{t}^2$ exactly. The norm of each rule sum is therefore at most the rule applied to the norm bound of its integrand. This bound is a multiple of 1, $\tilde{t}$ or $\tilde{t}^2$, so the rule returns its exact integral. Hence (43), and with it the Maze constants, also hold for the computed nose point.

With $L_{\mathrm{geom}} := \max_{c\in\Gamma} L_{\mathrm{geom}}(c)$ of [1, App. D-B], the constants of V-U and V-Δ are

$$L_x^{\mathrm{geom}} = L_{\mathrm{geom}} K_x^z, \qquad L_u^{\mathrm{geom}} = L_{\mathrm{geom}} K_u^z, \tag{45}$$

and Table 1 lists them.

1.4 Assumption Checks

As in [1, App. E-C], the checks test Assumptions 1 and 2 and the tightened condition along the recorded trajectories. They support the guarantees but do not prove them.

Each trajectory records the following quantities for the applied input, at every decision time or every executed transition.

Quantity	Recorded at	Flagged if
feasibility of $\mathcal{U}(x_t, c_{\max})$	decision times	$\mathcal{U}(x_t, c_{\max})$ is empty, which ends the trajectory
barrier-loss bound $Y_t \le c_{\max}$	transitions	$Y_t > c_{\max} + \varepsilon_{\mathrm{chk}}$
residual of the tightened condition (1)	transitions	residual below $-\varepsilon_{\mathrm{num}}$
CBF violation, indicator $I_{\mathrm{CBF},k,t}$ in (26)	transitions	$h(x_{k,t+1}) < (1-\gamma)h(x_{k,t}) - \varepsilon_{\mathrm{num}}$

The tolerances are $\varepsilon_{\mathrm{chk}} = 10^{-12}$ and the numerical tolerance $\varepsilon_{\mathrm{num}} := 10^{-10}$. Each trajectory also records its stopping reason.

Assumption 2 is checked at every decision time t with $c_t < c_{\max}$, for the test with the supporting test index r_t of (6). The check uses the bound excesses

$$\begin{aligned} e_{\mathrm{pw},t} &:= \max_{\sigma_{r_t} \le s < \tau_{r_t}(c_t)} [\widehat{p}_t(c_t) - \widehat{p}_s(c_t) - d_t^{\mathrm{V}}(c_t; s, s+1)], \\ e_{\mathrm{A2},t} &:= \widehat{p}_t(c_t) - \min_{\sigma_{r_t} \le s < \tau_{r_t}(c_t)} \widehat{p}_s(c_t) - d_t^{(r_t)}(c_t). \end{aligned} \tag{46}$$

Here $d_t^{\mathrm{V}}(c_t; s, s+1)$ is the risk-increase term of the deployed variant at s, capped at 1. For V-C the terms are at most $\bar{P}(c_t) \le 1/2$, so the cap does not apply.

Let $\sigma := \sigma_{r_t}$ and $\tau := \tau_{r_t}(c_t)$ bound the monitoring period. The two excesses test the bound at different resolutions. Since

$$e_{\mathrm{A2},t} = \max_{\sigma \le s < \tau} [\widehat{p}_t(c_t) - \widehat{p}_s(c_t)] - d_t^{(r_t)}(c_t),$$

the inequality $e_{\mathrm{A2},t} \le 0$ is (7) of Assumption 2 at t for every s of the monitoring period, with $\widehat{p}$ in place of p. The inequality $e_{\mathrm{pw},t} \le 0$ requires the same of each risk-increase term at its own s.

The pointwise check is the stricter one. Since $\max_s \min\{1, \cdot\} = \min\{1, \max_s \cdot\}$ and a failed required check sets both sides to 1, (5) and the definition of d_t^{V} (Variants and Baselines in the Overview) give

$$d_t^{(r_t)}(c_t) = d_t^{\mathrm{V}}(c_t; \sigma, \tau) = \max_{\sigma \le s < \tau} d_t^{\mathrm{V}}(c_t; s, s+1). \tag{47}$$

Let s^* minimize $\widehat{p}_s(c_t)$ over $\sigma \le s < \tau$. By the definition of $e_{\mathrm{A2},t}$ and (47),

$$\begin{aligned} e_{\mathrm{A2},t} &= \widehat{p}_t(c_t) - \widehat{p}_{s^*}(c_t) - \max_{\sigma \le s < \tau} d_t^{\mathrm{V}}(c_t; s, s+1) \\ &\le \widehat{p}_t(c_t) - \widehat{p}_{s^*}(c_t) - d_t^{\mathrm{V}}(c_t; s^*, s^*+1) \le e_{\mathrm{pw},t}. \end{aligned}$$

Hence $e_{\mathrm{A2},t} \le e_{\mathrm{pw},t}$.

For a bound factor $\phi < 1$, every term in (47) is scaled by ϕ, and these relations still hold. A check fails if an excess exceeds $\varepsilon_{\mathrm{chk}}$. The checks compare estimates, so a passed check supports Assumption 2 at the checked decisions but does not prove it. The frequencies $\widehat{q}_{\mathrm{pw}}$ and $\widehat{q}_{\mathrm{A2}}$ count the trajectories with a failed check.

2 Ablation Results

This section gives the ablation results of the simulations of [1, App. F] for the default setting $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$, S1–S5, S*, and the main grid (Table 3). Section 3 collects the tables and figures in the same order, under the numbers of these subsections. The data and the scripts are in the repository[1].

Section	Topic	Displays
2.1	table key and conventions	Table 3
2.2	default and selected settings	Tables 4–8, Figs. 1–30
2.3	α and α_0	Figs. 31–41
2.4	δ	Tables 9–10
2.5	K	Tables 11–12, Figs. 42–44
2.6	b	Tables 13–14
2.7	ϕ and L_μ	Tables 15–18
2.8	event and assumption checks	Tables 19–22
2.9	variation over time	Tables 23–24

2.1 Compared Parameters and Table Key

Table 3 gives the parameter values. The columns of the result tables are the quantities of Section 1.2. Tables 9–16 of the δ, K, b, and ϕ comparisons omit the event counts, which the sweeps (Section 2.2.3) show. For ϕ, Tables 17 and 18 also give them for S1–S5 with their variants.

[1] `https://github.com/omidmirza/ecbf-ablation`

2.1.1 Grid Cells, Cases and Series

The results are counted over the units below, built from the settings of Table 3.

Term	Meaning
Grid cell	one of the 375 main-grid settings with one variant in one scenario
Case	the default setting or one of S1–S5, with one variant, one scenario, and one value per compared parameter
Series	grid cells or cases that share the variant, the scenario, and all parameters except those varied
Matched baselines	the baselines at the same α_0 and L_μ as a case
Detailed checks	$192 = 6 \times 4 \times 2 \times 2 \times 2$ cases: the default and the five selected settings, four variants, two scenarios, two values of L_μ, and two test periods (K of the setting and $K = \infty$)

S1–S5 with their variants are the five pairs of the Overview, and a figure that shows one variant at the default setting shows V-C. The detailed checks (Section 2.8) compare the event counts with the bounds of Theorems 1–3 and evaluate the assumption checks of Section 1.4.

2.1.2 Compared Parameters

The table states how each parameter acts. Sections 2.3–2.7 report the effects.

Parameter	Enters through	Effect of a larger value
margin insufficiency α	each factor $1 + \lambda(\alpha - Z_{t-1}(c))$ of (3) and the exceedance buffer $\alpha_0 - \alpha$ of (5)	shrinks the null hypothesis $H_0(c)$ and speeds up certification, but narrows the exceedance buffer
CBF-violation probability α_0	the exceedance buffer $\alpha_0 - \alpha$ and the exceedance risk limit $p_t(c_t) \leq \alpha_0$	at fixed α, widens the exceedance buffer and relaxes the exceedance risk limit
error confidence δ	the error levels $\delta_r = \delta/(b(r+1)\ln^2(r+1))$ (4)	lowers every threshold $1/\delta_r$, and changes the expected-certification-time bound of [1, Sec. IV-B] only through $\log(\lvert\Lambda\rvert/\delta_r)$
test period K	the starting times $\sigma_r = (r-1)K$	gives older starting times, but a test that starts at a given time has a smaller index r and a lower threshold $1/\delta_r$
budget normalizer b	the error levels δ_r, with $\sum_{r \geq 1} \delta_r = \delta\kappa/b$	lowers every δ_r by the same factor, so Theorem 3 holds with $\delta\kappa/b \leq \delta$ in place of δ, and changes the expected-certification-time bound only logarithmically
bound factor ϕ	the current risk-increase bound (5), ϕ times the certificate bound $d_t^{\mathrm{V}}(c; \sigma, \tau)$, including the value 1 of a failed check	raises the bound to the certificate bound at $\phi = 1$
distribution-drift bound L_μ	the disturbance (10) and the drift term $\Delta_{t,s} = \min\{1, L_\mu(t-s)\}$ in every risk-increase term	makes the magnitude distribution change faster and raises the drift term

A certificate of V-U, V-Δ or V-L supports deployment only within the window (49) (Section 2.5.1), and (50) does not exclude deployment by a V-C certificate for up to twice this window. Values $\phi < 1$ shrink the bound, which then need not satisfy Assumption 2, so their results carry no guarantee. At $\phi = 0$, every such bound is zero, and deployment requires only a certificate.

2.1.3 Display Conventions

The key below explains the columns of the tables. The following conventions complement it.

- For QT at $\alpha_0 = 0.15$ in the Maze (Tables 5 and 21), an event count ranges from the confirmed to the possible events.
- Every upper confidence bound $\overline{q}$ (28) is rounded up and is not adjusted for the number of settings or for the selection of S1–S5. The text writes an event count as $n/200$, followed by $\overline{q}$ in parentheses.
- With $N_{\mathrm{MC}} = 200$, zero possible events give $\overline{q} = 1.5\%$, so no count is consistent with a bound below 1.5%. For $\phi < 1$, no guarantee applies, and δ and δ_1 serve only for comparison.
- Counts of trajectories with a CBF or safe-set violation carry no bound.
- Box plots show the median and the quartiles. Whiskers reach the most extreme values within 1.5 interquartile ranges of the quartiles in the distribution figures and the 5th and 95th percentiles in the sweeps.

Key to all tables. $\pm$: SD; $^{+}$: decision fraction; $\downarrow$/$\uparrow$: lower/higher is better; counts: trajectories of N_{MC} with a possible event, or the range from confirmed to possible events ([1, App. E-A]), except $\widehat{q}_{\mathrm{cert}}$ of S*, which gives confirmed events (Table 8), 95% upper confidence bound (%) from possible events in parentheses; ✓: bound at most δ, or δ_1 for $K = \infty$; N-A: not applicable; bold: best mean of the four variants at one setting (ties all bold; never SDs, $\widehat{m}_{\mathrm{cert}}$, $\widehat{m}_{\tau^{\mathrm{use}}}$, $\widehat{m}_{p^{\max}}$); a/m: a trajectories with an excess above $\varepsilon_{\mathrm{chk}}$ among the m with a checked decision (a alone if $m = N_{\mathrm{MC}}$), bound over all N_{MC}.

2.2 Default and Selected Settings

This subsection compares the methods at the default setting $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$ and the four variants at S1–S5 and S*, and then shows trajectories, distributions, and sweeps. Tables 4 and 5 compare all nine methods at the default setting, and Tables 6 and 7 give S1–S5 with all four variants. Both pairs of tables include the matched baselines (Section 2.1.1) and the secondary metrics of Section 1.2.4. Section VI-C and Appendix F-C of [1] discuss S1–S5 in the Maze and the Corridor.

2.2.1 Setting S*

S* is an additional Maze setting with $(\alpha_0, \alpha, \delta, K) = (0.25, 0.225, 0.15, 50)$ and the default values of the other parameters. It lies outside the main grid, since $\alpha = 0.225$ is not a grid value. Table 8 gives S* with all four variants and has no baseline rows. All trajectories reach the goal without contact. The false-certification counts are confirmed events (Section 2.1.3). For V-C, 3 trajectories have a confirmed and 4 a possible false-certification event, and the upper confidence bound 4.6% of the possible count lies below $\delta = 0.15$, consistent with Theorem 3.

2.2.2 Trajectories and Distributions

Figs. 1–12 show the nose-point trajectories of the default setting with V-C and of S1–S5 with their variants. The panels show the following.

(a) Nose-point paths. "Contact" marks the contact boundaries of the body center, and "$h = 0$" (Maze: "$h = 0$ at $v = 0$") the zero level of the barrier h. The inset shows an obstacle passage where the example trajectory passes closer than that of the $c_{\max}$ baseline.

(b) Robustness margin c_t of the example trajectory, and its mean over the trajectories not yet at the goal.

(c) Barrier loss Y_t of the example trajectory, with c_t. The percentage is its share q_k^+ (25) of decisions with $Y_t > c_t$, for the indicator $I_{Y,k,t}$ (36).
(d) Estimated exceedance risk $\widehat{p}_t(c_t)$ (19) of the example trajectory.
(e) Supporting test index r_t (6) of the example trajectory, and its mean as in (b).

The example trajectory has the same index k in every figure, chosen independently of the outcomes. Figs. 13–18 show the distributions of the per-trajectory values $T_{\mathrm{mis},k}$, $m_{c,k}/c_{\max}$, $q^+_{\mathrm{CBF},k}$, $q^+_{\mathrm{unsafe},k}$ (25), (26), $\widehat{p}_k^{\max}$ (33), and $h_k^{\min}$ (35) at the default setting and at S1–S5. Points are trajectories, and the boxes and whiskers follow Section 2.1.3. Unfiltered control is omitted for scale.

2.2.3 One-Parameter Sweeps

Figs. 19–30 vary one parameter (Section 2.1.2) per column around the default setting and each of S1–S5: α, δ, K, b, and ϕ, the last at both values of L_μ. The rows show $m_{c,k}/c_{\max}$, $q^+_{\mathrm{CBF},k}$, $q^+_{\mathrm{unsafe},k}$, and $T_{\mathrm{mis},k}$, defined in (25) and (26), as boxes (Section 2.1.3) with the mean as a marker, and the event counts with bars to their upper confidence bounds (28). A row that is zero for every variant is omitted. The level line is N_{MC} times the bound, δ or δ_1, of the guarantee the count checks. ACI and QT are drawn at $L_\mu = 0.0005$ in every column, as lines where their values lie within the axis range (Tables 15 and 16 give them at $L_\mu = 0.005$). Right-hand labels give their values, with ▲/▼ for a value above/below the axis range. At $\phi = 1$, every count in these figures is consistent with its guarantee except at S4 with $K = \infty$ (Section 2.8).

In the Maze around S3, reducing δ from 0.20 to 0.025 raises the $\widehat{m}_c/c_{\max}$ of V-C from 0.607 to 0.688 and lowers its $\widehat{q}^+_{\mathrm{CBF}}$ from 5.38 to 3.60%, since every threshold $1/\delta_r$ rises (Section 2.1). Reducing $\alpha_0 - \alpha$ from 0.25 to 0.05 lowers its $\widehat{m}_c/c_{\max}$ from 0.928 to 0.517 and raises its $\widehat{q}^+_{\mathrm{CBF}}$ from 0.27 to 7.40% (Section 2.3).

2.3 Margin Insufficiency and CBF-Violation Probability

This subsection shows how α and α_0 act on the deployed margins and the violations (Section 2.1), with $\phi = 1$ and $L_\mu = 0.0005$. Figs. 31–41 show heatmaps of the four variants over the main grid (Table 3).

	$\alpha_0 = 0.15$	$\alpha_0 = 0.30$	$\alpha_0 = 0.25$
Corridor	$\widehat{q}^+_{\mathrm{CBF}}$, $\widehat{m}_c/c_{\max}$: Figs. 31, 33	$\widehat{m}_c/c_{\max}$, $\widehat{q}^+_{\mathrm{CBF}}$, $\widehat{q}^+_{\mathrm{unsafe}}$: Figs. 35–37	$\widehat{q}^+_{\mathrm{unsafe}}$: Fig. 41
Maze	$\widehat{q}^+_{\mathrm{CBF}}$, $\widehat{m}_c/c_{\max}$: Figs. 32, 34	$\widehat{m}_c/c_{\max}$, $\widehat{q}^+_{\mathrm{CBF}}$, $\widehat{q}^+_{\mathrm{unsafe}}$: Figs. 38–40	–

Each cell gives the mean with one sample standard deviation below it, with bold as in the table key (Section 2.1.3). Each color scale covers the range of its metric and scenario over the main grid.

- Over the 1500 grid cells of each scenario, the Spearman rank correlation of $\widehat{m}_c/c_{\max}$ is at least 0.99 with $\widehat{T}_{\mathrm{mis}}$ and at most -0.98 with $\widehat{q}^+_{\mathrm{CBF}}$. Smaller margins thus accompany shorter missions and more CBF violations, almost monotonically, so $\widehat{m}_c/c_{\max}$ alone summarizes this trade-off.
- The values of $\widehat{m}_c/c_{\max}$ are ordered like the certificate bounds in (9), V-C lowest and V-U highest, for 731 of the 750 pairs of a main-grid setting and a scenario. Those of V-Δ and V-U, which differ only in the geometric correction and the required check, differ by at most 0.003. For the same history, the deployable sets are nested in this order, so V-C deploys the smallest margin (5). The deployed margins change the inputs, however, so the histories differ, and the other 19 pairs deviate.
- Across α_0, $\widehat{m}_c/c_{\max}$ of V-C changes by at most 0.016 at fixed (α, δ, K), and its $\widehat{q}^+_{\mathrm{nodep}}$ (32) stays between 1.0% and 3.3% over the main grid. For V-U and V-Δ, $\widehat{q}^+_{\mathrm{nodep}}$ reaches 84% in the Corridor and 89% in the Maze, and at $(\alpha, \delta, K) = (0.10, 0.05, 50)$, their $\widehat{m}_c/c_{\max}$ in the Corridor falls from 1 to 0.84

as α_0 grows from 0.15 to 0.30. A larger α_0 widens the exceedance buffer $\alpha_0 - \alpha$, which their larger bounds then meet more often.

- For every $\alpha_0 \geq 0.15$, δ, K, and scenario, V-C has its smallest $\widehat{m}_c/c_{\max}$ at the largest $\alpha = \alpha_0 - 0.05$, while at $\alpha_0 = 0.30$, V-U, V-Δ, and V-L have it at $\alpha \in \{0.15, 0.20\}$. A larger α speeds up certification but narrows the exceedance buffer. For V-C, whose bound meets the smallest exceedance buffer, faster certification prevails. With $\alpha_0 - \alpha = 0.05$ fixed, $\widehat{m}_c/c_{\max}$ is nonincreasing in α_0 in all 200 series, as α grows with α_0.
- For $\alpha_0 \leq 0.20$ in the Corridor and $\alpha_0 \leq 0.25$ in the Maze, $\widehat{q}^{+}_{\text{unsafe}}$ of every variant is below 0.004%. In the 200 series with $\alpha_0 - \alpha = 0.05$ fixed, the number of trajectories with a safe-set violation at $\alpha_0 = 0.30$ is never below that at $\alpha_0 = 0.10$. A larger α_0 relaxes the exceedance risk limit $p_t(c_t) \leq \alpha_0$ and so permits more CBF violations and, through them, more safe-set violations.

2.4 Error Confidence

Tables 9 and 10 vary the error confidence δ over the values of the main grid at the fixed (α_0, α, K) of the default setting and of S1–S5, with $L_\mu = 0.0005$. S1 and S2 share one series, since they differ only in δ.

A larger δ lowers every threshold $1/\delta_r$ in (4), so on the same observations a test certifies no later, while the expected-certification-time bound changes only through $\log(|\Lambda|/\delta_r)$ (Section 2.1.2). The eightfold range of δ changes this term by $\ln 8 \approx 2.1$.

- From $\delta = 0.025$ to 0.20, $\widehat{m}_c/c_{\max}$ decreases in 38 of the 40 series, by at most 0.095, and the other two remain equal to one (V-U and V-Δ in the Maze at the default (α_0, α, K)). These two variants never deploy a margin below $c_{\max}$ there.
- At the default (α_0, α, K) with $\delta = 0.20$, V-C has deployment events in both scenarios.

The small decreases match the logarithmic dependence on δ_r. The events at $\delta = 0.20$ reflect the weaker evidence that a lower threshold requires. Theorem 3 bounds the probability of a deployment event by δ, so a larger δ permits more events.

2.5 Test Period

Tables 11 and 12 vary the test period K over the values of the main grid at the fixed $(\alpha_0, \alpha, \delta)$ of the default setting and of S1–S5, with $L_\mu = 0.0005$. Section 2.5.1 derives how K and L_μ limit deployment, and Section 2.5.2 relates the results to this limit.

2.5.1 Deployment Time

Deployment requires a current risk-increase bound $d_t^{(r)}(c) \leq \alpha_0 - \alpha$ (5). With $L_\mu > 0$, the drift term in this bound grows with the time since the test started. A certificate of V-U, V-Δ or V-L therefore supports deployment for at most $(\alpha_0 - \alpha)/L_\mu$ decisions after its starting time (49), and the corresponding lower bound for V-C can exclude deployment only after at least twice this time. The notation is that of the Overview.

Fix a certificate (c, r) with starting time $\sigma = \sigma_r$ and monitoring period $[\sigma, \tau)$, and a decision time $t \geq \tau$. For $\mathrm{V} \in \{\mathrm{U}, \Delta, \mathrm{L}\}$, the certificate bound $d_t^{\mathrm{V}}(c; \sigma, \tau)$ equals 1 if a required check fails. Otherwise, it is the minimum of 1 and the maximum over $\sigma \leq s < \tau$ of the risk-increase term of the variant (Overview). Each term contains the drift term through an effective drift in (8).

The geometric corrections and all other summands are nonnegative, and $\Delta_{t,\sigma} \leq 1$. The term at $s = \sigma$ thus gives the following bound, which also holds when a check fails:

$$d_t^{\mathrm{V}}(c; \sigma, \tau) \geq \min\{1, L_\mu(t - \sigma)\}, \qquad \mathrm{V} \in \{\mathrm{U}, \Delta, \mathrm{L}\}. \tag{48}$$

Deployment at t in (5) requires $d_t^{(r)}(c) \le \alpha_0 - \alpha$. For $\phi = 1$, $d_t^{(r)}(c)$ equals $d_t^{\mathrm{V}}(c;\sigma,\tau)$. Since $\alpha_0 - \alpha < 1$, (48) shows that this holds only if

$$L_\mu(t - \sigma) \le \alpha_0 - \alpha. \tag{49}$$

The table gives this window, $(\alpha_0 - \alpha)/L_\mu$ decisions, for the two exceedance buffers of the default and the selected settings.

Exceedance buffer $\alpha_0 - \alpha$	$L_\mu = 0.0005$	$L_\mu = 0.005$
0.05 (default, S1, S2, S4, S5)	100 decisions	10 decisions
0.10 (S3)	200 decisions	20 decisions

For V-C, the term at $s = \sigma$ satisfies $(1 - \Delta_{t,\sigma})B_{t,\sigma}(c) + \Delta_{t,\sigma}\bar{P}(c) \ge \Delta_{t,\sigma}\bar{P}(c)$, since $B_{t,\sigma}(c) \ge 0$ and $\Delta_{t,\sigma} \le 1$. Hence

$$d_t^{\mathrm{C}}(c;\sigma,\tau) \ge \bar{P}(c)\min\{1, L_\mu(t - \sigma)\}, \tag{50}$$

with the fraction bound $\bar{P}(c) \le 1/2$. The right-hand side of (50) is therefore at most $\alpha_0 - \alpha$ whenever $L_\mu(t - \sigma) \le 2(\alpha_0 - \alpha)$. Hence (50) does not exclude deployment by a V-C certificate for up to twice the window of (49), and does not limit $t - \sigma$ at all if $\bar{P}(c) \le \alpha_0 - \alpha$. In every case, deployment still requires the certificate bound $d_t^{\mathrm{C}}(c;\sigma,\tau)$ to be at most $\alpha_0 - \alpha$.

2.5.2 Results

The results of Tables 11 and 12 are explained by the bounds (49) and (50) of Section 2.5.1. They give three findings.

- $K = \infty$ gives the largest $\widehat{m}_c/c_{\max}$, or ties for it, in all 48 series.
- Over the 48 series, the smallest unrounded mean $\widehat{m}_c/c_{\max}$ occurs at $K = 50$ in 27, at $K = 10$ in 17, and at $K = 100$ in two. The other two, V-U and V-Δ in the Maze at the default setting, remain equal to one. In 30 series, two or more values of K share this minimum at two decimals.
- The $\widehat{m}_c/c_{\max}$ of V-C varies by at most 0.025 over K, and its mean mission time by at most 0.2% of its smallest value over K.

With $K = \infty$, only test 1 starts, at $\sigma_1 = 0$. The certificates of V-U, V-Δ, and V-L then support deployment only during the first $(\alpha_0 - \alpha)/L_\mu$ decisions (table in Section 2.5.1). Afterwards, the controller uses $c_{\max}$ by (5), which gives the largest margins.

For $K = 200$, outside S3, each test period also ends with decisions beyond the window of the latest starting time, at which no certificate of these variants supports deployment. A smaller K removes these gaps, but a test that starts at a given time then has a larger index r and a higher threshold $1/\delta_r$, which delays its certification. The minima at $K = 50$ and $K = 10$ reflect this balance. V-C depends least on K, since (50) does not exclude its deployment for up to twice the window, and never when $\bar{P}(c) \le \alpha_0 - \alpha$.

Figs. 42–44 compare the event frequencies with the bounds of Theorems 1–3 for S3 with V-L in the detailed checks (Section 2.1.1) and for the default setting with V-C at the same values of K and L_μ.

2.6 Budget Normalizer

Tables 13 and 14 vary the budget normalizer b at the default setting and at S1–S5 over $b \in \{\kappa, 2.5, 3, 4\}$ (Table 3), with K of the setting, $\phi = 1$, and $L_\mu = 0.0005$.

The budget normalizer $b \ge \kappa$ replaces κ in the error levels $\delta_r = \delta/(b(r+1)\ln^2(r+1))$ of (4), so that $\sum_{r\ge1}\delta_r = \delta\kappa/b \le \delta$. A larger b raises every threshold $1/\delta_r$, so on the same observations a test certifies

no earlier, but the term $\log(|\Lambda|/\delta_r)$ of the expected-certification-time bound grows only by $\ln(4/\kappa) \approx 0.64$ from $b = \kappa$ to $b = 4$. Theorem 3 holds with $\delta\kappa/b$ in place of δ.

- In all 48 series, $\widehat{m}_c/c_{\max}$ is nondecreasing in b.
- From $b = \kappa$ to $b = 4$, $\widehat{m}_c/c_{\max}$ increases in 46 series, by at most 0.035, and the mean mission time increases in 44. The small increases match the logarithmic dependence on δ_r.
- No case has a deployment event.

2.7 Bound Factor and Distribution-Drift Bound

Tables 15 and 16 vary the bound factor ϕ and the distribution-drift bound L_μ of (10) for the default setting with V-C and S1–S5 with their variants. Values $\phi < 1$ need not satisfy Assumption 2 and carry no guarantee. The ϕ columns of the sweeps (Section 2.2.3) show all four variants. The series counts in this subsection cover 96 series, one for each combination of a setting (the default and S1–S5), a variant, a scenario, and a value of L_μ ($6 \times 4 \times 2 \times 2$). Tables 15 and 16 print the baselines, the default setting with V-C, and S1–S5 with their variants, and the repository contains the other cases.

A bound factor $\phi < 1$ scales down the current risk-increase bound (5), so a certificate meets the deployment condition earlier and for longer (Section 2.1). The scaled bound can fall below the actual risk increase and so violate (7) of Assumption 2, which Theorems 2 and 3 require. The table counts the series with a bound excess $e_{\mathrm{A2},t}$ (46) above $\varepsilon_{\mathrm{chk}}$ and with a deployment event.

ϕ	Series with $e_{\mathrm{A2},t} > \varepsilon_{\mathrm{chk}}$	Series with a deployment event
1	0	0
0.75	16	3
0.5	22	6
0.25	52	15
0	96	92

As Section 1.4 shows, such an excess indicates a violation of (7) with $\widehat{p}$ in place of p, so these counts show how often the scaled bounds fail this condition. Deployment events occur only at $\phi < 1$, where Theorems 2 and 3 no longer apply. Reducing ϕ from one to zero lowers $\widehat{m}_c/c_{\max}$ in all 96 series, raises the CBF-violation fraction $\widehat{q}^{+}_{\mathrm{CBF}}$ in 95, and shortens the mean mission time in 93, since the smaller bounds admit smaller margins earlier. At most 106 of the 200 trajectories of a case have a deployment event, with the upper confidence bound 59.0%.

With $\phi = 1$, the window (49) limits how long a certificate of V-U, V-Δ, or V-L supports deployment. At $L_\mu = 0.005$, it is 10 decisions for $\alpha_0 - \alpha = 0.05$ and 20 at S3 (Section 2.5.1). Since $t \geq \tau_r(c)$, the certificate must also be issued within this window.

- At $\phi = 1$ and $L_\mu = 0.005$, V-U, V-Δ, and V-L therefore stay at $\widehat{m}_c/c_{\max} \geq 0.998$.
- V-C has $\widehat{m}_c/c_{\max}$ of 0.81–0.96 at the selected settings, since (50) does not exclude its deployment for up to twice this window, or longer when $\bar{P}(c) \leq \alpha_0 - \alpha$. Its mean mission time, paired by initial state and disturbance realization, is 0.4–2.5% below that of the $c_{\max}$ baseline.

Tables 17 and 18 give the event counts and the bound-excess counts of S1–S5 with their variants.

2.8 Certification, Deployment, and Assumption Checks

Tables 19–22 give the results of the detailed checks (Section 2.1.1) for S1–S5 with their variants, the default setting with V-C, and the baselines. The repository contains the cases with the other three variants. The tables contain the following quantities.

Quantity	Definition
False-certification and deployment events	(27)
Event frequencies and upper confidence bounds	(28)
CBF-violation and safe-set-violation counts	(26)
Bound excesses	(46)

The rows with $K = \infty$ check Theorems 1 and 2 against δ_1, and the rows with the setting's K check Theorem 3 against δ.

A decision is checked if $c_t < c_{\max}$ (Section 1.4). V-U and V-Δ in the Maze at the default setting never deploy such a margin, so they have no checked decision at either L_μ. At $L_\mu = 0.005$, 26 of the 40 cases of S1–S5 and 6 of the 8 of the default setting have none at each test period, consistent with the short window of Section 2.5.1.

In the detailed checks, every count is consistent with its guarantee except at S4 with $K = \infty$. There, $\delta_1 = 1.23\%$ lies below the 1.5% upper confidence bound at zero possible events, although none of its 16 cases has an event.

Over the main grid, 154 of the 3000 grid cells have a deployment event, all at $\alpha_0 \leq 0.25$. In three groups of grid cells, a count has an upper confidence bound above the bound of its guarantee.

- 9 grid cells with finite K, each with a false-certification count of $2/200$ (3.2%) at $\delta = 0.025$.
- Every grid cell with $K = \infty$ and $\delta = 0.025$, where δ_1 lies below the 1.5% upper confidence bound at zero possible events.
- 40 Corridor grid cells with $K = \infty$ and $\alpha = 0.05$, with counts $2/200$ (3.2%) at $\delta = 0.05$ and $5/200$ (5.2%) at $\delta = 0.10$.

These groups reflect the limited resolution of 200 trajectories, at which the upper confidence bound is at least 1.5% and reaches 3.2% at 2 events and 5.2% at 5 events. Every event frequency itself lies below the bound, δ or δ_1, that it checks. Over the main grid, the false-certification frequencies reach at most 40% of δ and 51% of δ_1, as [1, App. F-D] reports.

In all simulations with $\phi = 1$, Assumption 1 holds, every applied input satisfies the tightened condition in (1), and no bound excess (46) is positive at a checked decision. Accordingly, every entry of $\widehat{q}_{\mathrm{A2}}$ and $\widehat{q}_{\mathrm{pw}}$ in Tables 19–22 is zero, or N-A where no decision is checked.

2.9 Variation in Robustness Margin and Estimated Exceedance Risk

Tables 23 and 24 measure how the normalized robustness margin $c_{k,t}/c_{\max}$ and the estimated exceedance risk $\widehat{p}_{k,t}(c_{k,t})$ vary along a trajectory over its first $N_{\mathrm{var}} = 200$ decisions. For each quantity, the temporal variance $\widehat{V}_\upsilon$ gives the spread about the trajectory mean and the root-mean-square change $\widehat{J}_\upsilon$ the change between consecutive decisions. Both are means over the trajectories in $\mathcal{K}_\upsilon$, and the tables also give $|\mathcal{K}_\upsilon|$. Section 1.2.5 defines them in (37).

References

[1] P. S. Trakas, O. Mirzaeedodangeh, and L. Lindemann, "Anytime-valid certification of robustness margins for safe control via e-values," arXiv preprint, 2026.

3 Figures and Tables

Section 1.3 Parameters, Nominal Input and Lipschitz Constants

Table 1: Parameters of both scenarios.

Quantity	Corridor	Maze
Start; goal; goal tolerance	$[0.3, 1.1] \times [0.5, 1.5]$ m, $\lvert\theta\rvert \le 15°$; $(14.4, 1.0)$ m; 0.25 m	$(1.0, 0.75 \pm 0.08)$ m, $\theta = 90°$; $(8.06, 8.25)$ m; 0.15 m
Input bounds	$\lvert v\rvert \le 0.6$ m/s, $\lvert\omega\rvert \le 2$ rad/s	$\lvert a\rvert \le 1.5$ m/s^2, $\lvert\nu\rvert \le 8$ rad/s^2
State bounds; braking a_{br}	–	$v \in [-0.3, 0.8]$ m/s, $\lvert\omega\rvert \le 1.5$ rad/s; 1.2 m/s^2
Cone radius ρ_j; body radius ρ_{body}	0.305 m; 0.22 m	same
Nose distance l	0.15 m	same
Nominal input	$k_\theta = 2$, $v_{\mathrm{c}} = 0.5$ m/s	look-ahead 0.5 m, $k_\theta = 2.5$, $k_\omega = 4$, $k_v = 2$, $v_{\mathrm{c}} = 0.6$ m/s
Filter weight	$w_\omega = 0.0225$	$w_\nu = 0.0225$
Gains Λ	$\{0.2, 0.4, 0.6, 0.8, 1.05\}$	same
K_x^z; K_u^z in [1, (63)], [1, (64)]	1.022; 0.102	1.067; 0.0204
$L_{\mathrm{geom}}(c)$ in [1, (57)] over Γ	$[3.09, 6.69] \times 10^{-3}$ m^{-1}	$[5.13, 11.14] \times 10^{-3}$ m^{-1}
L_x^{geom}; L_u^{geom} in [1, (67)]	6.84×10^{-3}; 6.80×10^{-4}	1.19×10^{-2}; 2.27×10^{-4}
ACI k_{ACI}; N^{win}; initial margin	0.02; 500; $c_{\max}$	0.005; 500; $c_{\max}$
QT k_{QT}; initial margin	$0.3\, c_{\max}$; $c_{\max}$	$0.02\, c_{\max}$; $c_{\max}$
Tolerances $\varepsilon_{\mathrm{num}}$; $\varepsilon_{\mathrm{chk}}$	10^{-10}; 10^{-12}	same

Table 2: Obstacle coordinates in meters: cone centers and walls $[\underline{z}_1, \bar{z}_1] \times [\underline{z}_2, \bar{z}_2]$.

Object	Coordinates
Corridor cones	$(2.40, 0.45)$, $(4.49, 1.51)$, $(6.20, 0.55)$, $(7.55, 1.55)$, $(8.90, 0.55)$, $(10.40, 1.35)$, $(11.85, 0.46)$, $(13.25, 1.45)$
Maze walls	$[2.650, 3.350] \times [0, 6.750]$, $[5.650, 6.350] \times [2.250, 9]$, $[8.650, 9] \times [0, 6.750]$, $[3.350, 4.400] \times [3.975, 5.025]$, $[7.600, 8.650] \times [4.931, 5.269]$, $[7.600, 8.650] \times [6.281, 6.619]$
Maze cones	$(1.62, 1.90)$, $(1.58, 4.87)$, $(1.80, 6.10)$, $(1.15, 7.80)$, $(4.20, 8.40)$, $(4.90, 7.25)$, $(5.08, 2.72)$, $(5.03, 2.03)$, $(7.77, 0.93)$, $(6.98, 2.47)$, $(7.83, 4.05)$, $(8.05, 6.95)$, $(6.70, 8.30)$

Section 2.1 Compared Parameters and Table Key

Table 3: Compared parameter values.

Comparison	Varied values	Fixed values
Main grid	$\alpha_0 = 0.10$: $\alpha = 0.05$ $\alpha_0 = 0.15$: $\alpha \in \{0.05, 0.10\}$ $\alpha_0 = 0.20$: $\alpha \in \{0.05, 0.10, 0.15\}$ $\alpha_0 = 0.25$: $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ $\alpha_0 = 0.30$: $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ $\delta \in \{0.025, 0.05, 0.10, 0.15, 0.20\}$ $K \in \{10, 50, 100, 200, \infty\}$ methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$
Baselines	methods: Unfiltered, $c = 0$, $c_{\max}$, ACI, QT $\alpha_0 \in \{0.15, 0.30\}$ ACI also at $\alpha_0 \in \{0.10, 0.20, 0.25\}$ with $L_\mu = 0.0005$ (in the repository) $L_\mu \in \{0.0005, 0.005\}$	no certification parameters
Default setting	$(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$ methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$ (part of the main grid)
Selected settings	settings $(\alpha_0, \alpha, \delta, K)$: S1: $(0.30, 0.25, 0.20, 50)$, V-C S2: $(0.30, 0.25, 0.15, 50)$, V-C S3: $(0.30, 0.20, 0.20, 50)$, V-L S4: $(0.30, 0.25, 0.025, 100)$, V-Δ S5: $(0.30, 0.25, 0.10, 10)$, V-C methods: V-U, V-Δ, V-L, V-C	$\phi = 1$ $L_\mu = 0.0005$ (part of the main grid)
Budget normalizer b	$b \in \{\kappa, 2.5, 3, 4\}$ settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	K of the setting $\phi = 1$ $L_\mu = 0.0005$
Bound factor ϕ and distribution-drift bound L_μ	$\phi \in \{0, 0.25, 0.5, 0.75, 1\}$ $L_\mu \in \{0.0005, 0.005\}$ (ϕ and L_μ crossed) settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	K of the setting
Test period K (detailed checks)	K of the setting and $K = \infty$ $L_\mu \in \{0.0005, 0.005\}$ (K and L_μ crossed) settings: default setting, S1–S5 methods: V-U, V-Δ, V-L, V-C	$\phi = 1$

Section 2.2 Default and Selected Settings

Table 4: Corridor, all metrics of Section 1.2 for the baselines and for the default setting with all four variants. $N_{\mathrm{MC}} = 200$ trajectories per method.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92	0 (1.5)
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87	200 (100.0)
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89	200 (100.0)

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
default setting: $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$						
V-U	1.00±0.00 **0.17**±0.01 1.00±0.08	**0.01**±0.07 **0.03**±0.10 **0.00**±0.00	1.9±1.5 5.83±2.25 2.1±1.7	**17.9**±6.5 80.2±5.4 9.65±2.33	**0.05**±0.01 **0.17**±0.01 35.91±0.91	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.01 1.00±0.08	**0.01**±0.07 **0.03**±0.10 **0.00**±0.00	1.9±1.5 5.83±2.25 2.1±1.7	**17.9**±6.5 80.2±5.4 9.65±2.33	**0.05**±0.01 **0.17**±0.01 35.91±0.91	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.96±0.02 **0.17**±0.01 0.98±0.08	0.30±0.32 0.49±0.40 **0.00**±0.00	25.3±9.6 6.64±2.97 3.4±2.0	18.0±6.5 56.7±8.6 9.53±2.18	0.04±0.01 **0.17**±0.01 35.75±0.92	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.83**±0.05 **0.17**±0.01 **0.93**±0.09	2.12±1.18 3.37±1.63 **0.00**±0.00	**79.2**±6.2 6.45±2.37 5.7±2.2	18.3±6.6 **2.5**±0.7 9.39±2.38	0.03±0.01 0.16±0.01 **35.14**±0.91	0 (1.5) ✓ 0 (1.5) ✓

Table 5: Maze, as in Table 4.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32	0 (1.5)
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40	200 (100.0)
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33	85–87 (49.6)

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)						
V-U	1.00±0.00 **0.17**±0.02 0.81±0.02	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.0±0.0 N-A N-A	**24.4**±8.9 75.6±8.9 6.59±1.93	**0.03**±0.00 **0.22**±0.01 54.44±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.02 0.81±0.02	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.0±0.0 N-A N-A	**24.4**±8.9 75.6±8.9 6.59±1.93	**0.03**±0.00 **0.22**±0.01 54.44±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.96±0.02 **0.17**±0.02 0.77±0.03	0.17±0.28 0.55±0.51 **0.00**±0.00	26.2±12.2 13.80±6.99 2.9±1.8	24.5±8.9 49.3±9.2 6.57±1.96	**0.03**±0.00 0.21±0.01 54.15±0.38	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.86**±0.05 **0.17**±0.02 **0.70**±0.04	0.77±0.73 2.36±1.25 **0.00**±0.00	**72.6**±8.8 13.28±4.84 4.7±1.9	24.8±9.0 **2.6**±0.7 6.49±1.88	0.02±0.00 0.21±0.01 **53.40**±0.44	0 (1.5) ✓ 0 (1.5) ✓

Table 6: Corridor, all metrics of Section 1.2 for the baselines and for S1–S5 with all four variants. $N_{\mathrm{MC}} = 200$ trajectories per method.

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92	0 (1.5)
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81	200 (100.0)
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83	200 (100.0)

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
S1: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$						
V-U	0.79±0.03 **0.17**±0.01 0.93±0.08	**2.57**±0.88 4.71±1.12 **0.01**±0.15	67.7±3.3 1.53±0.31 16.2±4.0	**4.3**±0.9 28.0±3.2 31.50±3.27	**0.02**±0.01 **0.16**±0.01 35.06±0.83	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.79±0.03 **0.17**±0.01 0.93±0.08	**2.57**±0.88 **4.70**±1.12 **0.01**±0.15	67.7±3.3 1.53±0.31 16.2±4.0	**4.3**±0.9 28.0±3.2 31.48±3.21	**0.02**±0.01 **0.16**±0.01 35.06±0.83	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.63±0.05 **0.17**±0.02 0.85±0.08	5.86±1.20 9.48±1.53 0.04±0.21	92.2±1.5 1.53±0.31 17.8±3.7	4.4±0.9 3.4±1.2 30.97±3.33	0.01±0.01 0.14±0.01 34.38±0.87	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.50**±0.06 **0.17**±0.02 **0.80**±0.09	9.74±2.12 15.10±2.94 0.18±0.41	**92.3**±1.1 1.53±0.31 21.1±3.3	4.5±0.9 **3.2**±0.8 30.46±3.36	0.01±0.01 0.13±0.01 **33.86**±0.89	0 (1.5) ✓ 3 (3.9) ✓
S2: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$						
V-U	0.81±0.03 **0.17**±0.01 0.94±0.08	**2.32**±0.81 **4.29**±1.08 **0.01**±0.15	65.0±3.4 1.74±0.32 15.3±3.8	**4.9**±0.9 30.1±3.3 30.95±3.27	**0.03**±0.01 **0.16**±0.01 35.12±0.84	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.81±0.03 **0.17**±0.01 0.94±0.08	**2.32**±0.81 **4.29**±1.08 **0.01**±0.15	65.0±3.4 1.74±0.32 15.3±3.8	**4.9**±0.9 30.1±3.3 30.95±3.26	**0.03**±0.01 **0.16**±0.01 35.12±0.84	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.64±0.05 **0.17**±0.02 0.86±0.09	5.54±1.19 8.94±1.47 0.03±0.21	**91.7**±1.2 1.74±0.32 17.0±3.6	5.0±0.9 3.3±0.9 30.41±3.49	0.02±0.01 0.14±0.01 34.43±0.88	0 (1.5) ✓ 1 (2.4) ✓
V-C	**0.52**±0.06 **0.17**±0.02 **0.81**±0.09	9.35±1.96 14.52±2.74 0.14±0.36	**91.7**±1.2 1.74±0.32 20.3±3.3	5.1±0.9 **3.2**±0.8 29.80±3.27	0.01±0.01 0.13±0.01 **33.92**±0.88	0 (1.5) ✓ 1 (2.4) ✓

Table 6 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)						
V-U	0.67±0.05 **0.17**±0.02 0.87±0.09	**5.03**±1.15 **7.89**±1.54 **0.02**±0.20	**90.9**±1.6 2.04±0.54 14.0±3.6	**5.9**±1.6 3.3±0.8 24.57±3.19	**0.02**±0.01 **0.15**±0.01 34.53±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-Δ	0.67±0.05 **0.17**±0.02 0.87±0.09	5.04±1.19 7.91±1.57 **0.02**±0.20	**90.9**±1.7 2.04±0.54 14.0±3.6	**5.9**±1.6 3.3±0.8 24.56±3.17	**0.02**±0.01 **0.15**±0.01 34.53±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-L	0.62±0.06 **0.17**±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	**90.9**±1.7 2.04±0.54 15.3±3.6	**5.9**±1.6 **3.2**±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90	0 (1.5) ✓ 7 (6.5) ✓
V-C	**0.58**±0.07 **0.17**±0.02 **0.83**±0.09	7.44±2.16 11.73±3.19 0.07±0.27	**90.9**±1.7 2.04±0.54 16.6±3.8	**5.9**±1.6 **3.2**±0.8 24.28±3.09	0.01±0.01 0.14±0.01 **34.18**±0.91	0 (1.5) ✓ 7 (6.5) ✓
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)						
V-U	0.92±0.02 **0.17**±0.01 0.97±0.08	**0.75**±0.52 **1.44**±0.71 **0.00**±0.00	29.8±3.3 2.74±0.47 11.2±3.4	**7.7**±1.3 62.5±3.0 16.38±1.82	**0.03**±0.01 **0.17**±0.01 35.62±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.92±0.02 **0.17**±0.01 0.97±0.08	**0.75**±0.52 **1.44**±0.71 **0.00**±0.00	29.8±3.3 2.74±0.47 11.2±3.4	**7.7**±1.3 62.5±3.0 16.38±1.82	**0.03**±0.01 **0.17**±0.01 35.62±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.82±0.03 **0.17**±0.01 0.92±0.08	2.48±0.91 4.12±1.18 **0.00**±0.02	58.7±2.7 2.74±0.47 13.8±3.2	7.8±1.3 33.6±2.5 15.88±1.88	0.02±0.01 0.15±0.01 35.07±0.90	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.58**±0.05 **0.17**±0.02 **0.83**±0.09	7.74±1.71 11.99±2.17 0.05±0.19	**88.9**±1.4 2.74±0.47 18.3±2.9	8.0±1.3 **3.1**±0.7 15.15±1.83	0.01±0.01 0.14±0.01 **34.14**±0.88	0 (1.5) ✓ 0 (1.5) ✓
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)						
V-U	0.77±0.04 **0.17**±0.01 0.91±0.08	**2.87**±0.82 **4.77**±0.91 **0.01**±0.15	89.1±3.4 1.99±0.40 13.6±4.0	**5.7**±1.1 5.3±3.2 122.99±13.66	**0.03**±0.01 **0.15**±0.01 34.92±0.89	0 (1.5) ✓ 1 (2.4) ✓
V-Δ	0.77±0.04 **0.17**±0.01 0.91±0.08	**2.87**±0.82 **4.77**±0.91 **0.01**±0.15	89.1±3.4 1.99±0.40 13.6±4.0	**5.7**±1.1 5.3±3.2 122.99±13.67	**0.03**±0.01 **0.15**±0.01 34.92±0.89	0 (1.5) ✓ 1 (2.4) ✓
V-L	0.66±0.05 **0.17**±0.02 0.87±0.09	5.14±1.09 8.21±1.32 0.02±0.20	**91.0**±1.4 1.99±0.40 15.7±3.6	**5.7**±1.1 3.3±0.8 121.62±14.51	0.02±0.01 **0.15**±0.01 34.51±0.88	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.53**±0.06 **0.17**±0.02 **0.81**±0.09	8.94±1.88 13.84±2.67 0.12±0.36	**91.0**±1.4 1.99±0.40 19.6±3.2	5.8±1.2 **3.2**±0.8 119.71±13.69	0.01±0.01 0.13±0.01 **33.97**±0.88	0 (1.5) ✓ 0 (1.5) ✓

Table 7: Maze, as in Table 6.

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14	N-A
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27	200 (100.0)
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32	0 (1.5)
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32	199 (100.0)
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32	131 (71.1)

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$ $\widehat{m}_{\text{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$ $\widehat{T}_{\text{mis}}\downarrow(\text{s})$	$\widehat{q}_{\text{dep}}\downarrow$ $\widehat{q}_{\text{cert}}\downarrow$
S1: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$						
V-U	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.15**±0.25 **0.68**±0.53 **0.00**±0.00	11.3±2.0 3.04±0.63 11.1±5.3	**5.6**±1.2 83.1±1.6 23.50±2.71	**0.03**±0.00 **0.21**±0.01 54.36±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.15**±0.26 0.71±0.54 **0.00**±0.00	12.2±2.1 3.04±0.63 11.1±5.2	**5.6**±1.2 82.2±1.8 23.44±2.67	**0.03**±0.00 **0.21**±0.01 54.34±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.62±0.05 **0.17**±0.02 0.56±0.03	4.77±1.42 9.74±1.93 **0.00**±0.04	91.0±1.4 3.04±0.63 17.2±3.8	5.9±1.2 3.2±0.8 22.91±2.42	0.01±0.00 **0.21**±0.01 51.88±0.42	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.52**±0.07 **0.17**±0.02 **0.51**±0.04	7.40±2.27 14.52±3.57 0.01±0.06	**91.1**±1.4 3.04±0.63 19.7±3.5	5.9±1.2 **2.9**±0.8 22.83±2.54	0.01±0.00 **0.21**±0.01 **51.25**±0.52	0 (1.5) ✓ 1 (2.4) ✓
S2: $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$						
V-U	0.98±0.01 **0.17**±0.02 0.80±0.02	**0.13**±0.25 **0.54**±0.46 **0.00**±0.00	9.4±1.9 3.72±1.99 9.6±4.9	**6.4**±1.1 84.3±1.5 23.00±2.66	**0.03**±0.00 **0.21**±0.01 54.36±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.97±0.01 **0.17**±0.02 0.80±0.02	**0.13**±0.25 0.56±0.47 **0.00**±0.00	10.2±2.0 3.50±0.98 9.7±4.9	**6.4**±1.1 83.5±1.7 23.05±2.68	**0.03**±0.00 **0.21**±0.01 54.36±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.64±0.05 **0.17**±0.02 0.56±0.03	4.59±1.37 9.25±1.84 **0.00**±0.04	**90.3**±1.5 3.45±0.62 16.7±3.9	6.6±1.2 3.1±0.8 22.56±2.38	0.01±0.00 **0.21**±0.01 51.93±0.45	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.53**±0.08 **0.17**±0.02 **0.51**±0.04	7.28±2.37 13.89±3.46 0.01±0.07	**90.3**±1.5 3.45±0.62 19.2±3.7	6.7±1.2 **2.9**±0.8 22.37±2.50	0.01±0.00 **0.21**±0.01 **51.33**±0.53	0 (1.5) ✓ 0 (1.5) ✓

Table 7 (continued).

Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
Maze $\phi=1,\ L_\mu=0.0005,\ c_{\max}=0.01\,\mathrm{m},\ T_s=0.2\,\mathrm{s},\ N_{\mathrm{MC}}=200$						
S3: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.20, 0.20, 50)						
V-U	0.82±0.04 **0.17**±0.02 0.68±0.03	**1.49**±0.82 **3.70**±1.23 **0.00**±0.00	67.8±5.6 4.01±0.97 11.7±3.8	**7.5**±1.8 24.7±5.3 18.91±2.53	**0.02**±0.00 **0.21**±0.01 53.19±0.38	0 (1.5) ✓ 5 (5.2) ✓
V-Δ	0.82±0.04 **0.17**±0.02 0.68±0.03	1.54±0.84 3.76±1.23 **0.00**±0.00	69.4±5.5 4.01±0.97 11.6±3.9	**7.5**±1.8 23.1±5.1 18.83±2.57	**0.02**±0.00 **0.21**±0.01 53.17±0.37	0 (1.5) ✓ 5 (5.2) ✓
V-L	0.62±0.07 **0.17**±0.02 **0.55**±0.04	5.04±1.93 9.98±2.87 **0.00**±0.06	**89.1**±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 **0.21**±0.01 51.86±0.54	0 (1.5) ✓ 4 (4.6) ✓
V-C	**0.61**±0.08 **0.17**±0.02 **0.55**±0.04	5.38±2.19 10.63±3.31 **0.00**±0.06	**89.1**±1.9 4.01±0.97 15.3±3.8	7.7±1.8 **3.1**±0.9 18.26±2.33	0.01±0.00 **0.21**±0.01 **51.79**±0.58	0 (1.5) ✓ 4 (4.6) ✓
S4: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.025, 100)						
V-U	1.00±0.00 **0.17**±0.01 0.81±0.02	**0.00**±0.00 **0.04**±0.12 **0.00**±0.00	1.3±0.7 7.85±7.25 3.9±3.1	**10.0**±1.6 88.7±1.0 12.86±1.58	**0.03**±0.00 **0.22**±0.01 54.43±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	1.00±0.00 **0.17**±0.01 0.81±0.02	**0.00**±0.00 0.05±0.13 **0.00**±0.00	1.4±0.8 7.85±7.25 3.9±3.1	**10.0**±1.6 88.5±1.0 12.86±1.58	**0.03**±0.00 **0.22**±0.01 54.43±0.32	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.81±0.04 **0.17**±0.02 0.65±0.03	2.05±0.95 4.27±1.47 **0.00**±0.00	58.5±3.1 5.45±0.85 13.2±3.5	10.3±1.6 31.2±2.7 12.74±1.59	0.02±0.00 0.21±0.01 53.01±0.40	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.60**±0.06 **0.17**±0.02 **0.54**±0.04	5.75±1.88 11.17±2.50 0.01±0.05	**86.6**±1.8 5.45±0.85 17.2±3.2	10.5±1.6 **2.9**±0.8 12.20±1.44	0.01±0.00 0.21±0.01 **51.67**±0.47	0 (1.5) ✓ 0 (1.5) ✓
S5: $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.10, 10)						
V-U	0.99±0.01 **0.17**±0.02 0.80±0.02	**0.03**±0.12 **0.29**±0.33 **0.00**±0.00	5.4±1.7 3.96±1.03 7.4±4.6	**7.2**±1.4 87.4±1.0 90.06±11.54	**0.03**±0.00 **0.21**±0.01 54.43±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-Δ	0.99±0.01 **0.17**±0.02 0.80±0.02	**0.03**±0.12 **0.29**±0.33 **0.00**±0.00	5.6±1.7 3.96±1.03 7.4±4.6	**7.2**±1.4 87.2±1.0 90.06±11.54	**0.03**±0.00 **0.21**±0.01 54.43±0.34	0 (1.5) ✓ 0 (1.5) ✓
V-L	0.66±0.05 **0.17**±0.02 0.57±0.03	4.20±1.20 8.42±1.51 **0.00**±0.05	**89.3**±1.6 3.93±0.77 15.0±3.7	7.5±1.5 3.2±0.8 85.18±9.56	0.01±0.00 **0.21**±0.01 52.11±0.45	0 (1.5) ✓ 0 (1.5) ✓
V-C	**0.55**±0.07 **0.17**±0.02 **0.52**±0.04	6.76±2.17 13.03±3.13 0.01±0.09	**89.3**±1.7 3.93±0.77 18.5±3.3	7.6±1.5 **3.0**±0.8 83.58±9.71	0.01±0.00 **0.21**±0.01 **51.43**±0.52	0 (1.5) ✓ 0 (1.5) ✓

Table 8: Maze, S*, all four variants. False-certification counts are confirmed events, with upper confidence bounds based on possible events.

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ –: not available; a column keeps its recorded digits where rounding to those of Table 7 is ambiguous						
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\lVert u-u^{\mathrm{nom}}\rVert^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$	$\widehat{q}_{\mathrm{dep}}\downarrow$ $\widehat{q}_{\mathrm{cert}}\downarrow$
S*: $(\alpha_0, \alpha, \delta, K)$ = (0.25, 0.225, 0.15, 50)						
V-U	1.00±0.00 **0.170**±0.015 0.808±0.021	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.00±0.00 N-A N-A	**6.88**±1.59 93.1±1.6 20.9±2.5	**0.03**±0.00 – 54.44±0.32	0 (1.5) ✓ 1 (3.2) ✓
V-Δ	1.00±0.00 **0.170**±0.015 0.808±0.021	**0.00**±0.00 **0.00**±0.00 **0.00**±0.00	0.00±0.00 N-A N-A	**6.88**±1.59 93.1±1.6 20.9±2.5	**0.03**±0.00 – 54.44±0.32	0 (1.5) ✓ 1 (3.2) ✓
V-L	0.91±0.02 **0.170**±0.016 0.748±0.026	0.74±0.64 1.89±1.00 **0.00**±0.00	34.19±4.12 3.74±0.86 11.7±4.5	6.95±1.60 58.9±3.9 20.8±2.5	0.02±0.00 – 53.88±0.34	0 (1.5) ✓ 1 (3.2) ✓
V-C	**0.61**±0.06 **0.170**±0.016 **0.549**±0.035	5.17±1.66 10.35±2.21 <0.01±0.04	**89.75**±1.70 3.74±0.86 15.8±3.8	7.23±1.66 **3.0**±0.8 20.3±2.3	0.01±0.00 – **51.76**±0.43	1 (2.4) ✓ 3 (4.6) ✓

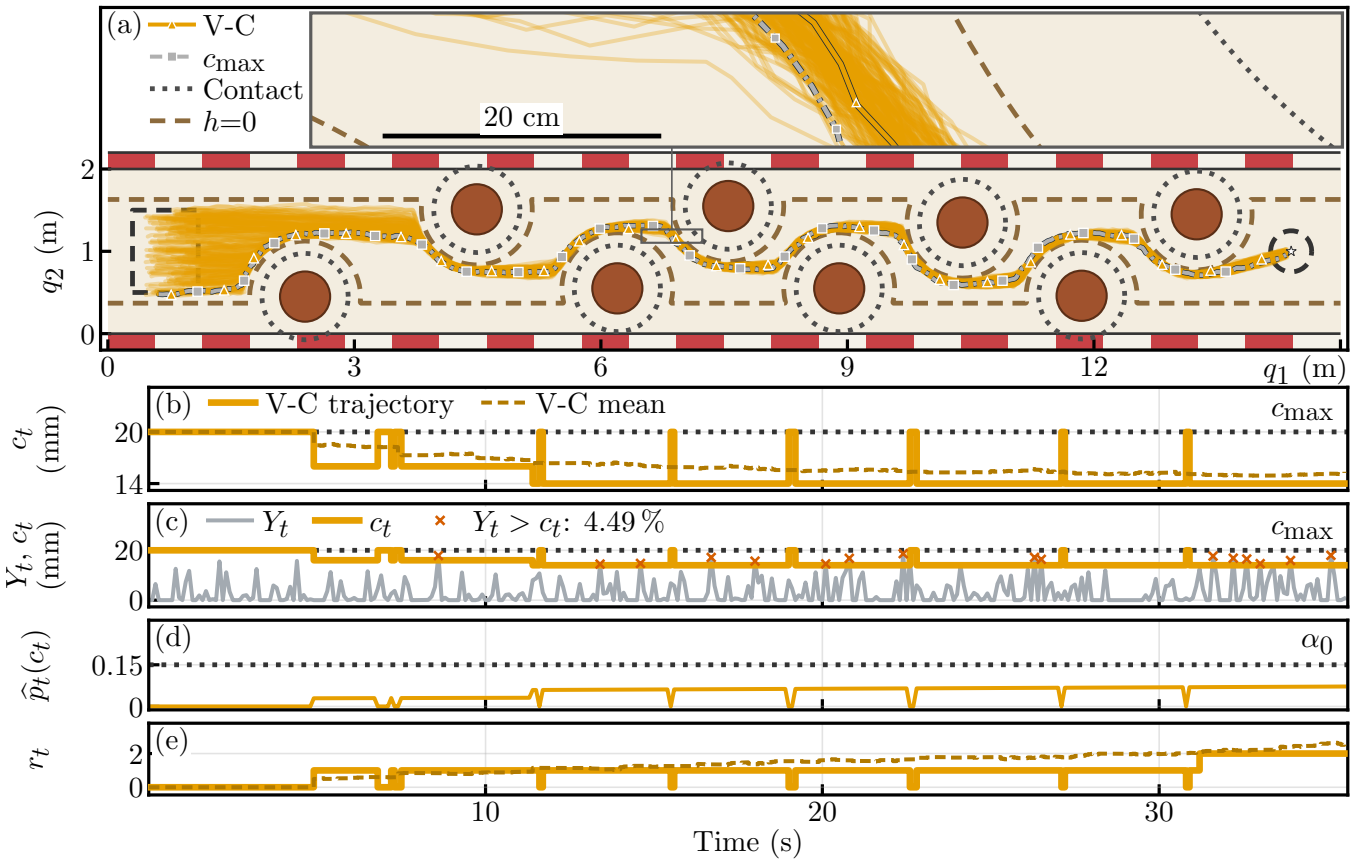


Figure 1: Corridor, default setting with V-C. Panels in Section 2.2.2.

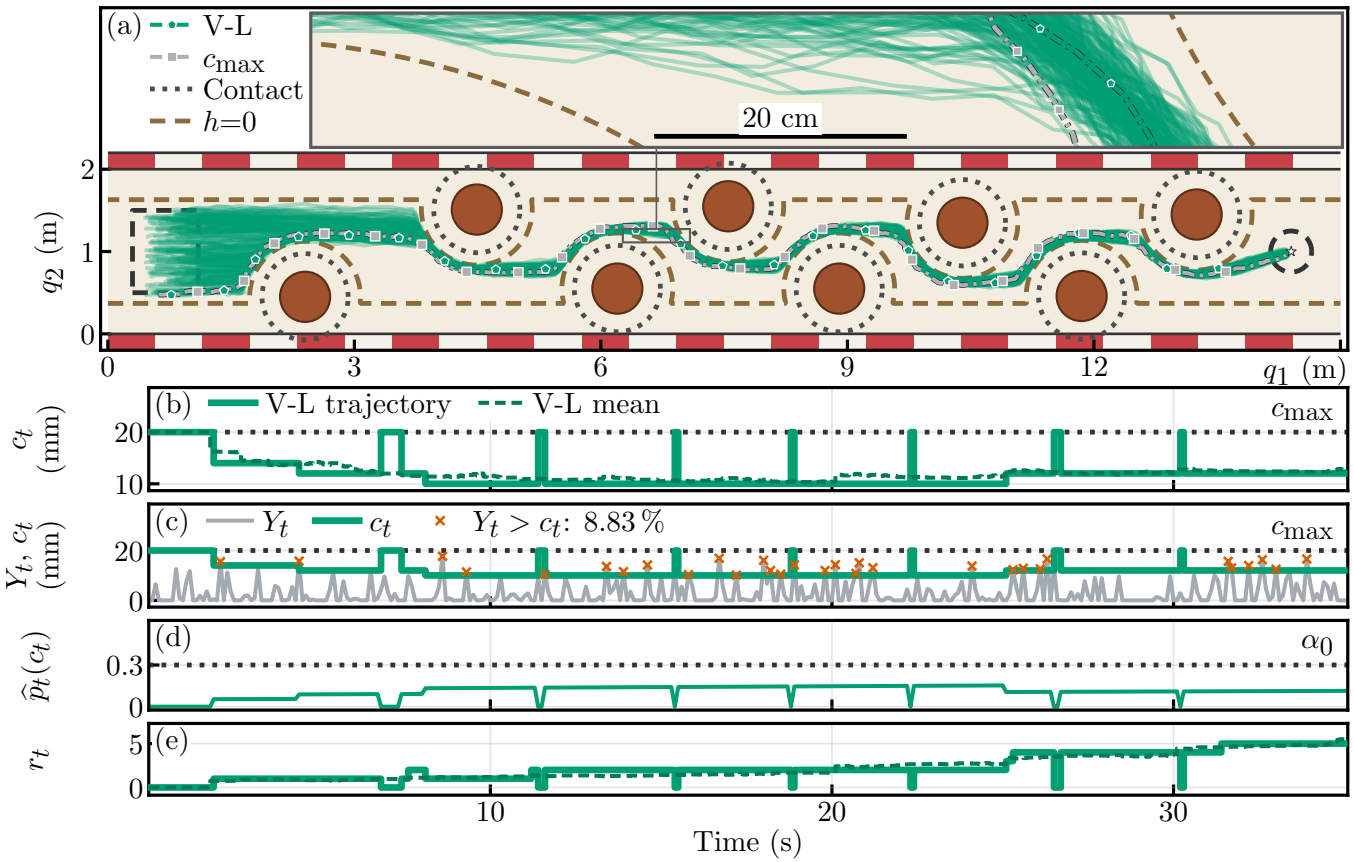


Figure 2: Corridor, S3 with V-L. Panels as in Fig. 1.

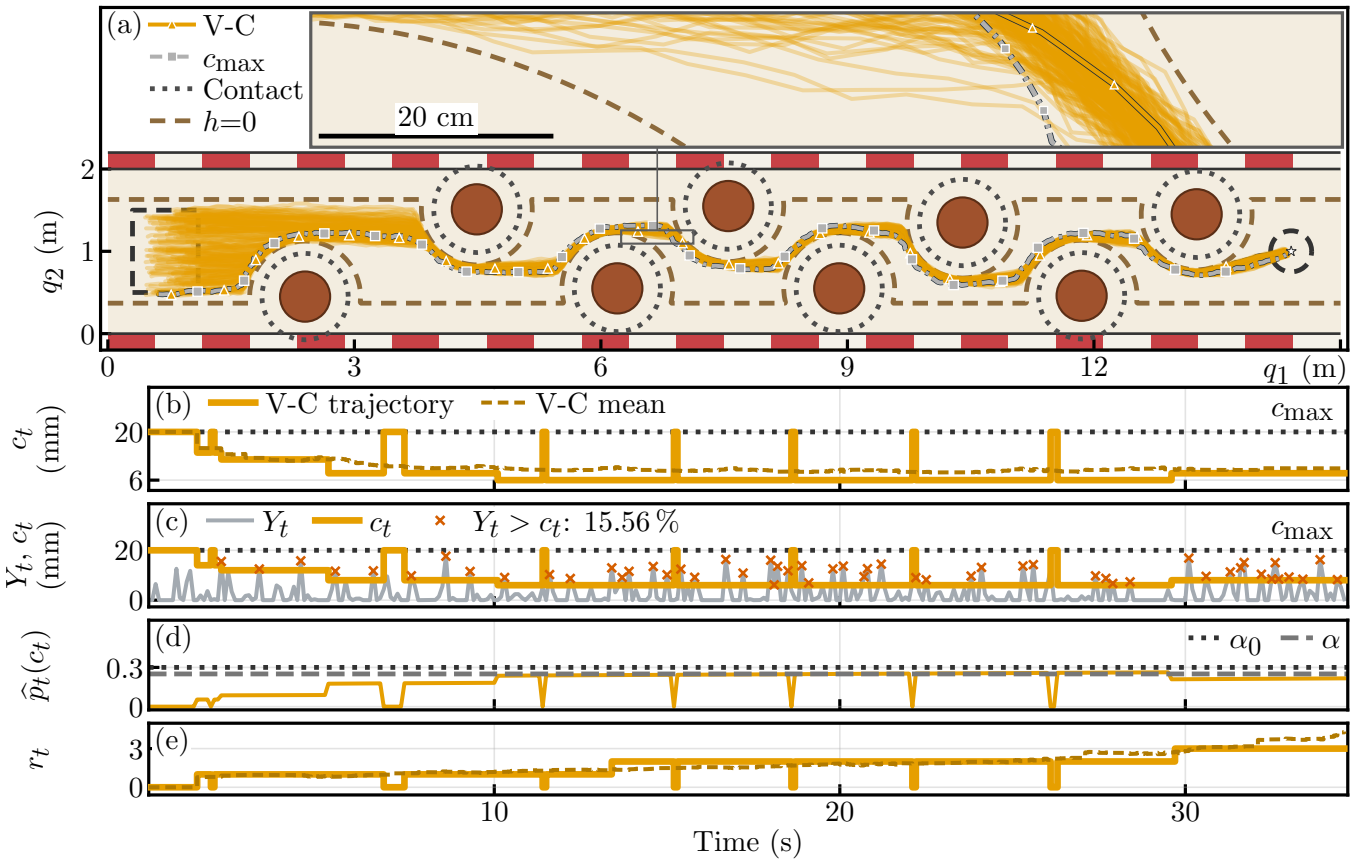


Figure 3: Corridor, S1 with V-C. Panels as in Fig. 1.

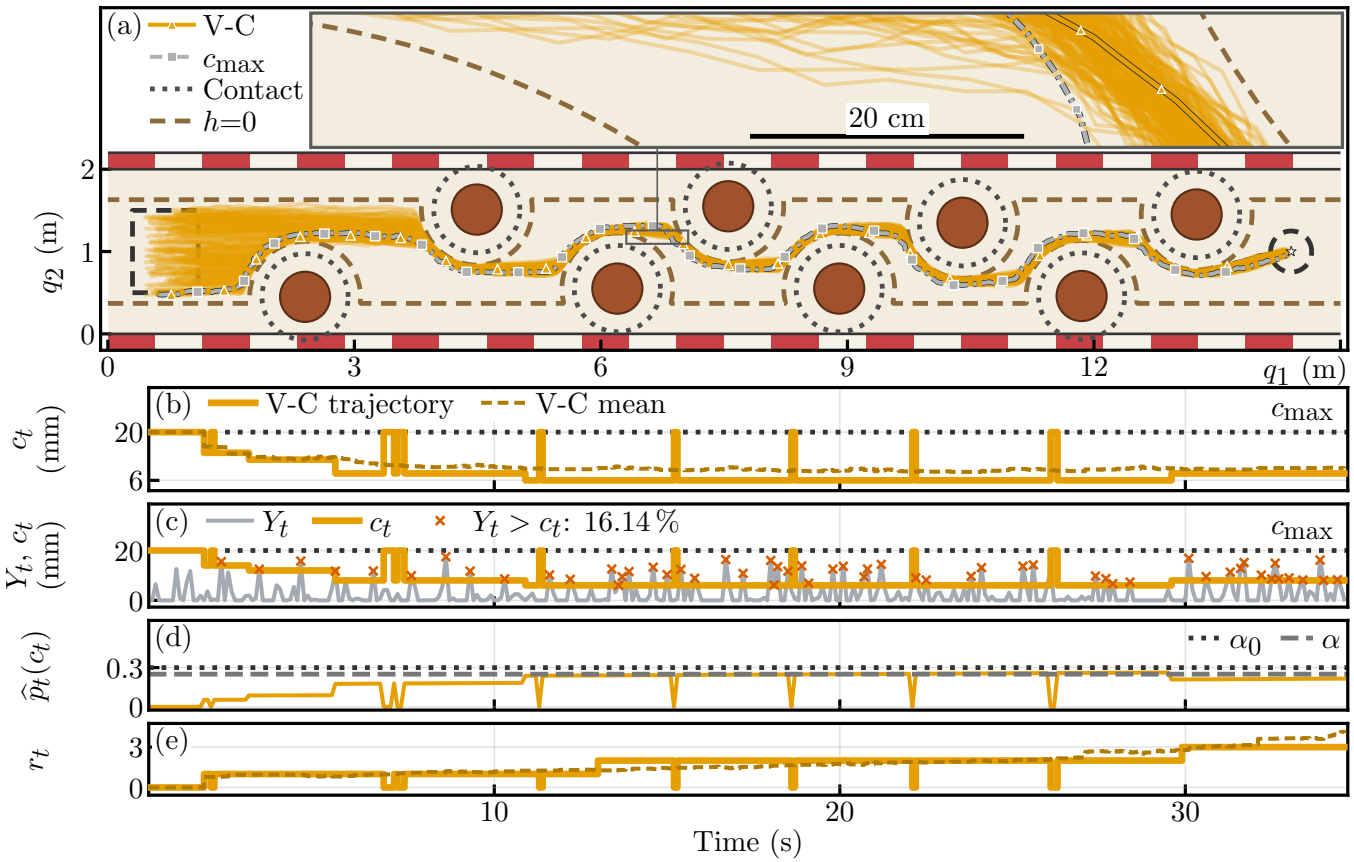


Figure 4: Corridor, S2 with V-C. Panels as in Fig. 1.

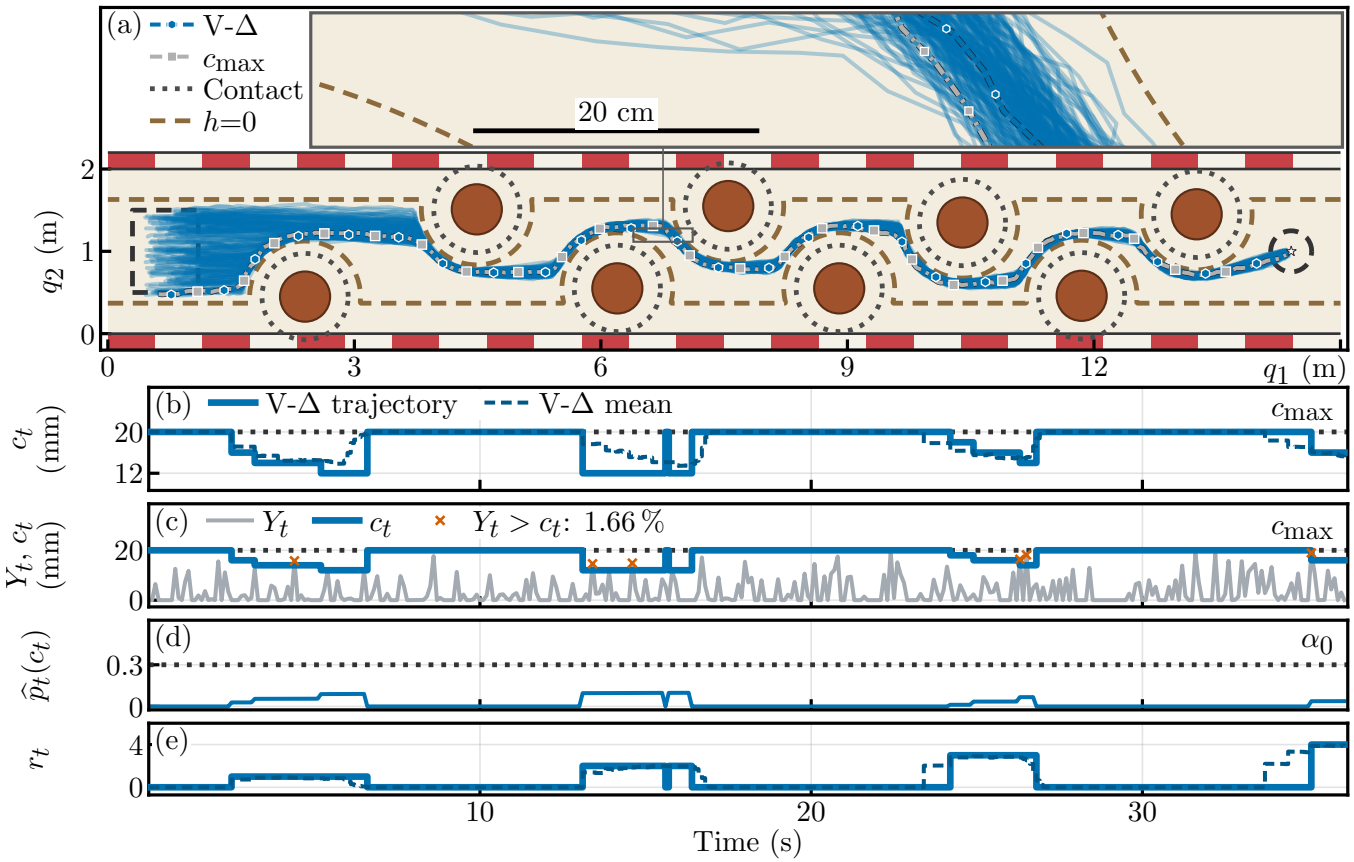


Figure 5: Corridor, S4 with V-Δ. Panels as in Fig. 1.

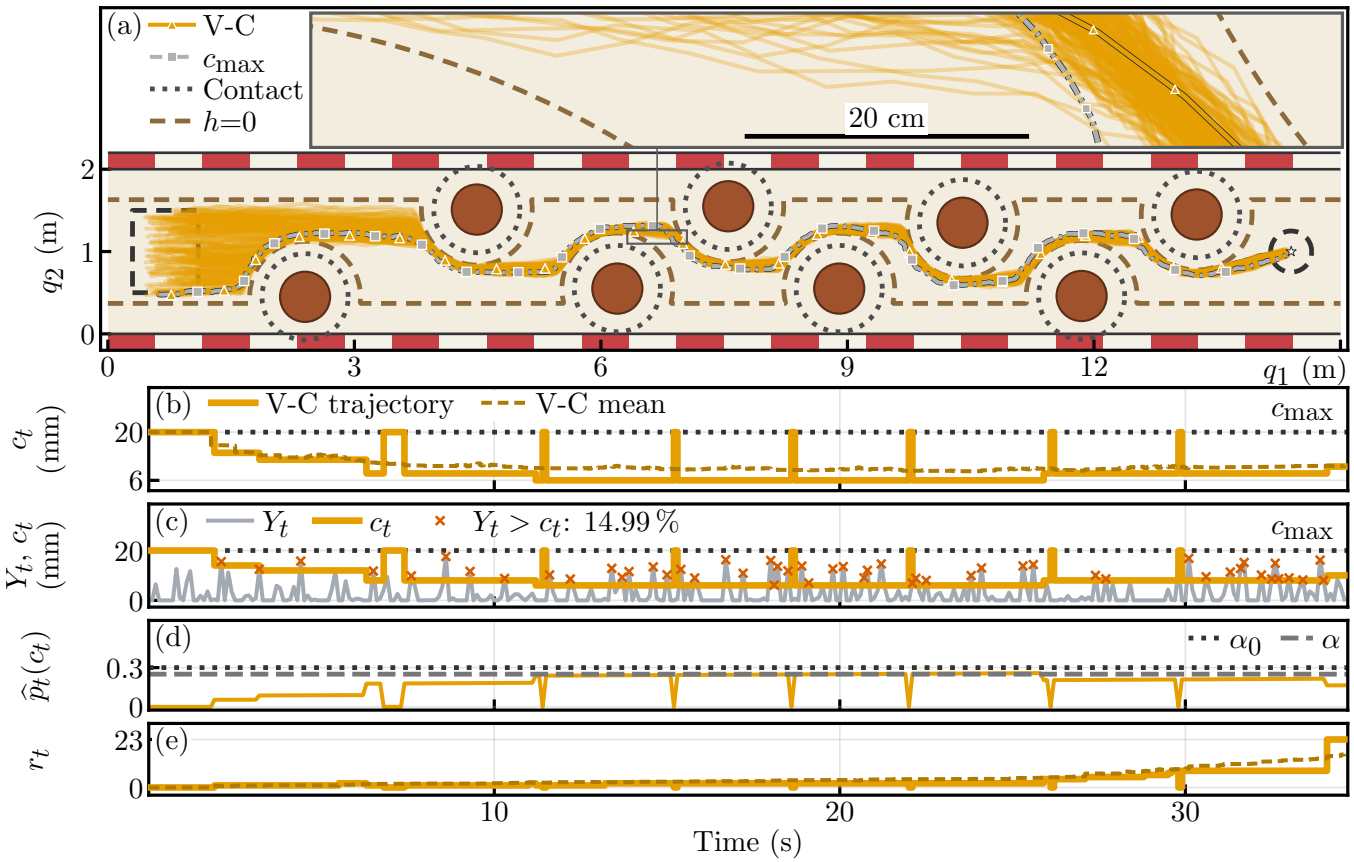


Figure 6: Corridor, S5 with V-C. Panels as in Fig. 1.

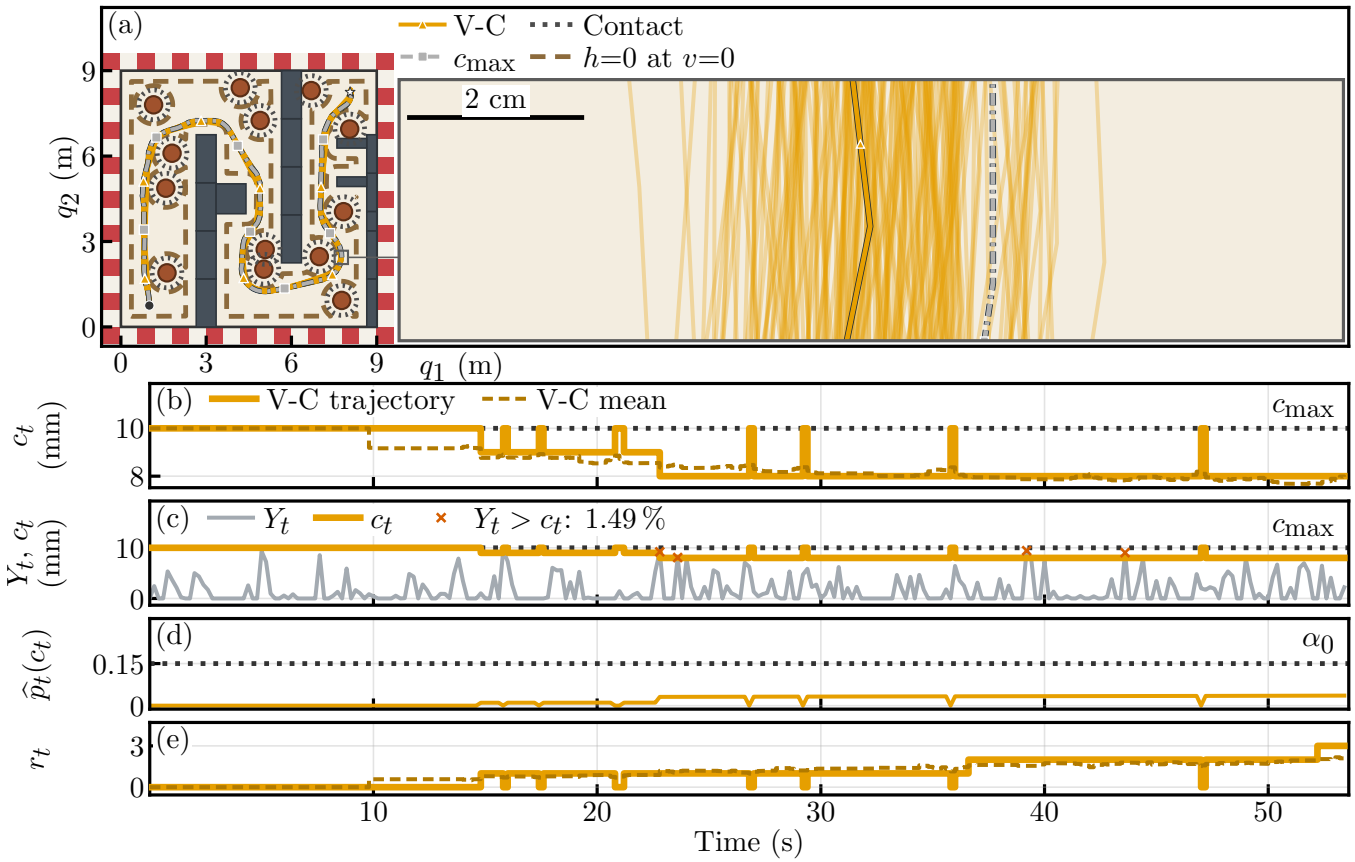


Figure 7: Maze, default setting with V-C. Panels as in Fig. 1.

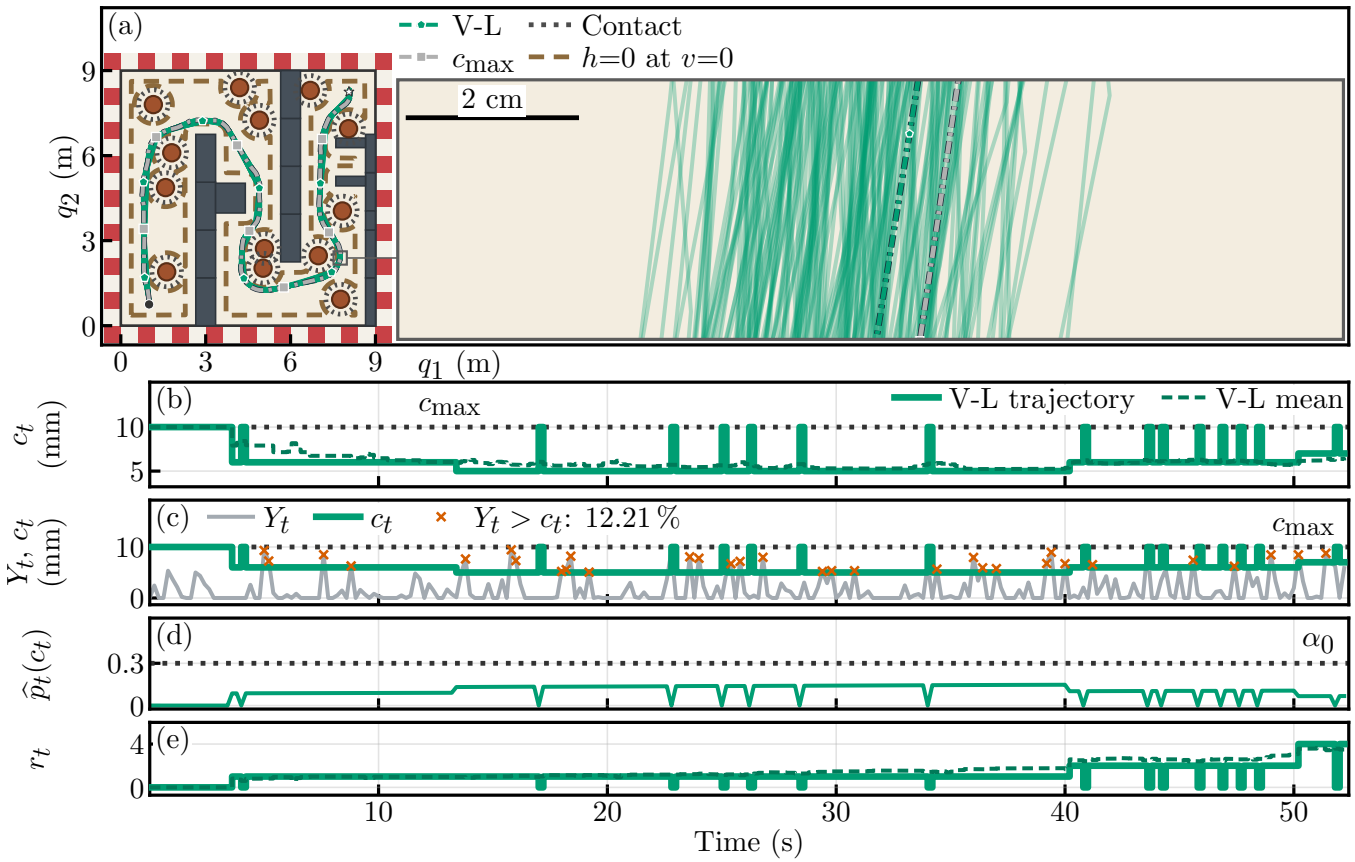


Figure 8: Maze, S3 with V-L. Panels as in Fig. 1.

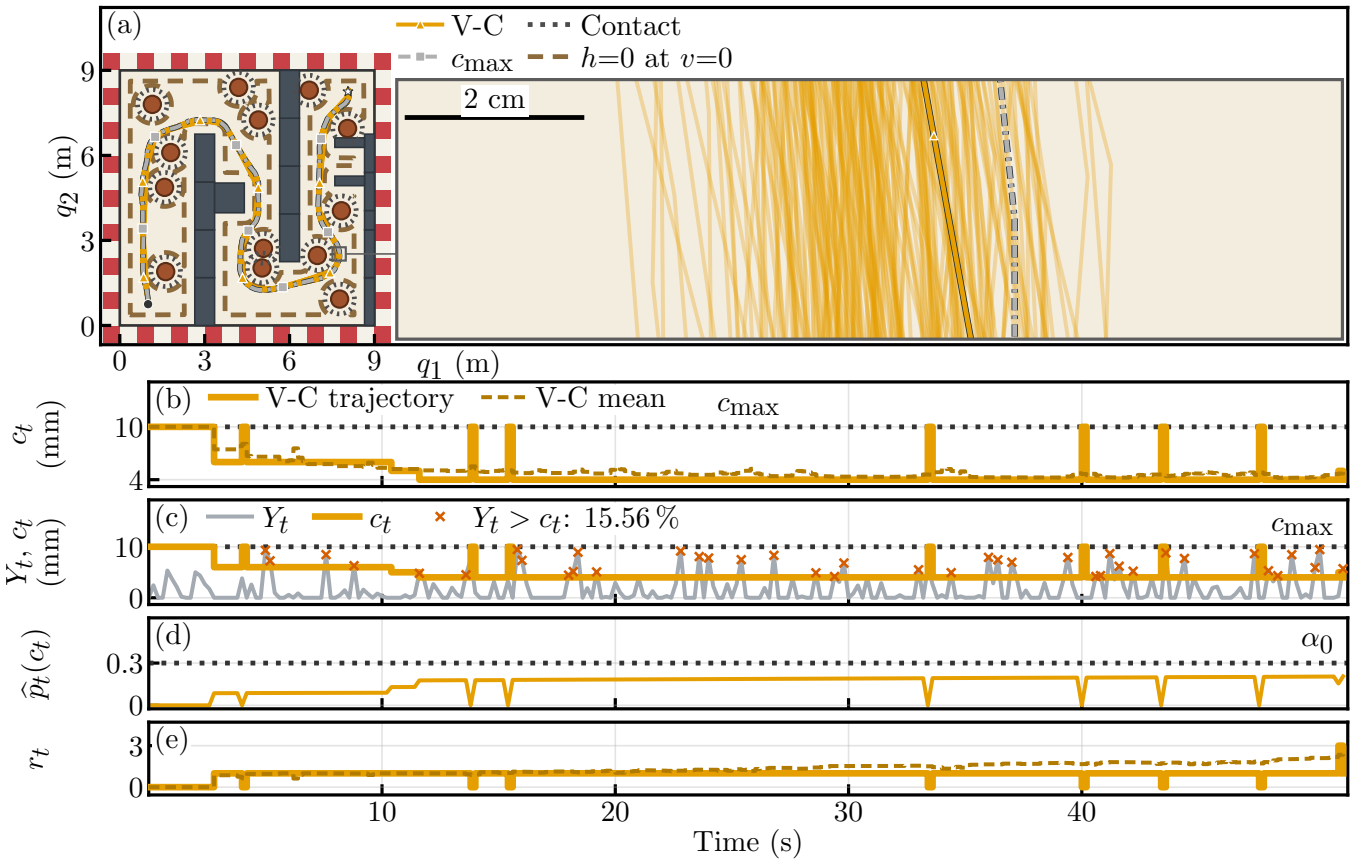


Figure 9: Maze, S1 with V-C. Panels as in Fig. 1.

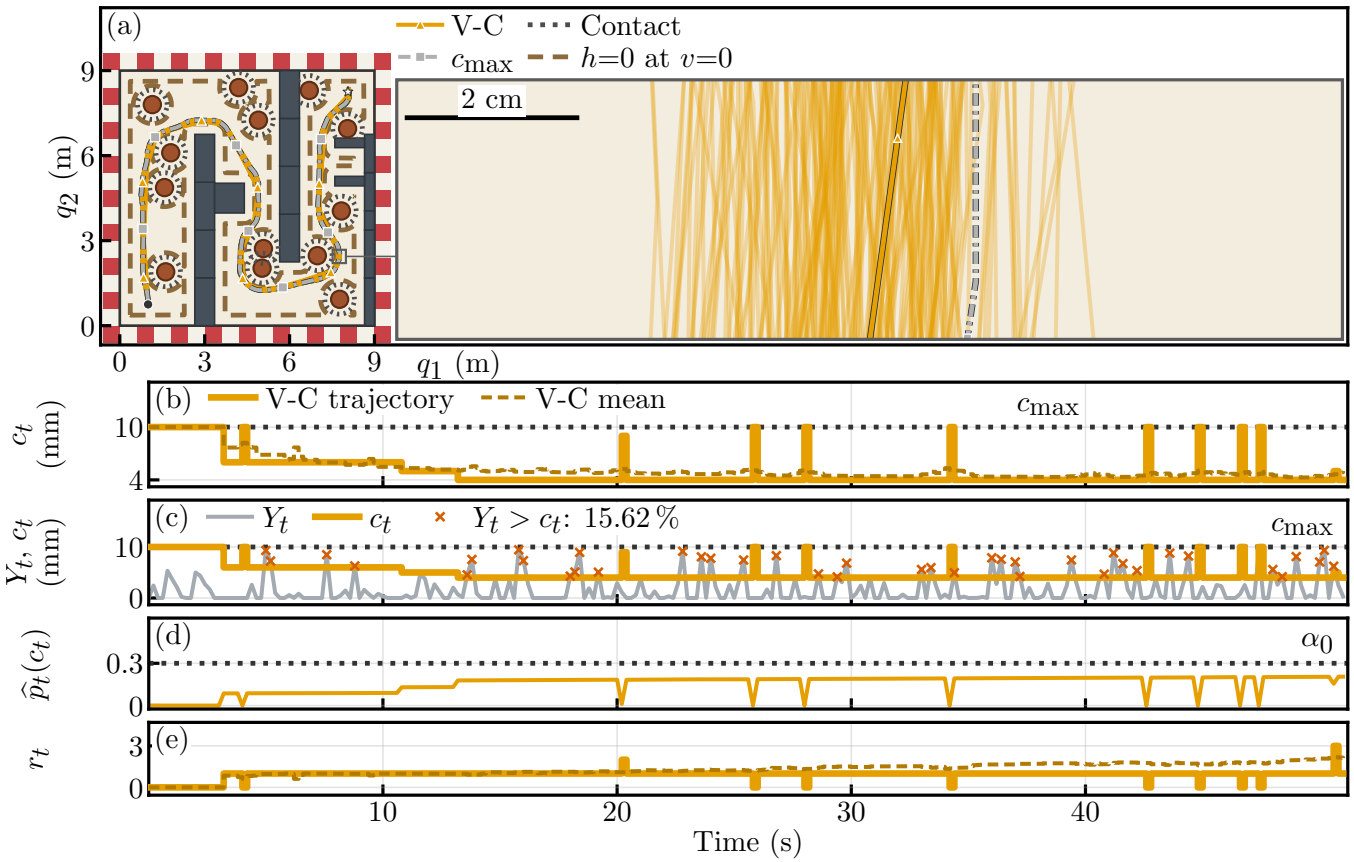


Figure 10: Maze, S2 with V-C. Panels as in Fig. 1.

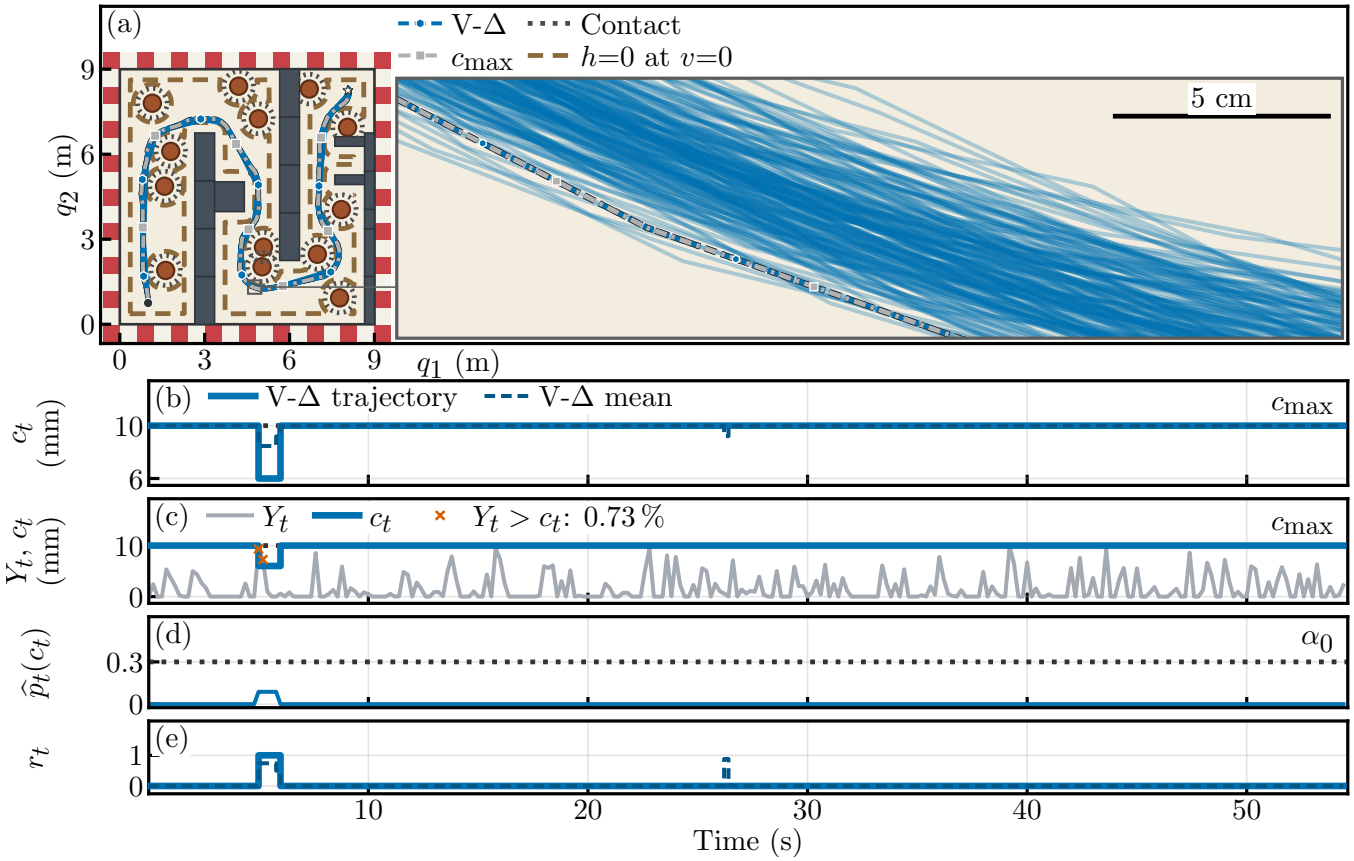


Figure 11: Maze, S4 with V-Δ. Panels as in Fig. 1. In (a), the example trajectory coincides with that of the $c_{\max}$ baseline.

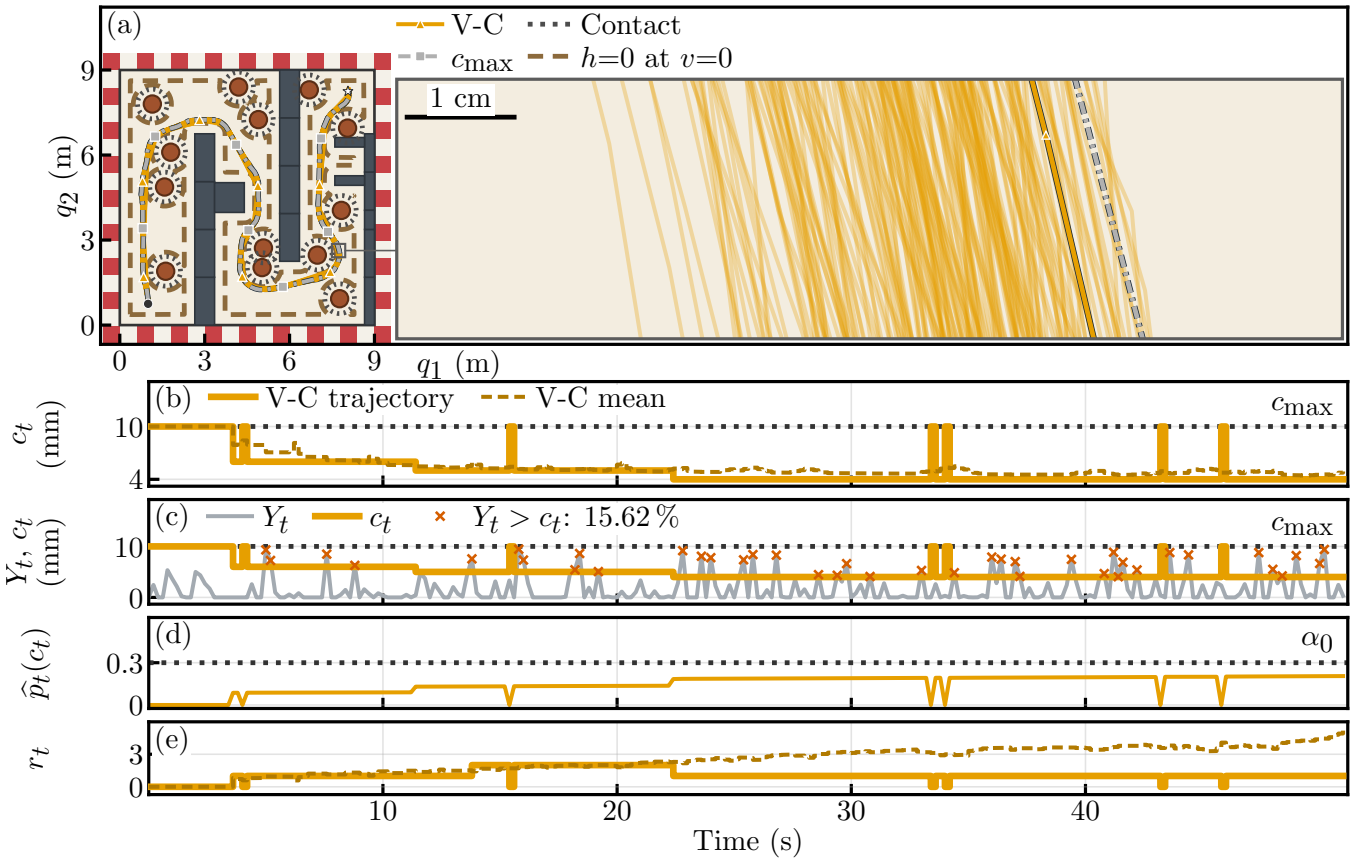


Figure 12: Maze, S5 with V-C. Panels as in Fig. 1.

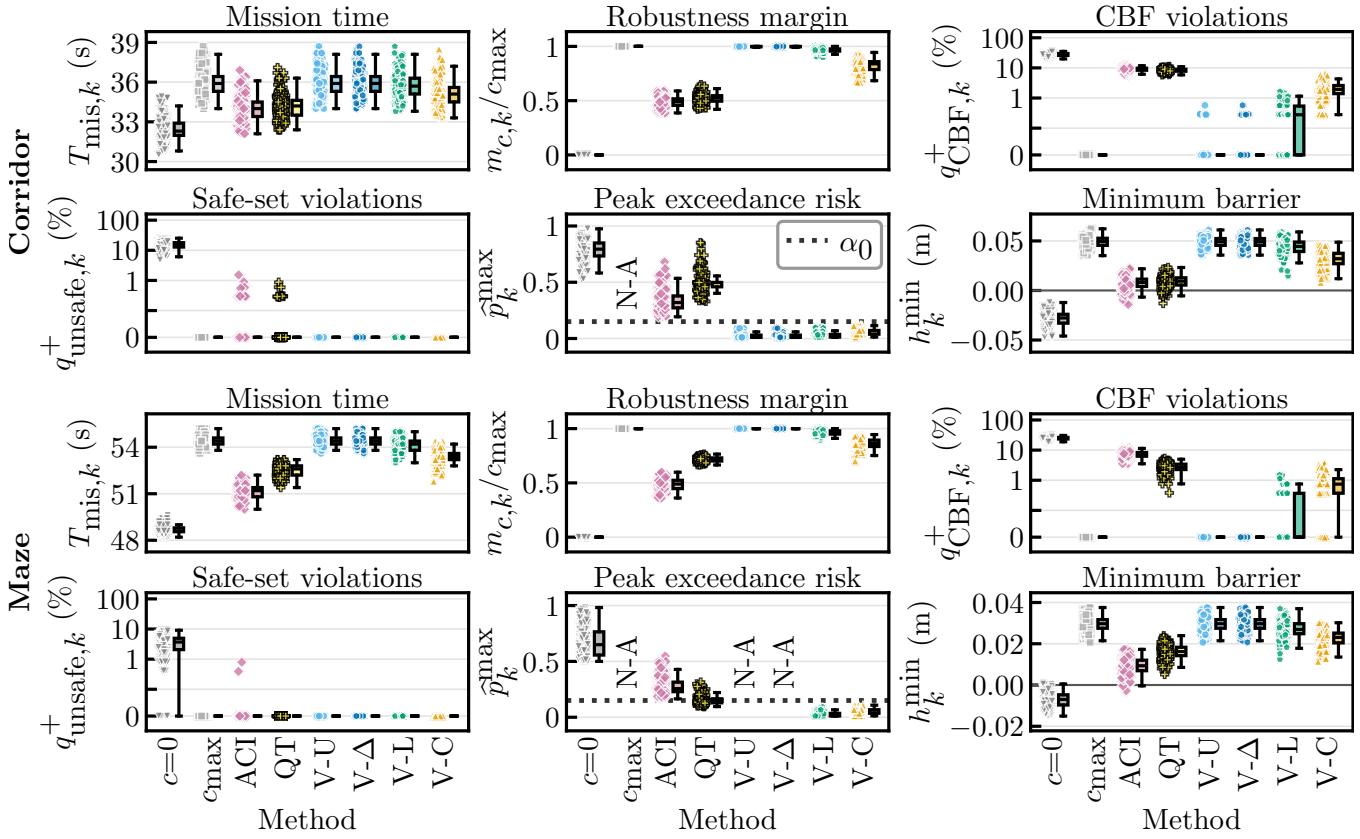


Figure 13: Corridor (top) and Maze (bottom), default setting: per-trajectory distributions.

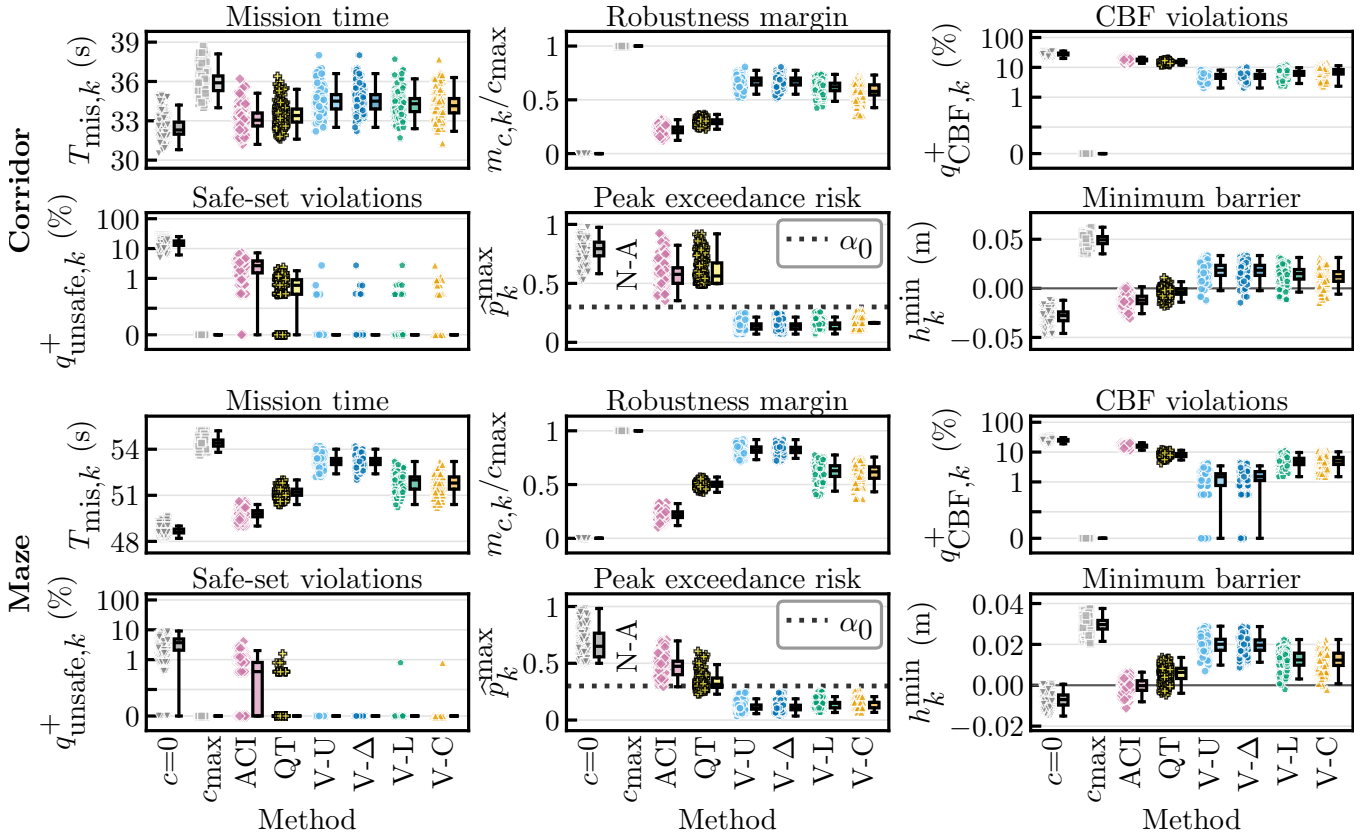


Figure 14: Corridor and Maze, S3, with panels as in Fig. 13.

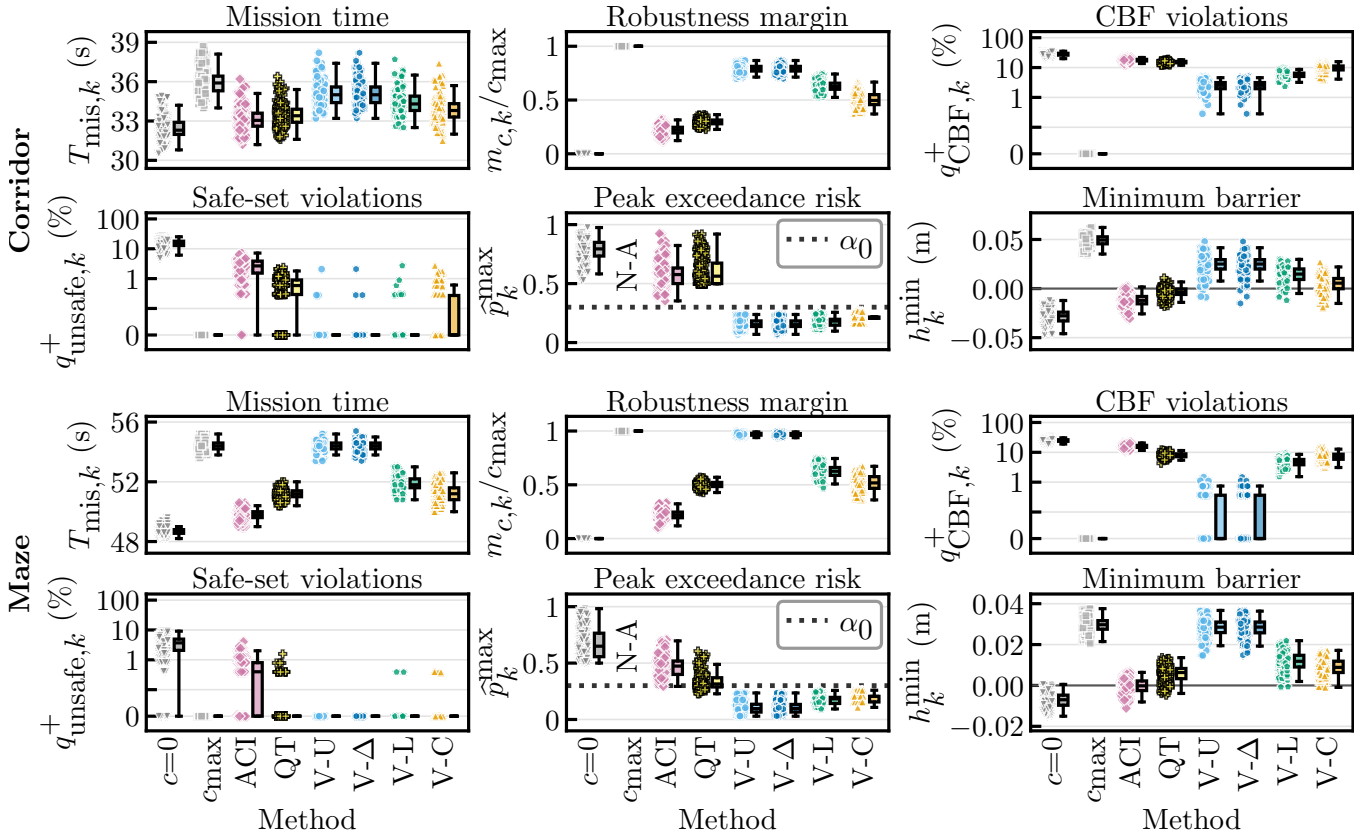


Figure 15: Corridor and Maze, S1, with panels as in Fig. 13.

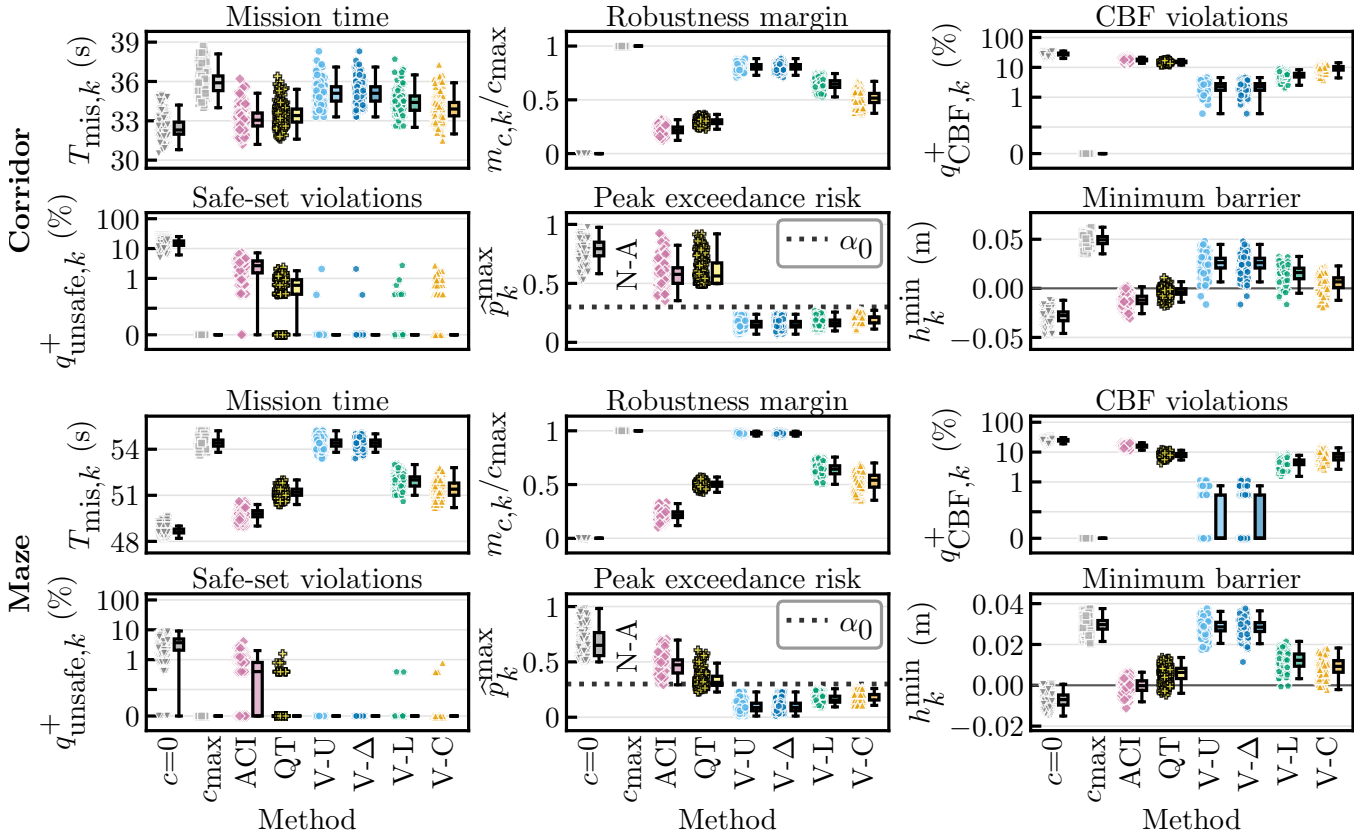


Figure 16: Corridor and Maze, S2, with panels as in Fig. 13.

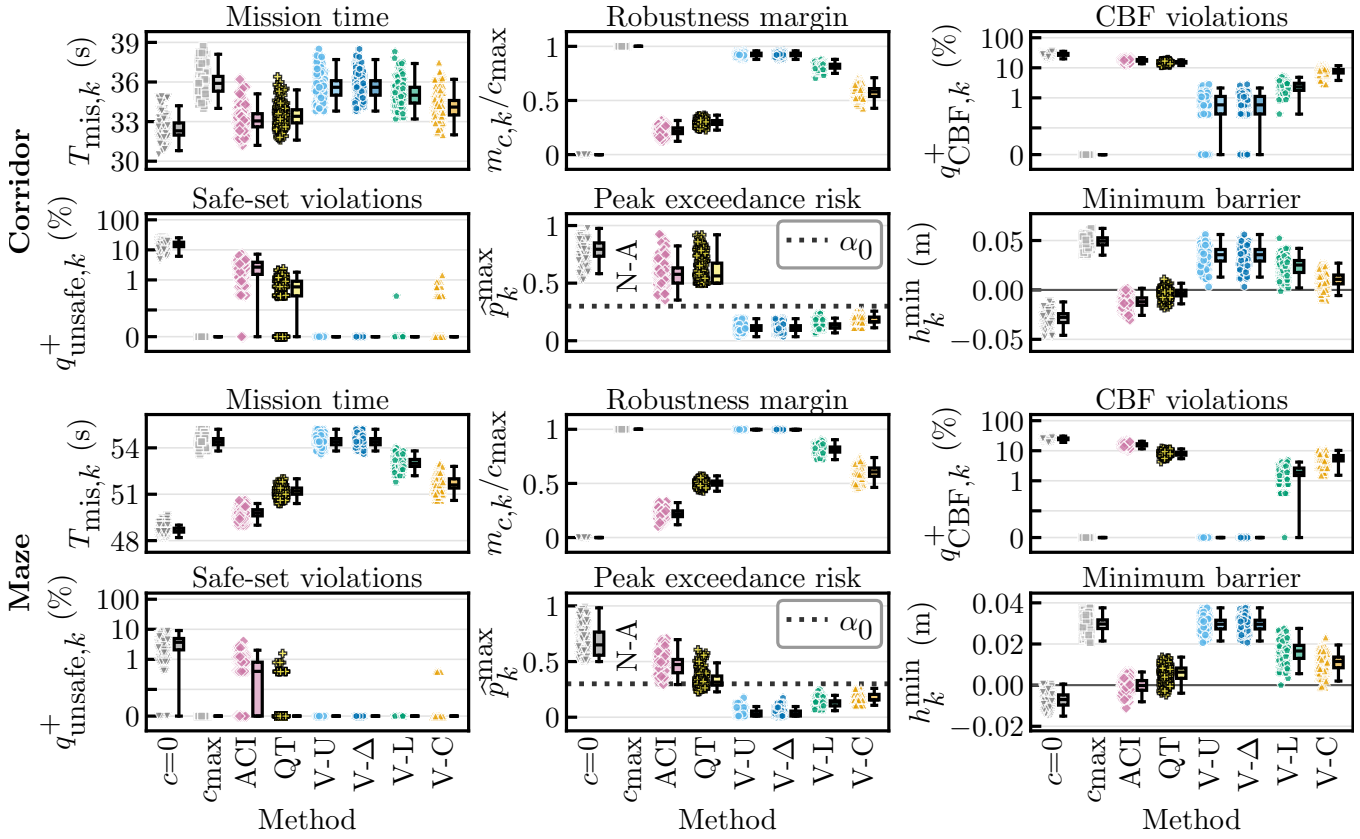


Figure 17: Corridor and Maze, S4, with panels as in Fig. 13.

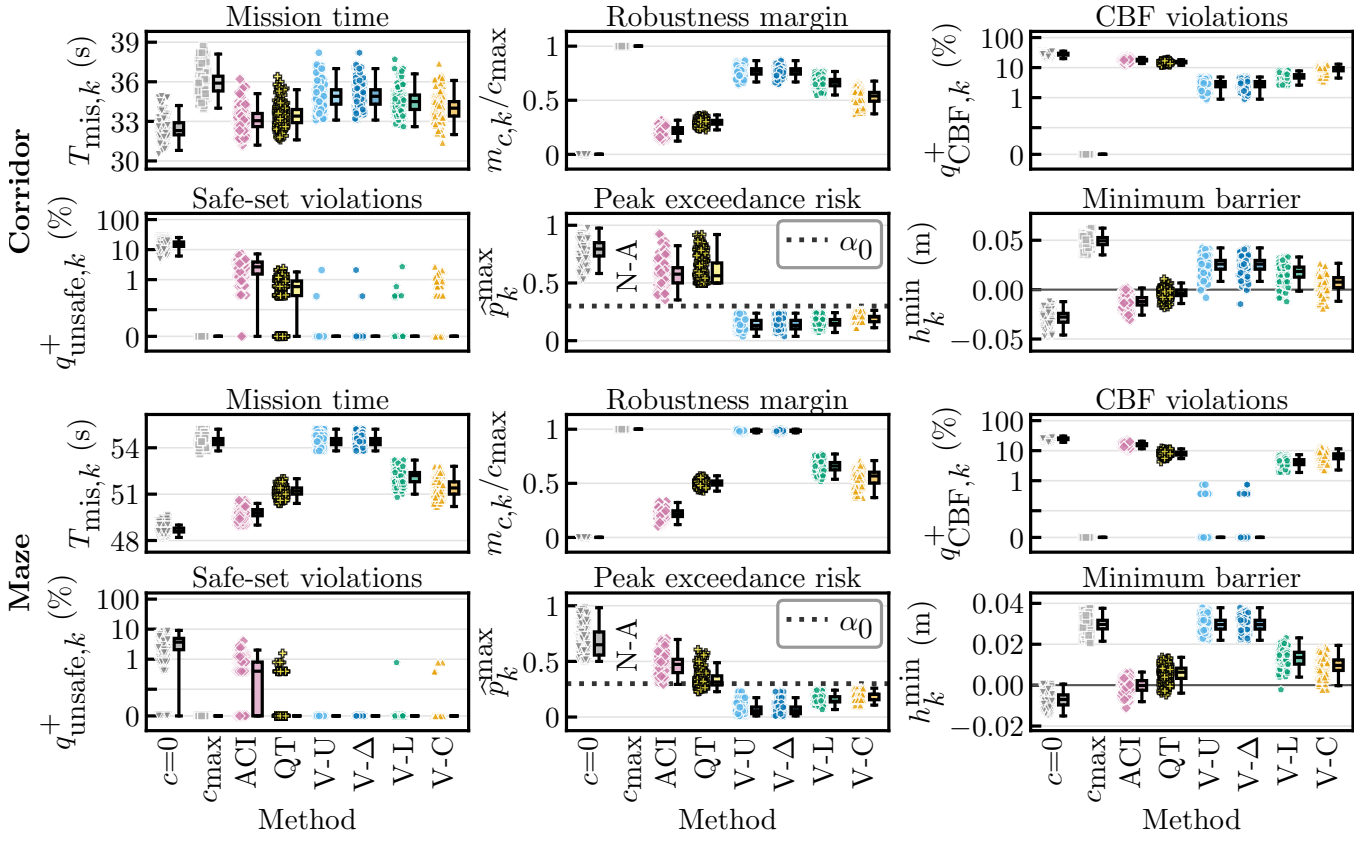


Figure 18: Corridor and Maze, S5, with panels as in Fig. 13.

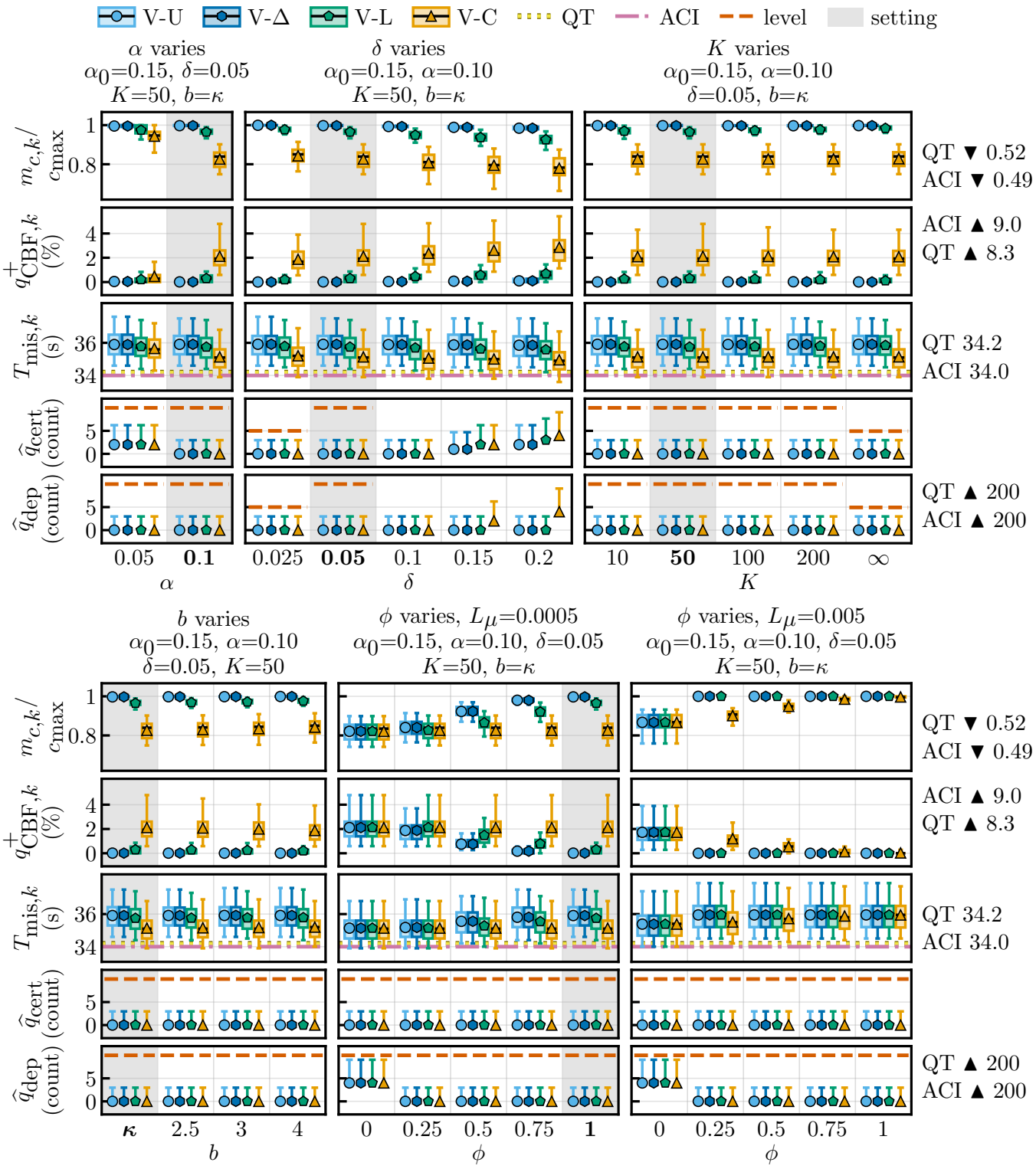


Figure 19: Corridor, default setting: sweeps, with layout in Section 2.2.3.

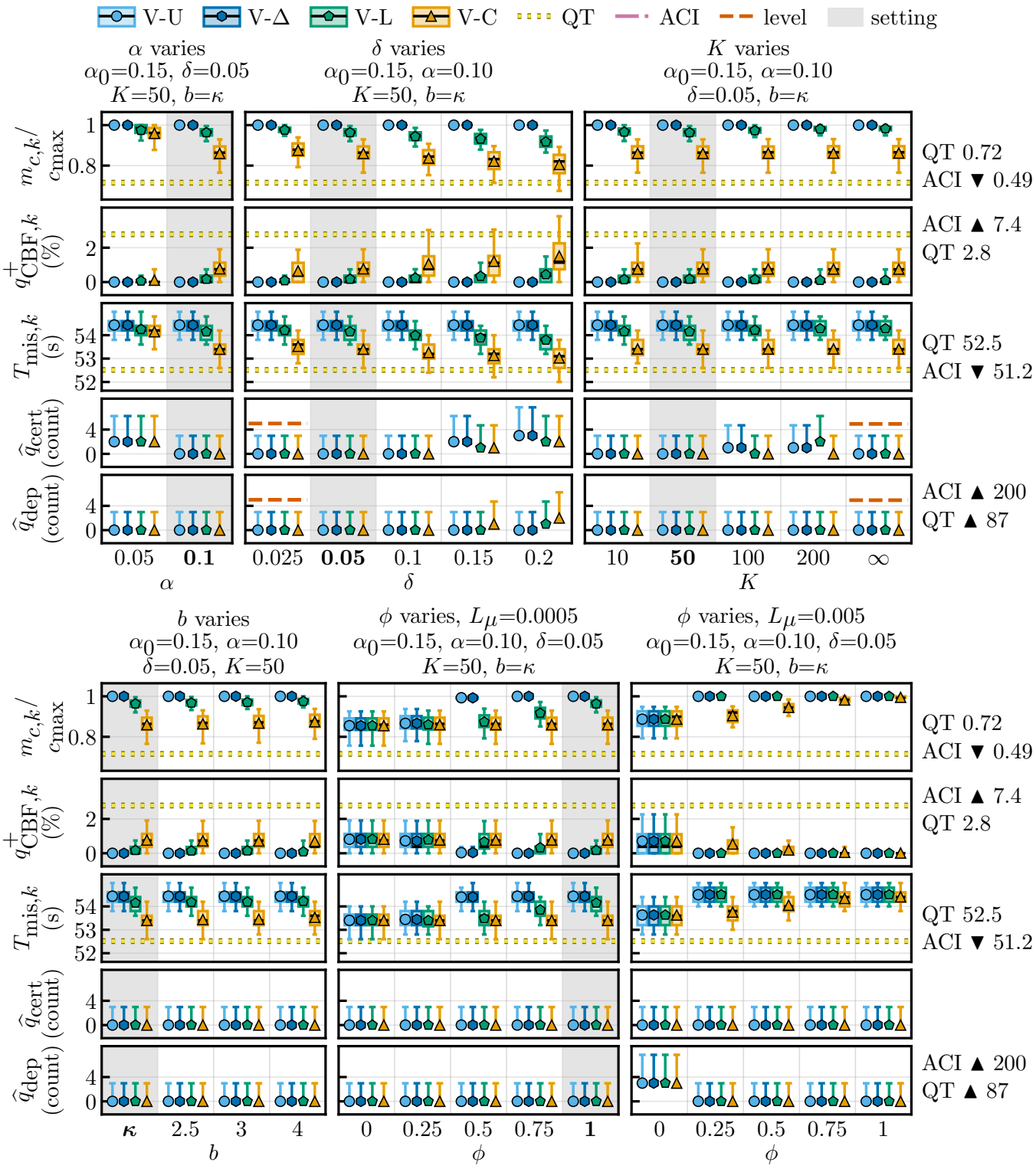


Figure 20: Maze, default setting: sweeps, with layout in Section 2.2.3.

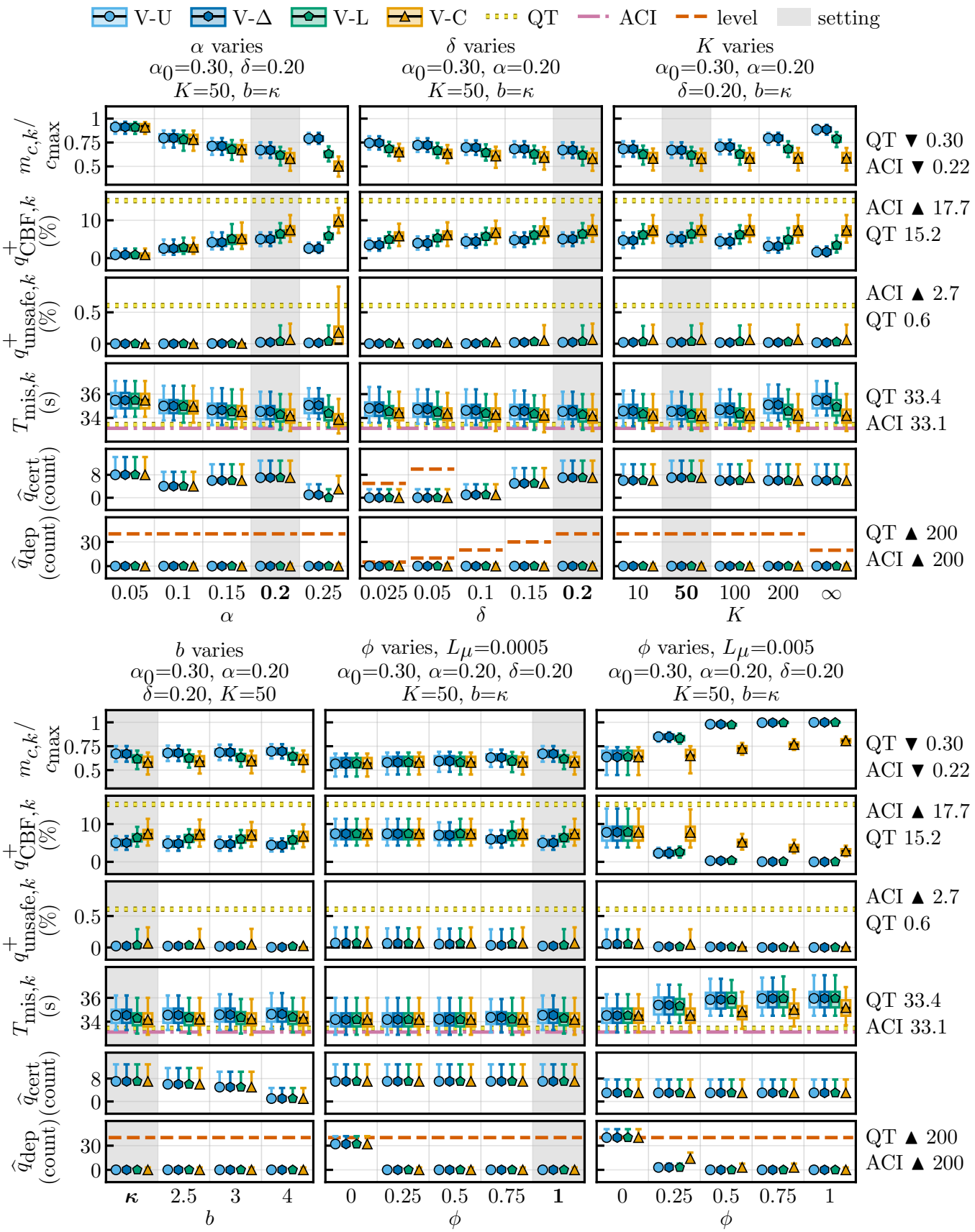


Figure 21: Corridor, S3: sweeps, with layout in Section 2.2.3.

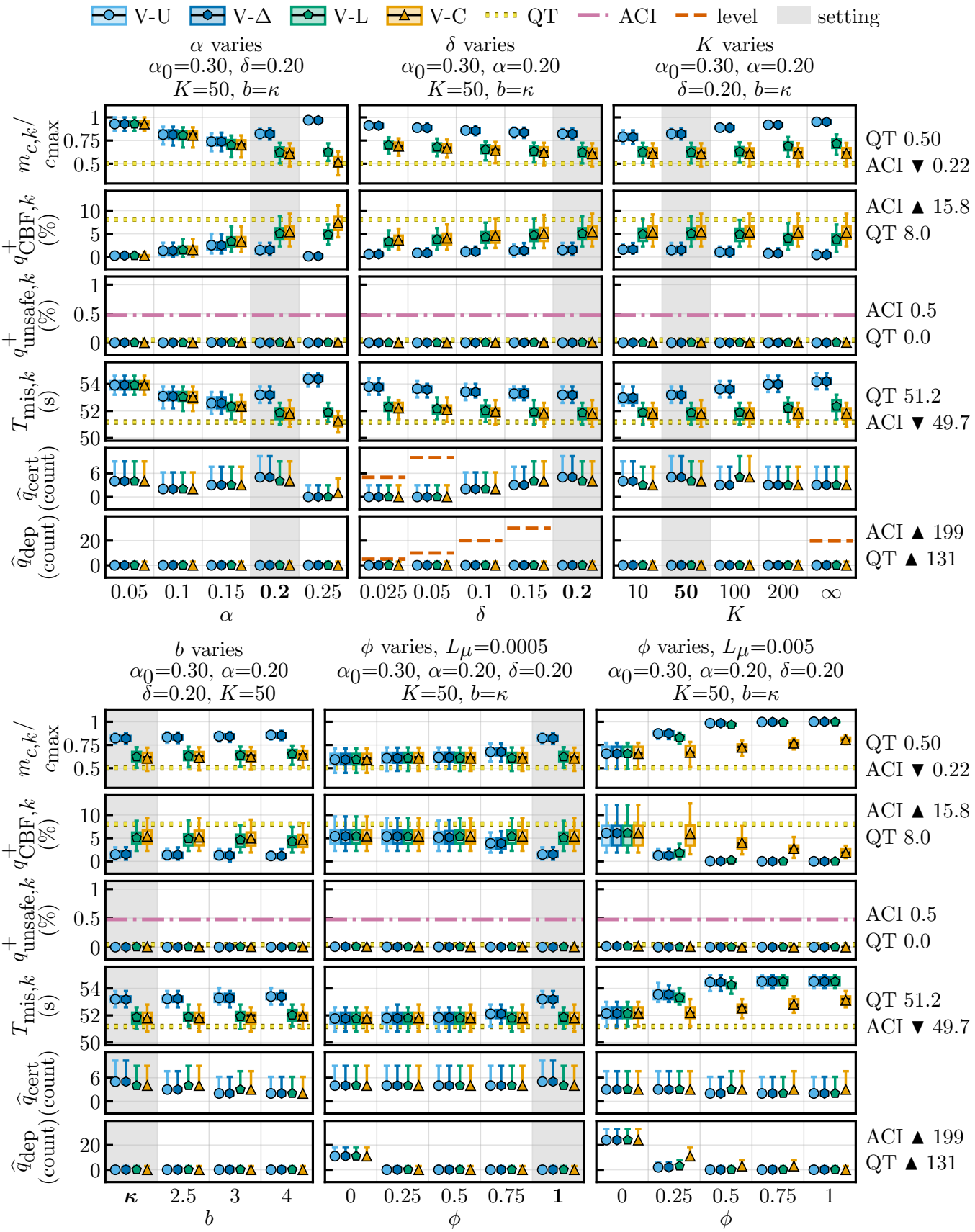


Figure 22: Maze, S3: sweeps, with layout in Section 2.2.3.

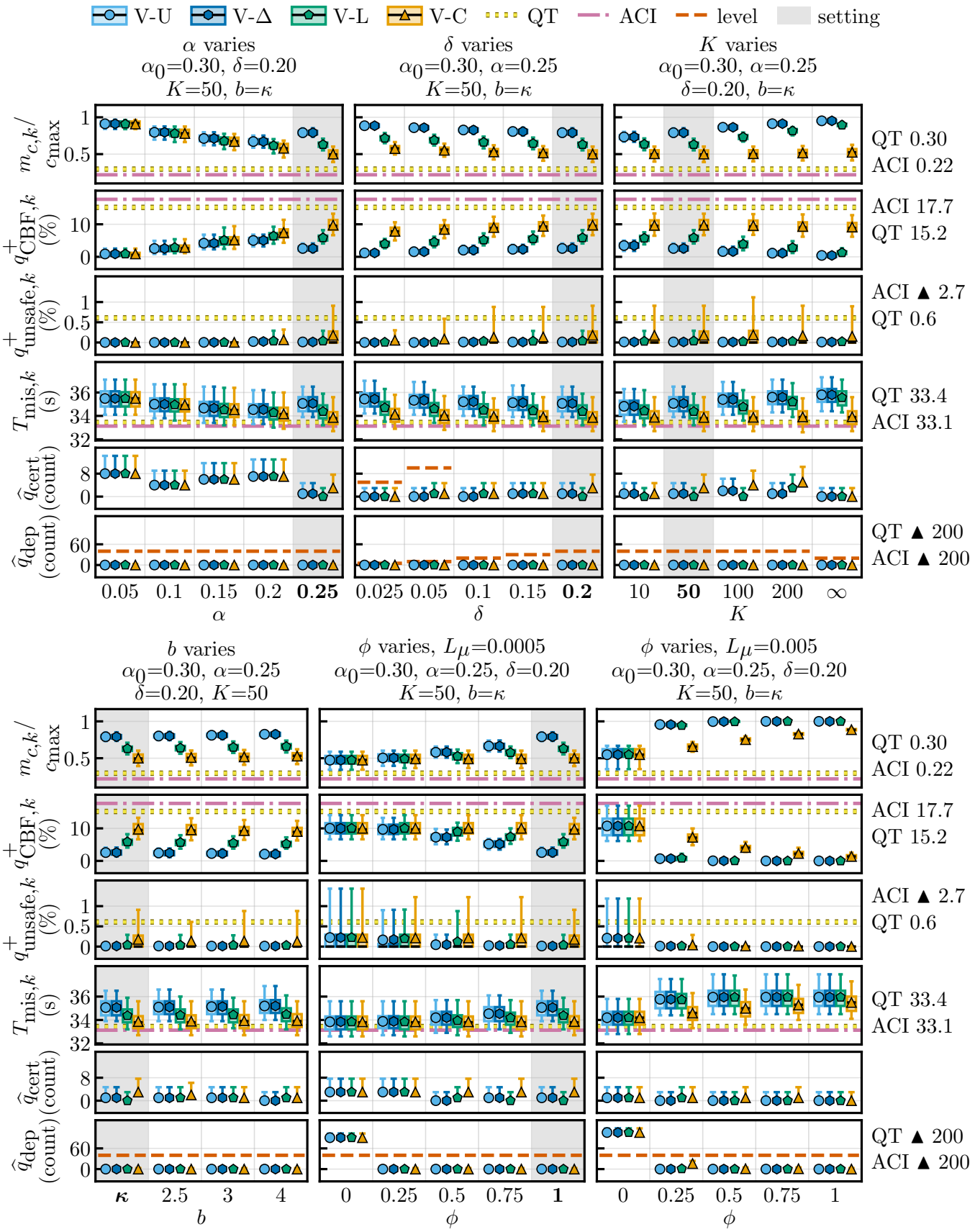


Figure 23: Corridor, S1: sweeps, with layout in Section 2.2.3.

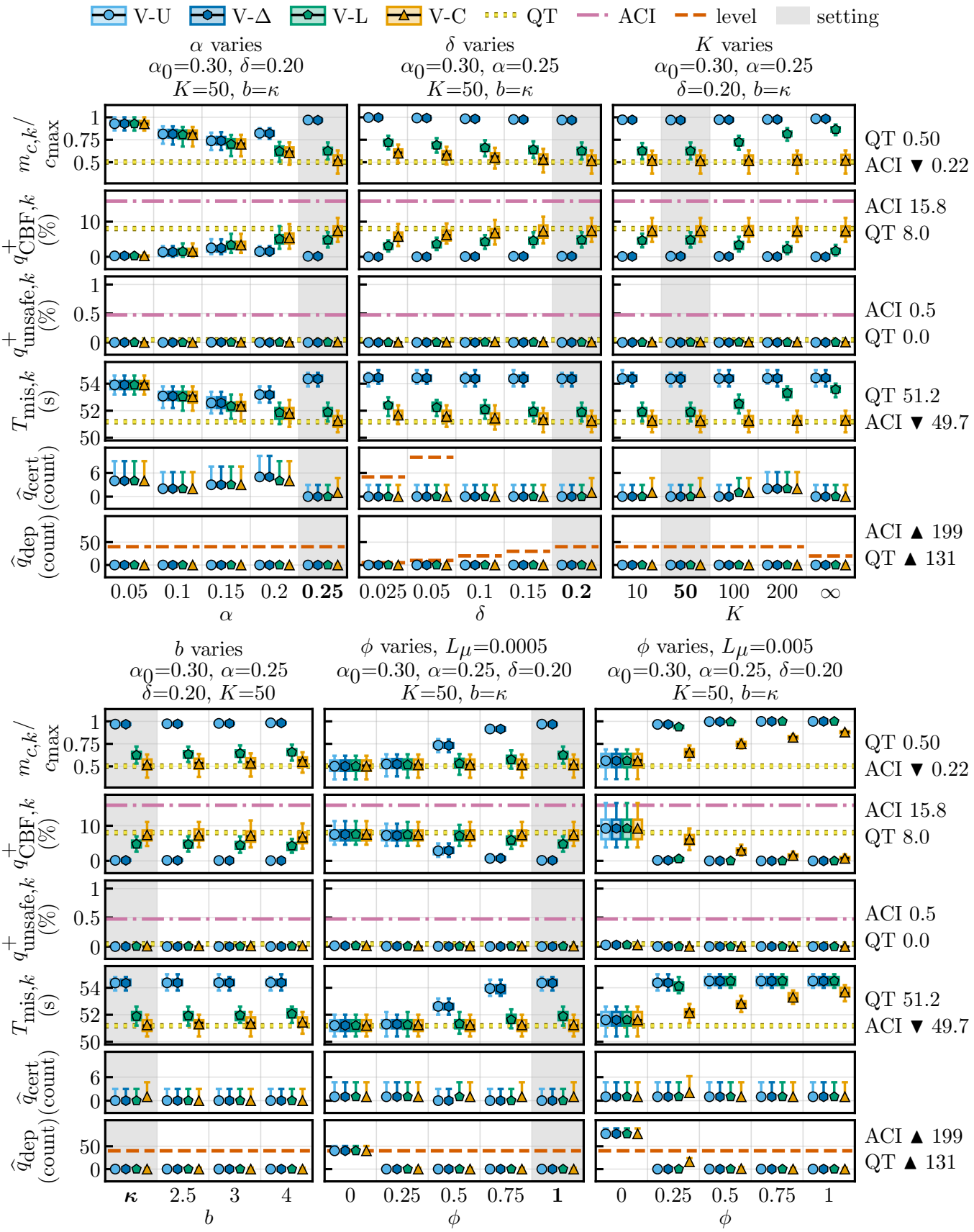


Figure 24: Maze, S1: sweeps, with layout in Section 2.2.3.

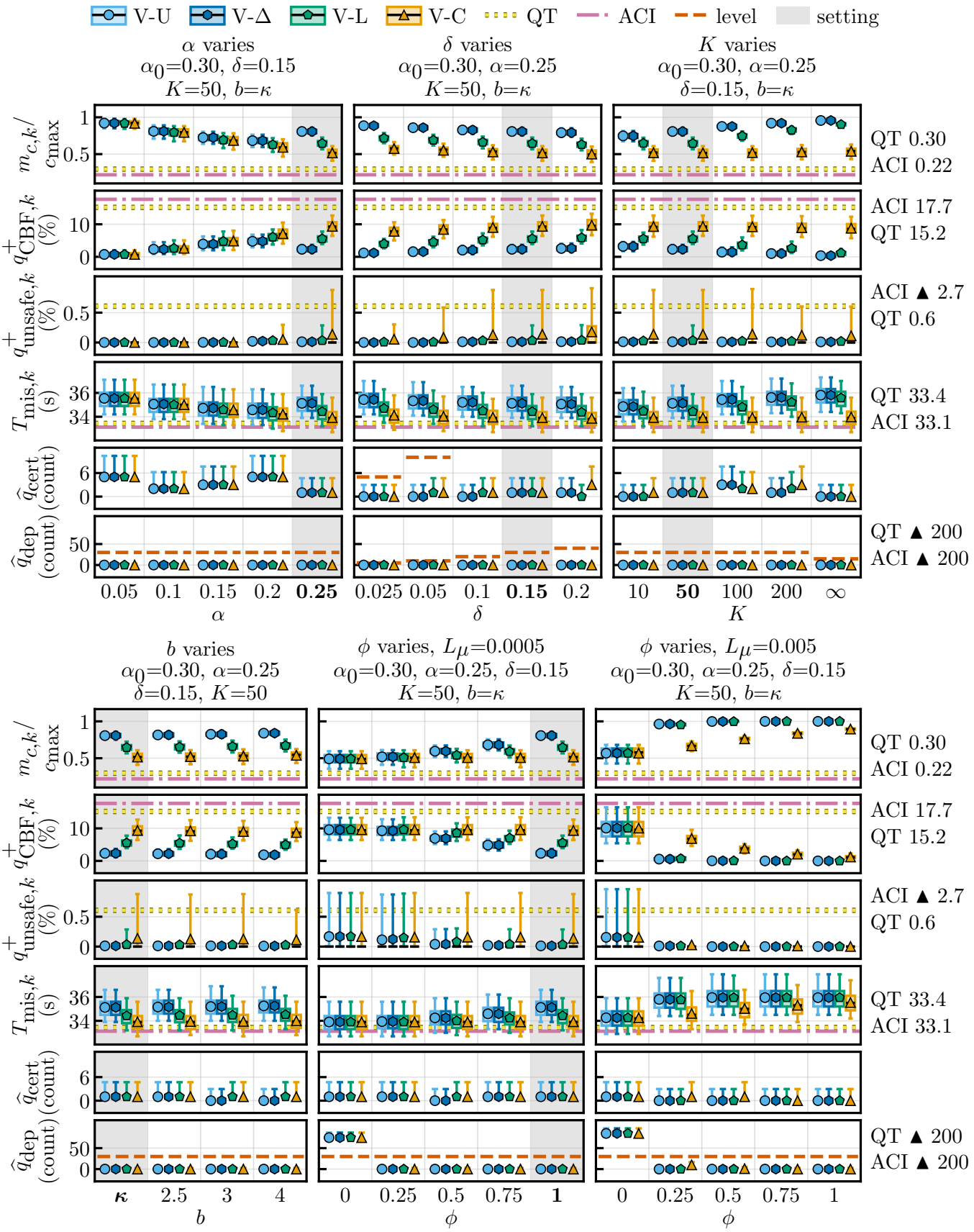


Figure 25: Corridor, S2: sweeps, with layout in Section 2.2.3.

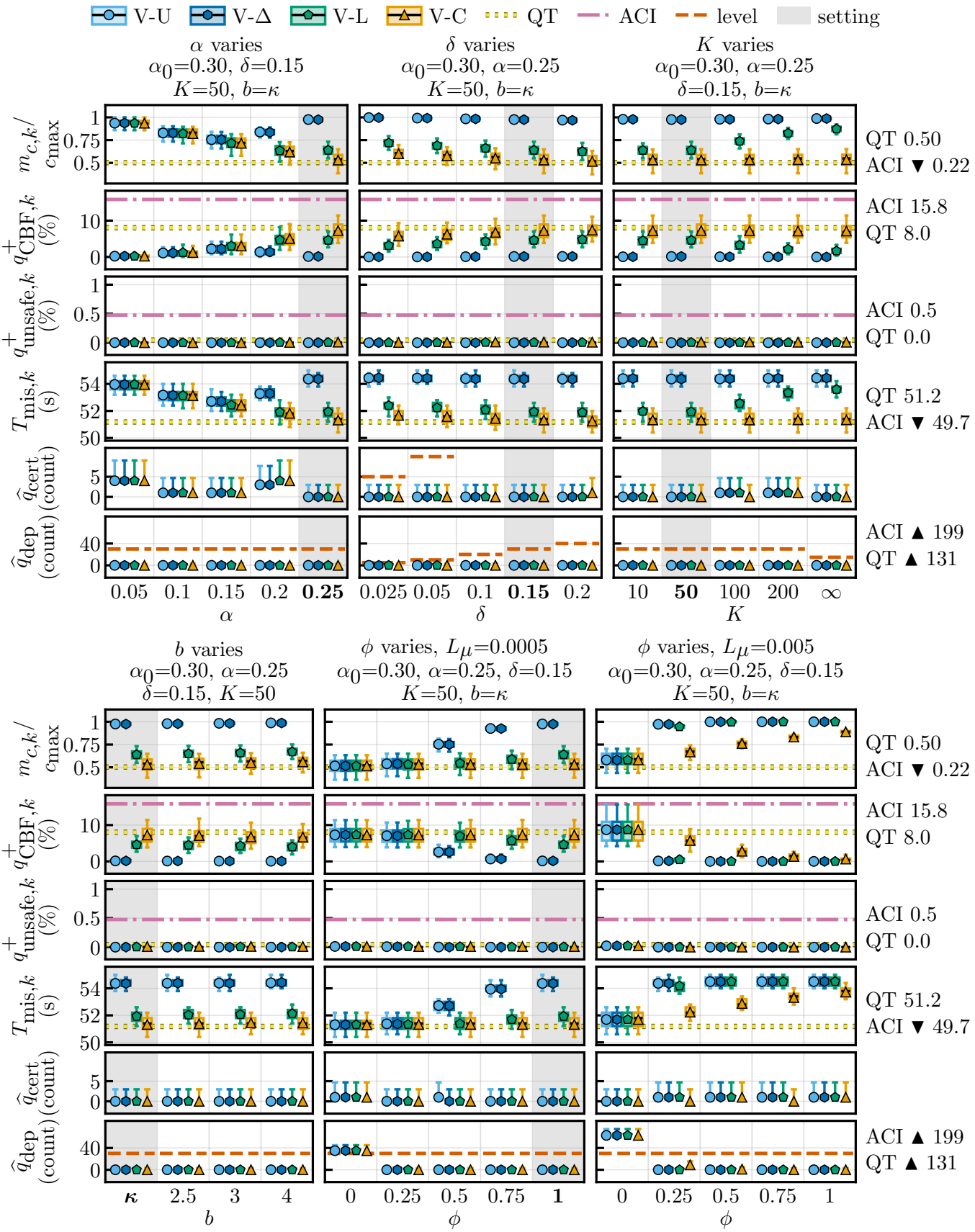


Figure 26: Maze, S2: sweeps, with layout in Section 2.2.3.

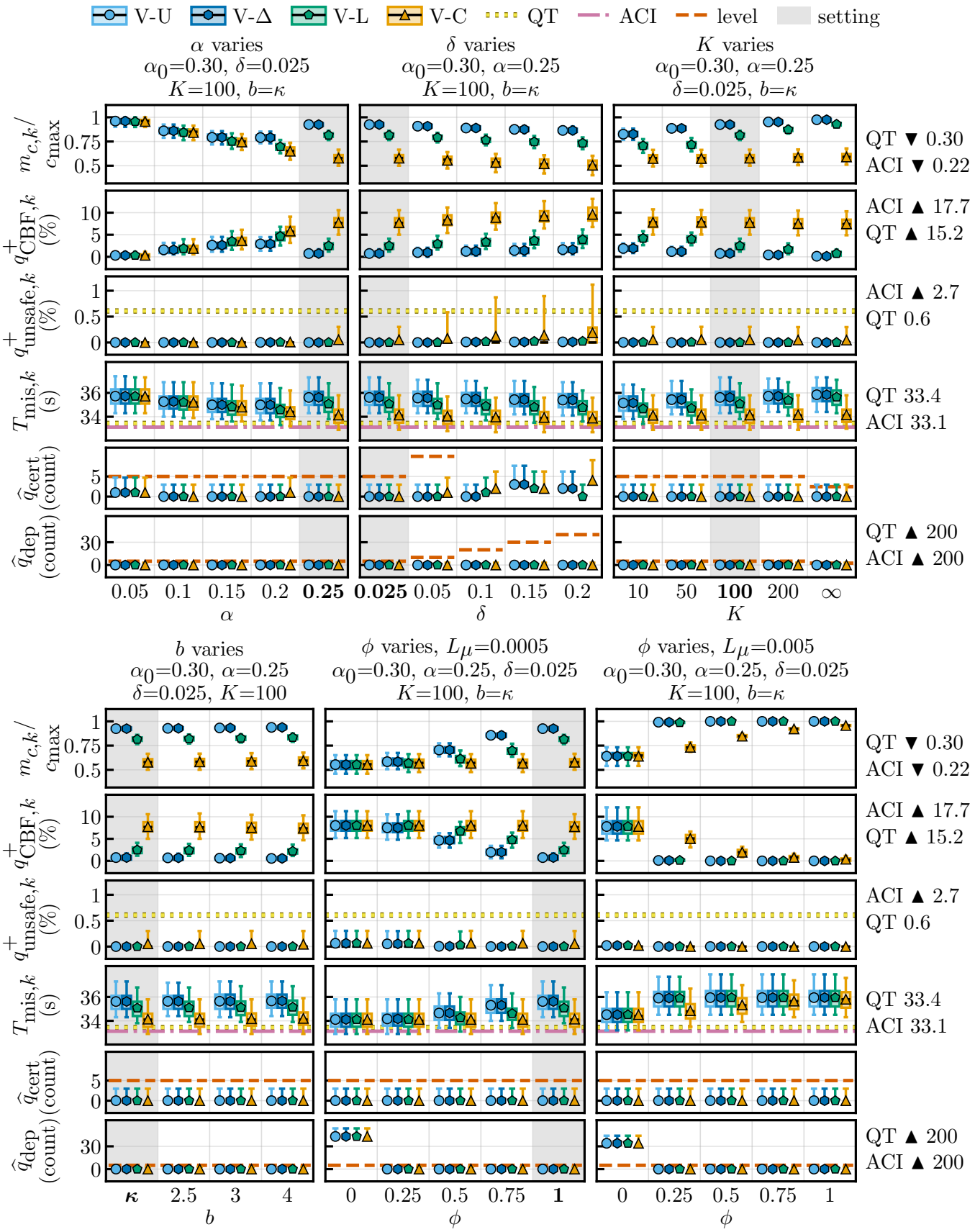


Figure 27: Corridor, S4: sweeps, with layout in Section 2.2.3.

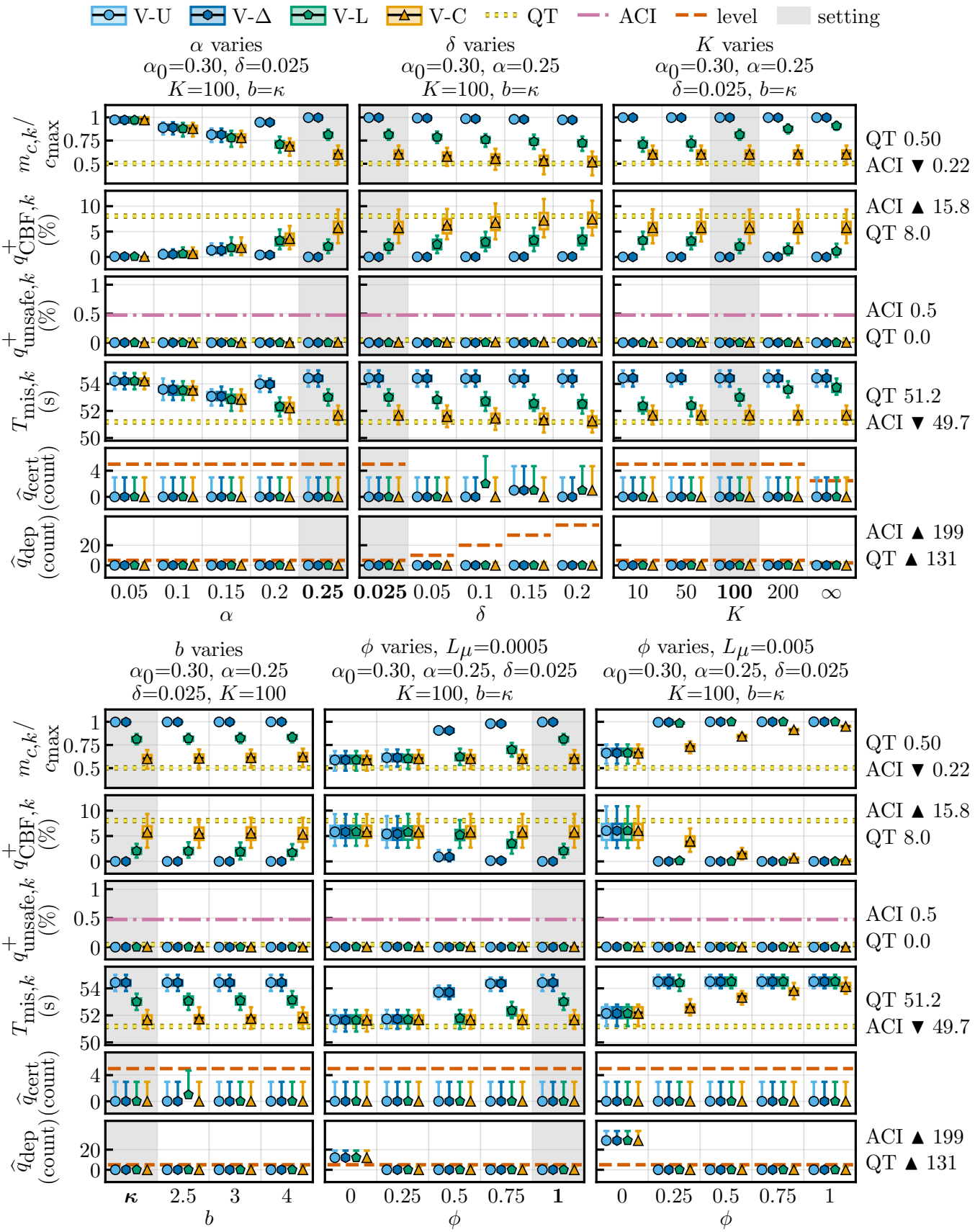


Figure 28: Maze, S4: sweeps, with layout in Section 2.2.3.

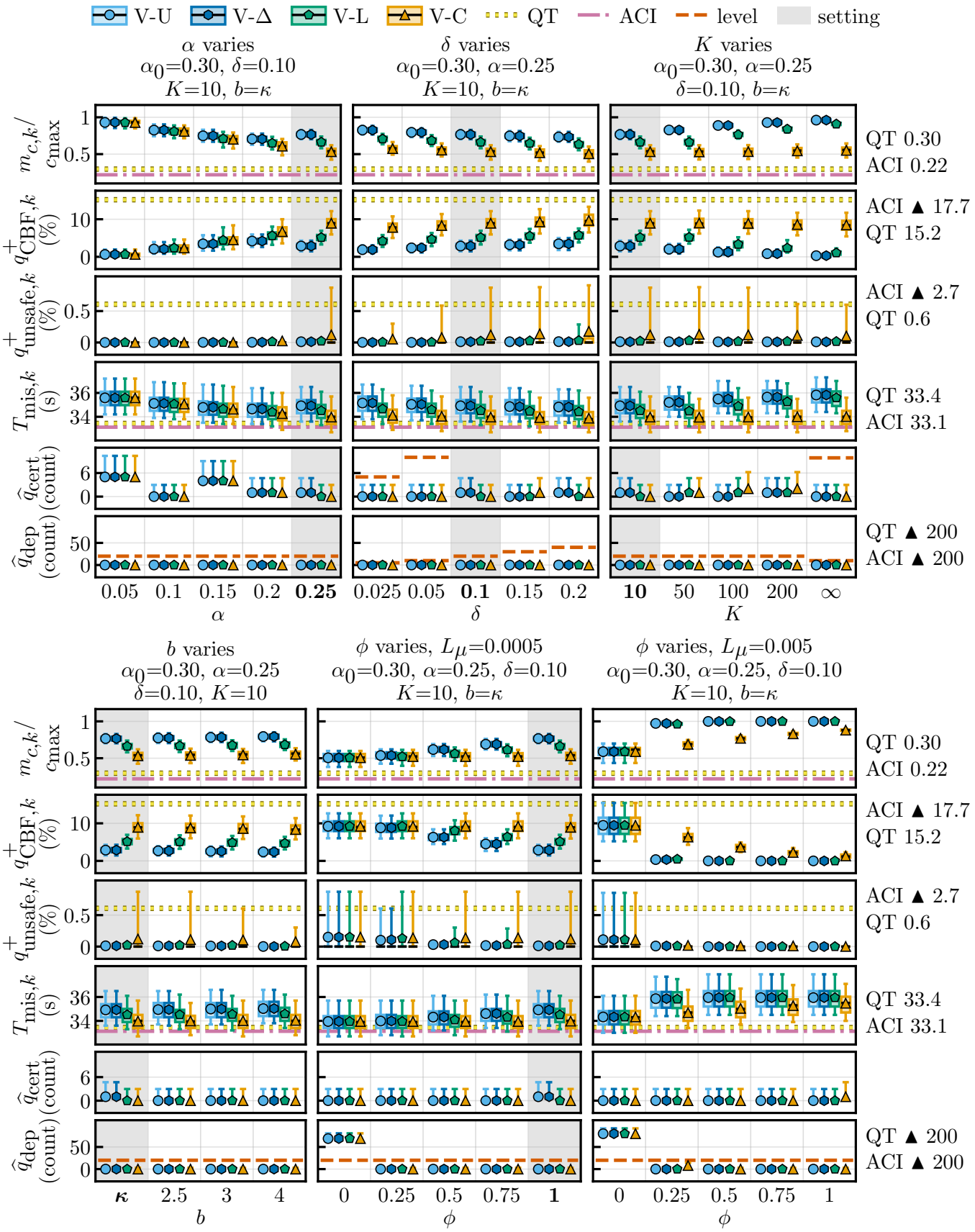


Figure 29: Corridor, S5: sweeps, with layout in Section 2.2.3.

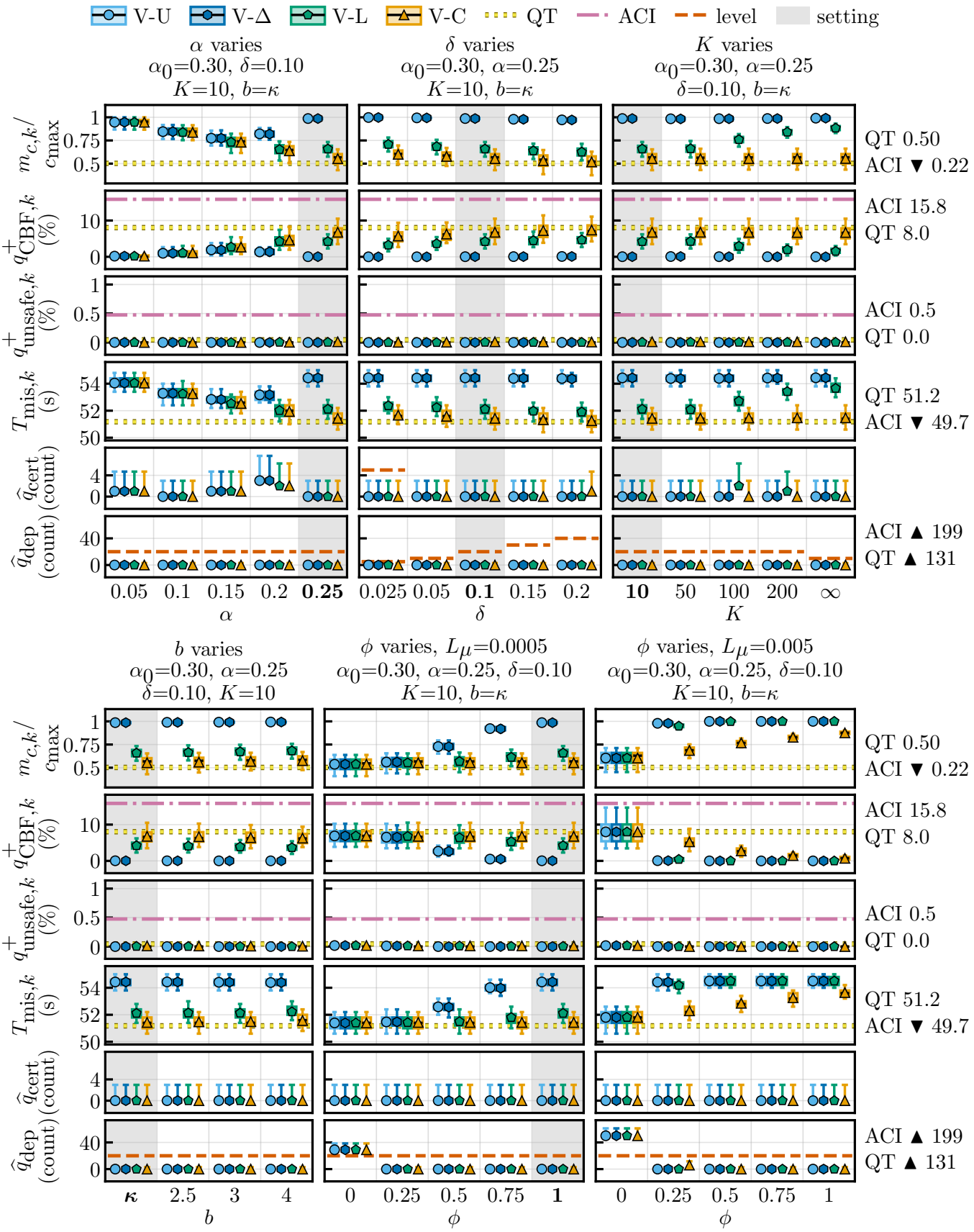


Figure 30: Maze, S5: sweeps, with layout in Section 2.2.3.

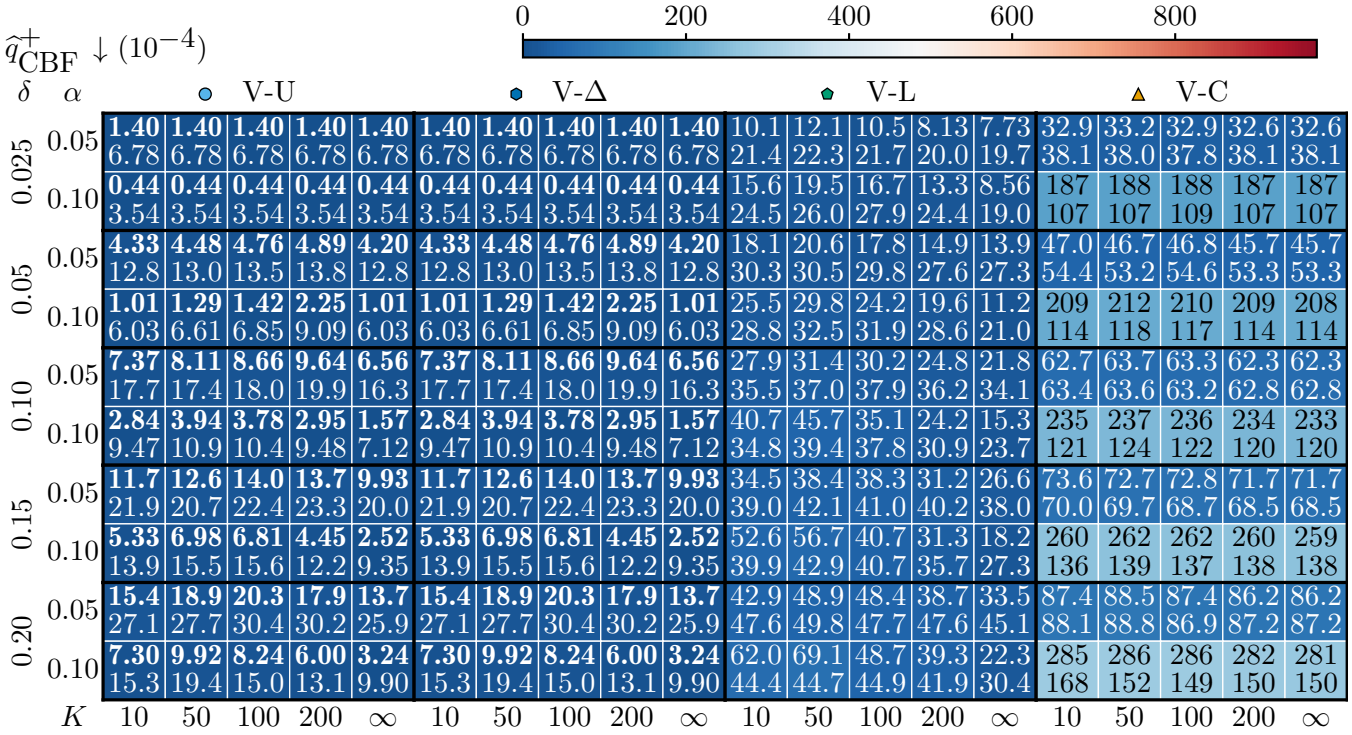


Figure 31: Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.15$.

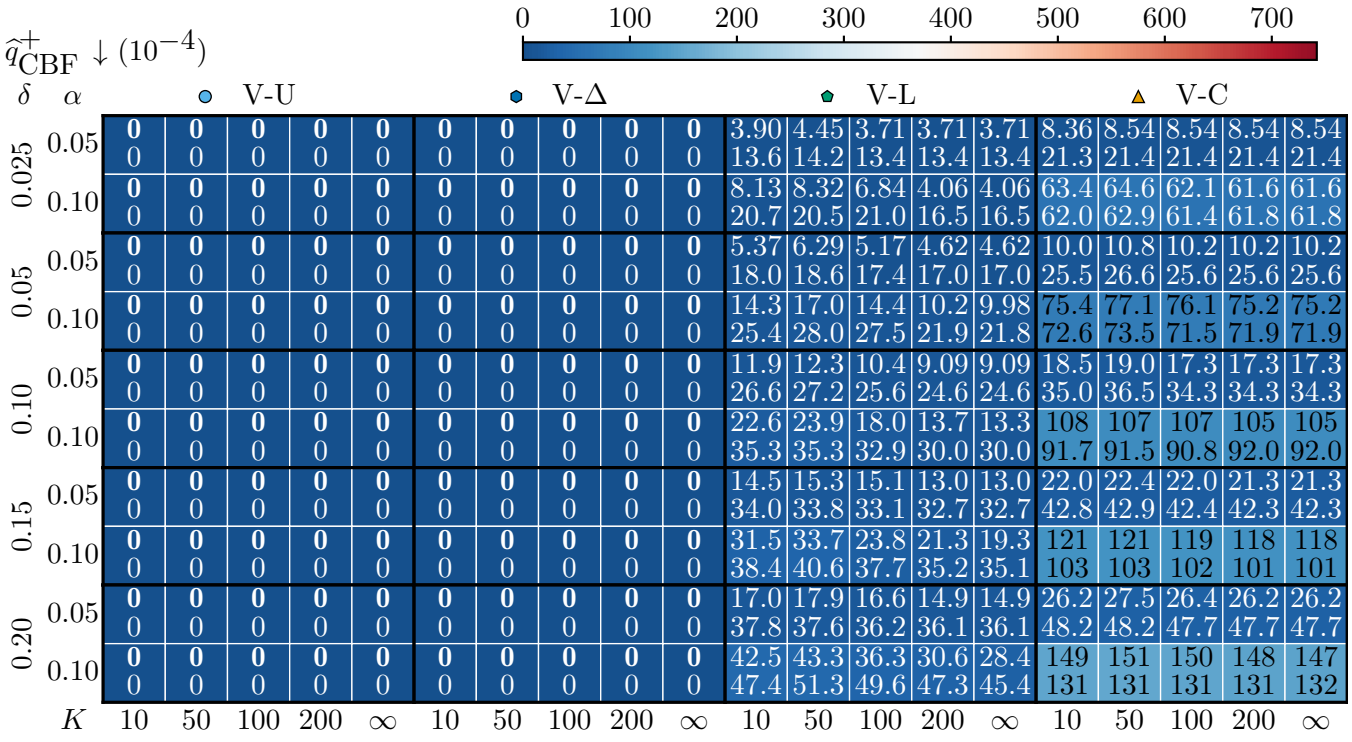


Figure 32: Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.15$.

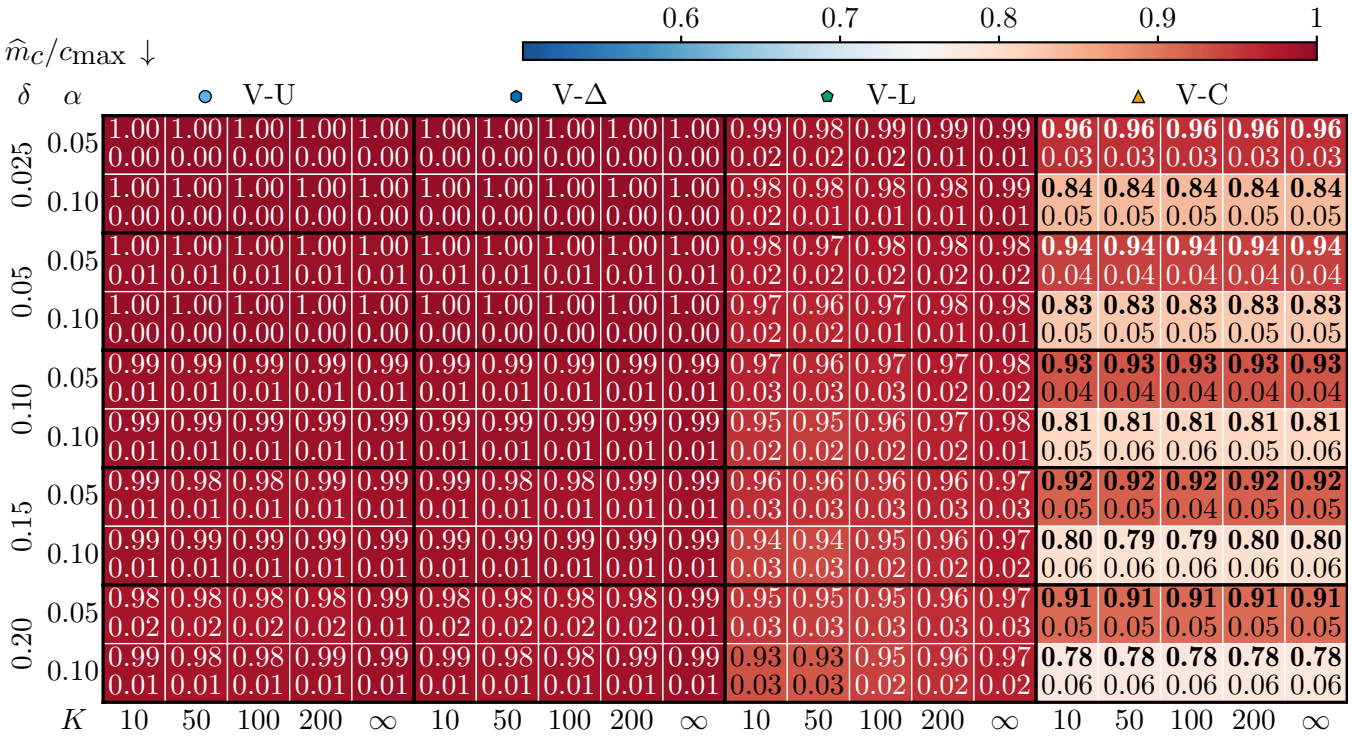


Figure 33: Corridor, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.15$.

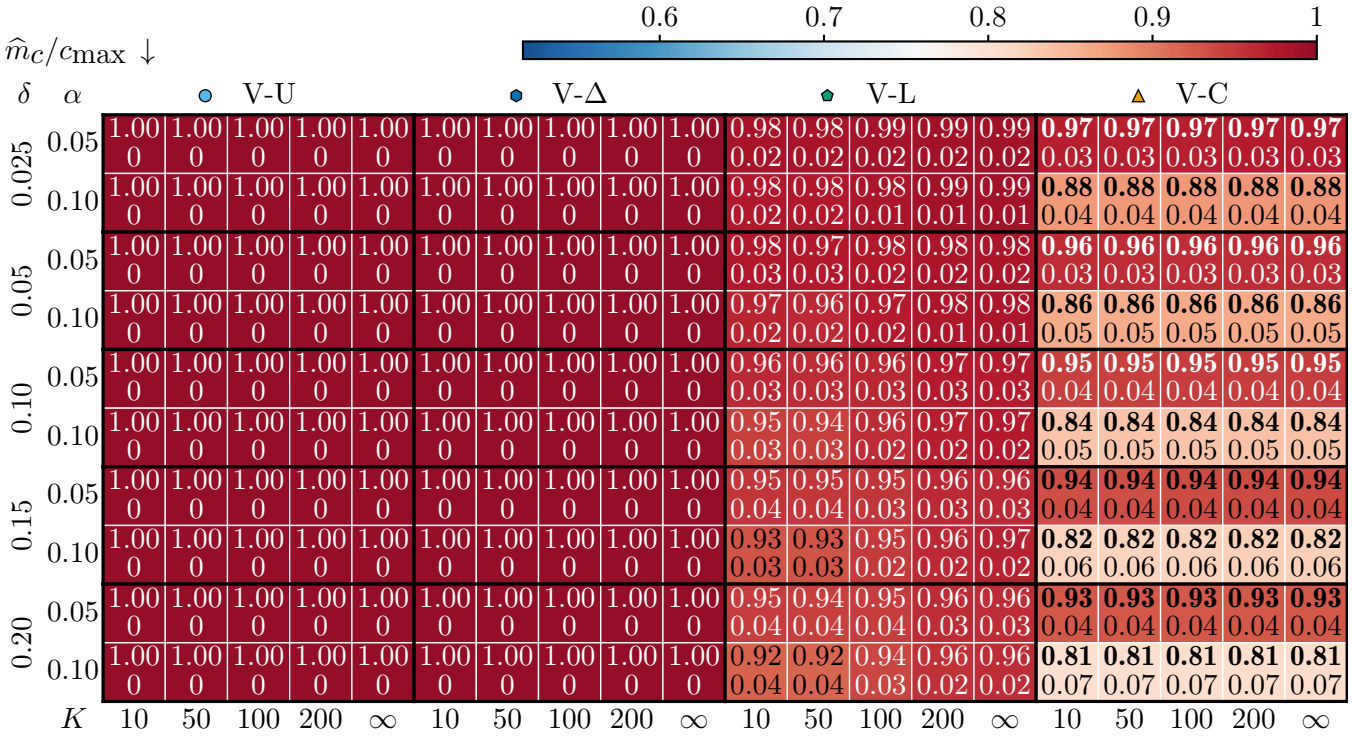


Figure 34: Maze, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.15$.

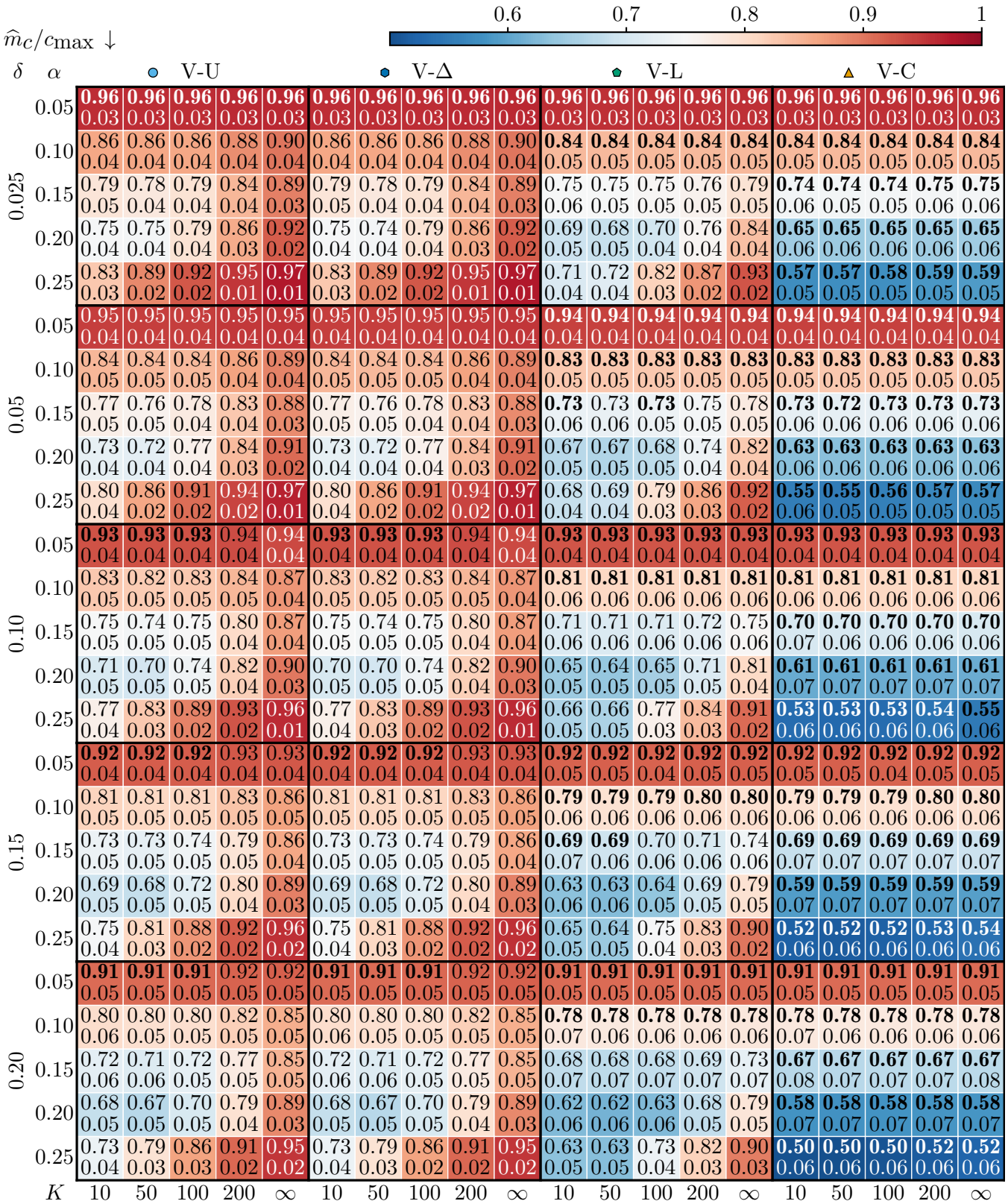


Figure 35: Corridor, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.30$.

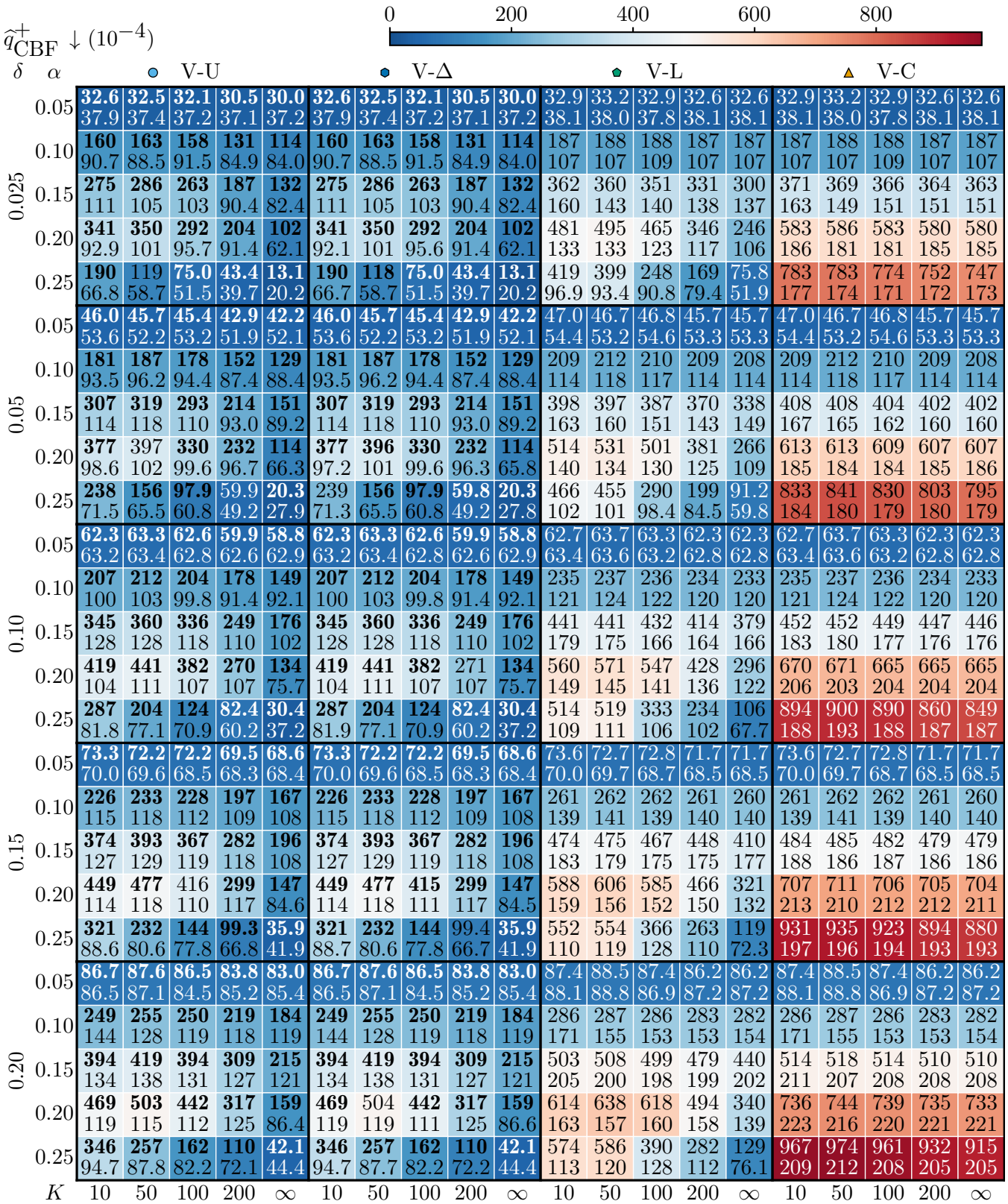
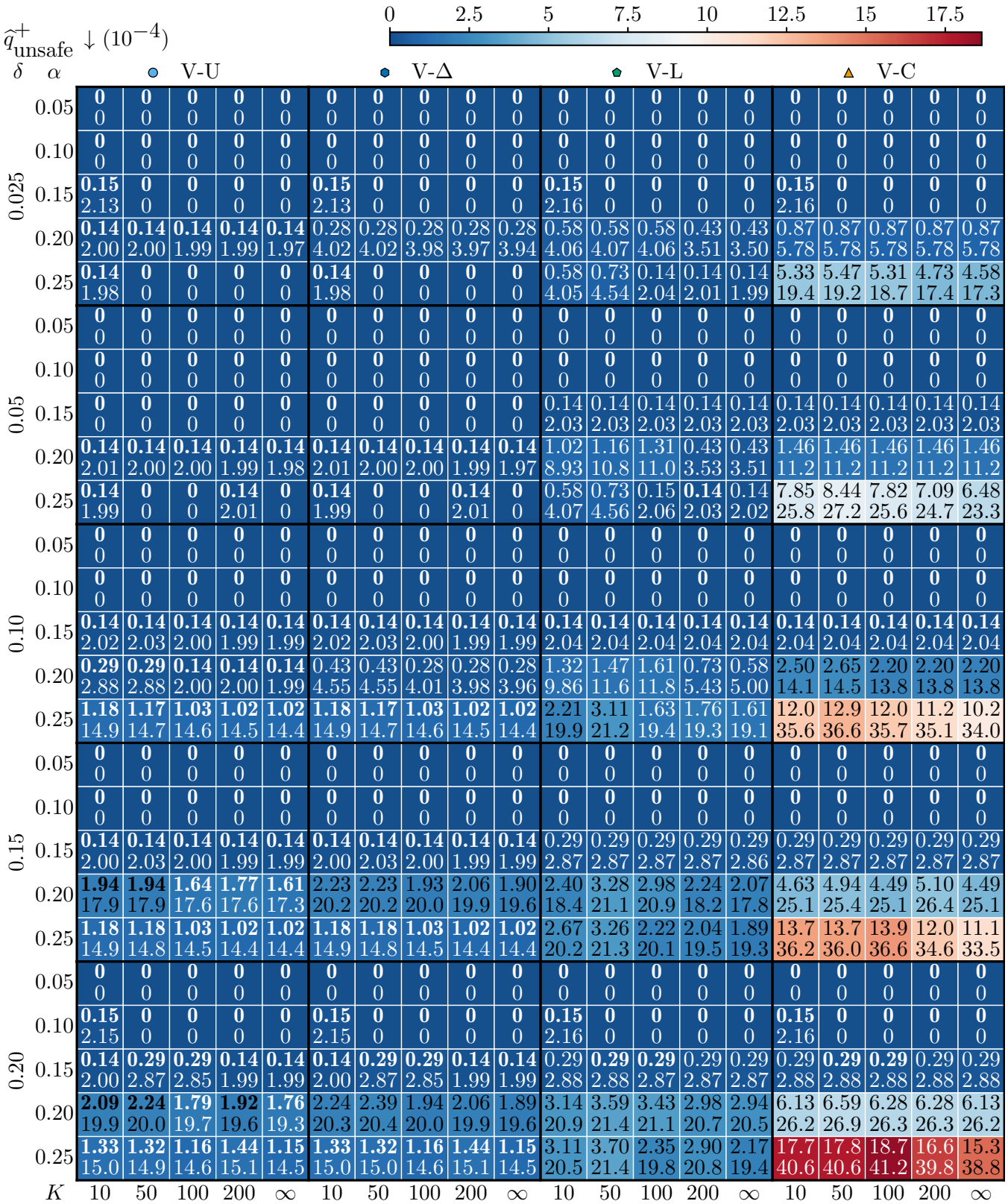


Figure 36: Corridor, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.30$.


$\widehat{q}^{+}_{\text{unsafe}} \downarrow (10^{-4})$
0 2.5 5 7.5 10 12.5 15 17.5
δ α
V-U V-Δ V-L V-C
0.025 0.05 0 0 0

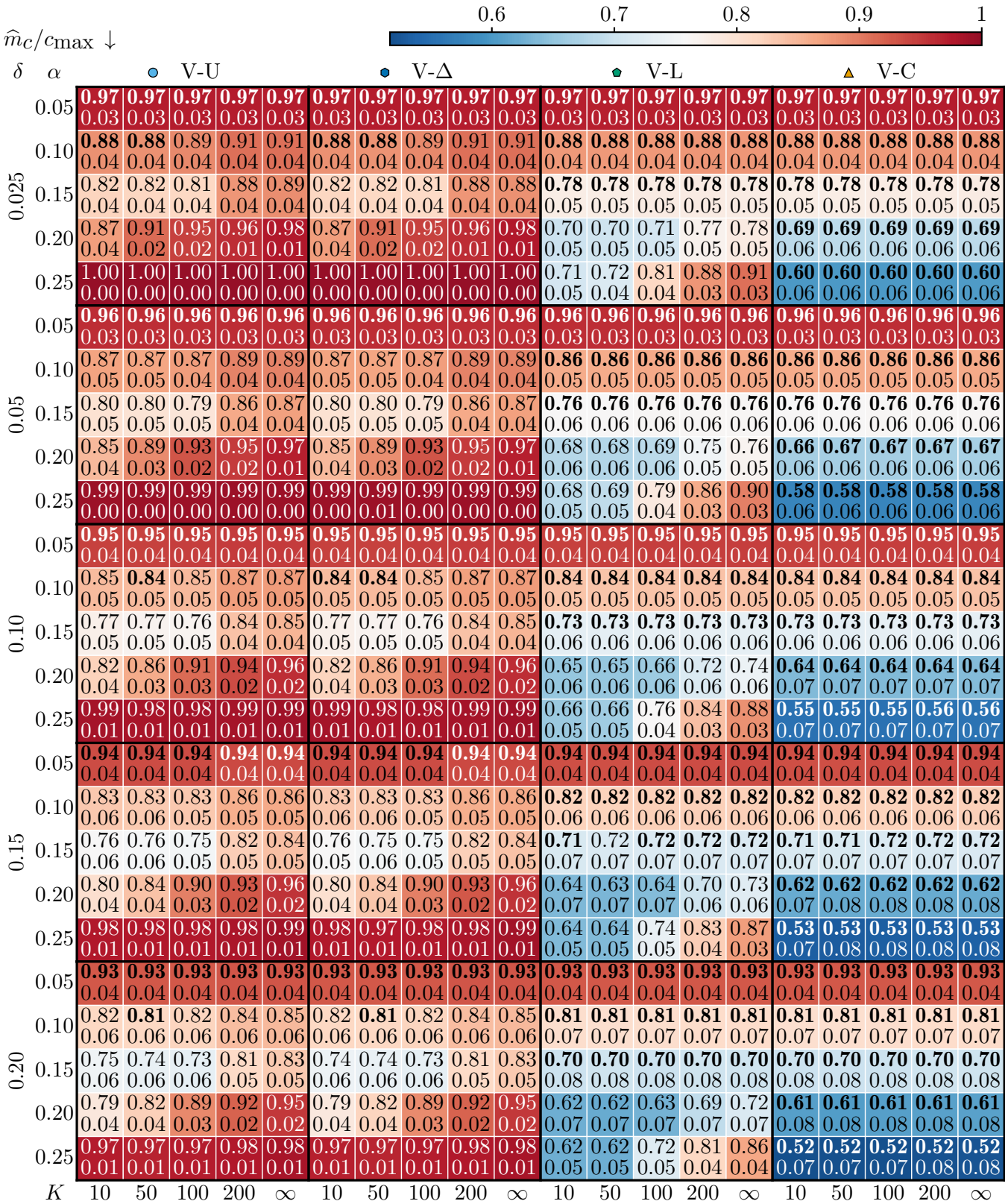
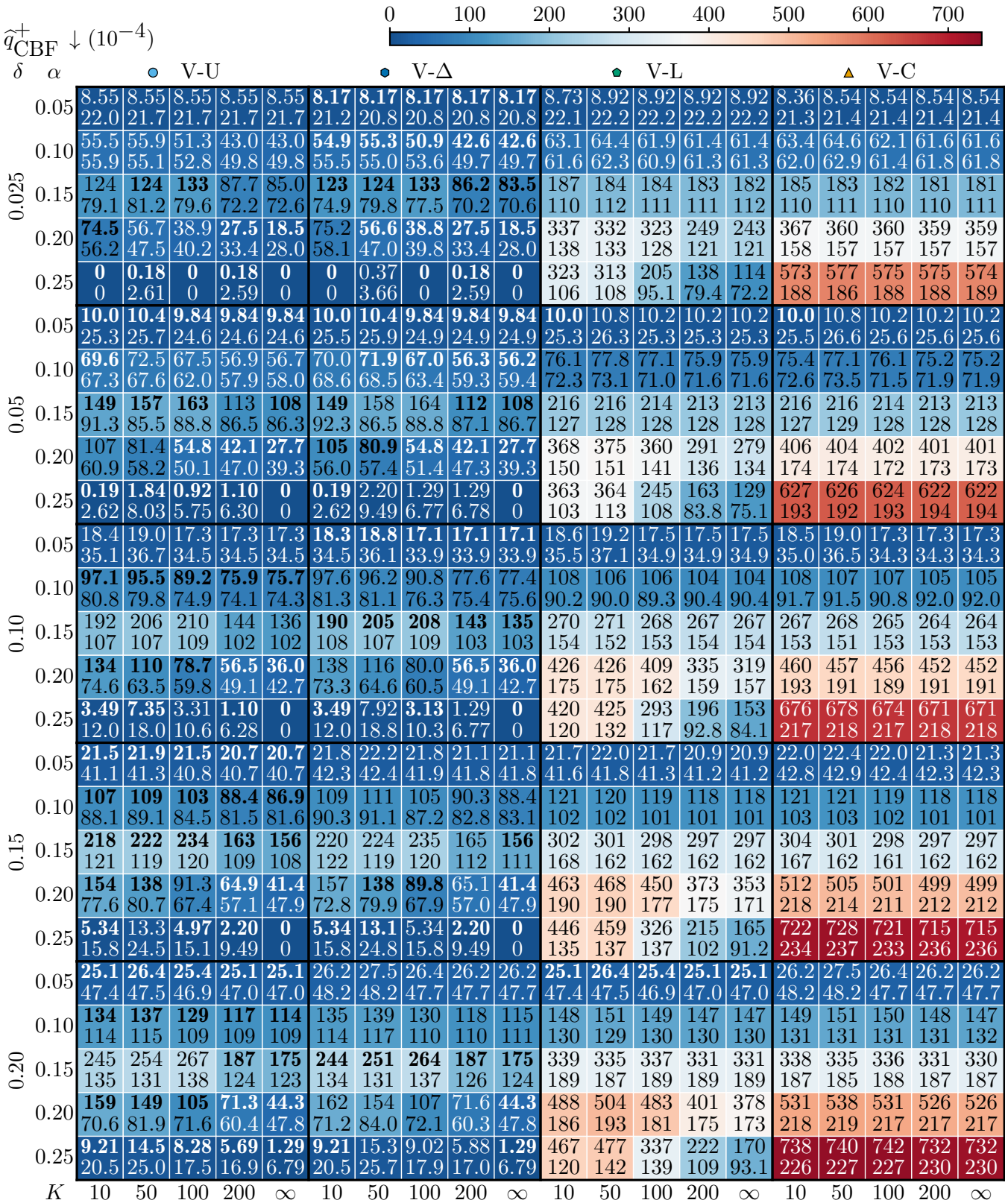


Figure 38: Maze, $\widehat{m}_c/c_{\max}$ at $\alpha_0 = 0.30$.



δ	α	V-U $K=10$	V-U $K=50$	V-U $K=100$	V-U $K=200$	V-U $K=\infty$	V-Δ $K=10$	V-Δ $K=50$	V-Δ $K=100$	V-Δ $K=200$	V-Δ $K=\infty$	V-L $K=10$	V-L $K=50$	V-L $K=100$	V-L $K=200$	V-L $K=\infty$	V-C $K=10$	V-C $K=50$	V-C $K=100$	V-C $K=200$	V-C $K=\infty$
0.025	0.05	8.55	8.55	8.55	8.55	8.55	**8.17**	**8.17**	**8.17**	**8.17**	**8.17**	8.73	8.92	8.92	8.92	8.92	8.36	8.54	8.54	8.54	8.54
		22.0	21.7	21.7	21.7	21.7	21.2	20.8	20.8	20.8	20.8	22.1	22.2	22.2	22.2	22.2	21.3	21.4	21.4	21.4	21.4
	0.10	55.5	55.9	51.3	43.0	43.0	**54.9**	**55.3**	**50.9**	**42.6**	**42.6**	63.1	64.4	61.9	61.4	61.4	63.4	64.6	62.1	61.6	61.6
		55.9	55.1	52.8	49.8	49.8	55.5	55.0	53.6	49.7	49.7	61.6	62.3	60.9	61.3	61.3	62.0	62.9	61.4	61.8	61.8
	0.15	124	**124**	**133**	87.7	85.0	**123**	**124**	**133**	**86.2**	**83.5**	187	184	184	183	182	185	183	182	181	181
		79.1	81.2	79.6	72.2	72.6	74.9	79.8	77.5	70.2	70.6	110	112	111	111	112	110	111	110	110	111
	0.20	**74.5**	56.7	38.9	**27.5**	**18.5**	75.2	**56.6**	**38.8**	**27.5**	**18.5**	337	332	323	249	243	367	360	360	359	359
		56.2	47.5	40.2	33.4	28.0	58.1	47.0	39.8	33.4	28.0	138	133	128	121	121	158	157	157	157	157
	0.25	**0**	**0.18**	**0**	**0.18**	**0**	**0**	0.37	**0**	**0.18**	**0**	323	313	205	138	114	573	577	575	575	574
		0	2.61	0	2.59	0	0	3.66	0	2.59	0	106	108	95.1	79.4	72.2	188	186	188	188	189
0.05	0.05	**10.0**	**10.4**	**9.84**	**9.84**	**9.84**	**10.0**	**10.4**	**9.84**	**9.84**	**9.84**	**10.0**	10.8	10.2	10.2	10.2	**10.0**	10.8	10.2	10.2	10.2
		25.3	25.7	24.6	24.6	24.6	25.5	25.9	24.9	24.9	24.9	25.3	26.3	25.3	25.3	25.3	25.5	26.6	25.6	25.6	25.6
	0.10	**69.6**	72.5	67.5	56.9	56.7	70.0	**71.9**	**67.0**	**56.3**	**56.2**	76.1	77.8	77.1	75.9	75.9	75.4	77.1	76.1	75.2	75.2
		67.3	67.6	62.0	57.9	58.0	68.6	68.5	63.4	59.3	59.4	72.3	73.1	71.0	71.6	71.6	72.6	73.5	71.5	71.9	71.9
	0.15	**149**	**157**	**163**	113	**108**	**149**	158	164	**112**	**108**	216	216	214	213	213	216	216	214	213	213
		91.3	85.5	88.8	86.5	86.3	92.3	86.5	88.8	87.1	86.7	127	128	128	128	128	127	129	128	128	128
	0.20	107	81.4	**54.8**	**42.1**	**27.7**	**105**	**80.9**	**54.8**	**42.1**	**27.7**	368	375	360	291	279	406	404	402	401	401
		60.9	58.2	50.1	47.0	39.3	56.0	57.4	51.4	47.3	39.3	150	151	141	136	134	174	174	172	173	173
	0.25	**0.19**	**1.84**	**0.92**	**1.10**	**0**	**0.19**	2.20	1.29	1.29	**0**	363	364	245	163	129	627	626	624	622	622
		2.62	8.03	5.75	6.30	0	2.62	9.49	6.77	6.78	0	103	113	108	83.8	75.1	193	192	193	194	194
0.10	0.05	18.4	19.0	17.3	17.3	17.3	**18.3**	**18.8**	**17.1**	**17.1**	**17.1**	18.6	19.2	17.5	17.5	17.5	18.5	19.0	17.3	17.3	17.3
		35.1	36.7	34.5	34.5	34.5	34.5	36.1	33.9	33.9	33.9	35.5	37.1	34.9	34.9	34.9	35.0	36.5	34.3	34.3	34.3
	0.10	**97.1**	**95.5**	**89.2**	**75.9**	**75.7**	97.6	96.2	90.8	77.6	77.4	108	106	106	104	104	108	107	107	105	105
		80.8	79.8	74.9	74.1	74.3	81.3	81.1	76.3	75.4	75.6	90.2	90.0	89.3	90.4	90.4	91.7	91.5	90.8	92.0	92.0
	0.15	192	206	210	144	136	**190**	**205**	**208**	**143**	**135**	270	271	268	267	267	267	268	265	264	264
		107	107	109	102	102	108	107	109	103	103	154	152	153	154	154	153	151	153	153	153
	0.20	**134**	**110**	**78.7**	**56.5**	**36.0**	138	116	80.0	**56.5**	**36.0**	426	426	409	335	319	460	457	456	452	452
		74.6	63.5	59.8	49.1	42.7	73.3	64.6	60.5	49.1	42.7	175	175	162	159	157	193	191	189	191	191
	0.25	**3.49**	**7.35**	3.31	**1.10**	**0**	**3.49**	7.92	**3.13**	1.29	**0**	420	425	293	196	153	676	678	674	671	671
		12.0	18.0	10.6	6.28	0	12.0	18.8	10.3	6.77	0	120	132	117	92.8	84.1	217	218	217	218	218
0.15	0.05	**21.5**	**21.9**	**21.5**	**20.7**	**20.7**	21.8	22.2	21.8	21.1	21.1	21.7	22.0	21.7	20.9	20.9	22.0	22.4	22.0	21.3	21.3
		41.1	41.3	40.8	40.7	40.7	42.3	42.4	41.9	41.8	41.8	41.6	41.8	41.3	41.2	41.2	42.8	42.9	42.4	42.3	42.3
	0.10	**107**	**109**	**103**	**88.4**	**86.9**	109	111	105	90.3	88.4	121	120	119	118	118	121	121	119	118	118
		88.1	89.1	84.5	81.5	81.6	90.3	91.1	87.2	82.8	83.1	102	102	101	101	101	103	103	102	101	101
	0.15	**218**	**222**	**234**	**163**	**156**	220	224	235	165	**156**	302	301	298	297	297	304	301	298	297	297
		121	119	120	109	108	122	119	120	112	111	168	162	162	162	162	167	162	161	162	162
	0.20	**154**	**138**	91.3	**64.9**	**41.4**	157	**138**	**89.8**	65.1	**41.4**	463	468	450	373	353	512	505	501	499	499
		77.6	80.7	67.4	57.1	47.9	72.8	79.9	67.9	57.0	47.9	190	190	177	175	171	218	214	211	212	212
	0.25	**5.34**	13.3	**4.97**	**2.20**	**0**	**5.34**	**13.1**	5.34	**2.20**	**0**	446	459	326	215	165	722	728	721	715	715
		15.8	24.5	15.1	9.49	0	15.8	24.8	15.8	9.49	0	135	137	137	102	91.2	234	237	233	236	236
0.20	0.05	**25.1**	**26.4**	**25.4**	**25.1**	**25.1**	26.2	27.5	26.4	26.2	26.2	**25.1**	**26.4**	**25.4**	**25.1**	**25.1**	26.2	27.5	26.4	26.2	26.2
		47.4	47.5	46.9	47.0	47.0	48.2	48.2	47.7	47.7	47.7	47.4	47.5	46.9	47.0	47.0	48.2	48.2	47.7	47.7	47.7
	0.10	**134**	**137**	**129**	**117**	**114**	135	139	130	118	115	148	151	149	147	147	149	151	150	148	147
		114	115	109	109	109	114	117	110	110	111	130	129	130	130	130	131	131	131	131	132
	0.15	245	254	267	**187**	**175**	**244**	**251**	**264**	**187**	**175**	339	335	337	331	331	338	335	336	331	330
		135	131	138	124	123	134	131	137	126	124	189	187	189	189	189	187	185	188	187	187
	0.20	**159**	**149**	**105**	**71.3**	**44.3**	162	154	107	71.6	**44.3**	488	504	483	401	378	531	538	531	526	526
		70.6	81.9	71.6	60.4	47.8	71.2	84.0	72.1	60.3	47.8	186	193	181	175	173	218	219	217	217	217
	0.25	**9.21**	**14.5**	**8.28**	**5.69**	**1.29**	**9.21**	15.3	9.02	5.88	**1.29**	467	477	337	222	170	738	740	742	732	732
		20.5	25.0	17.5	16.9	6.79	20.5	25.7	17.9	17.0	6.79	120	142	139	109	93.1	226	227	227	230	230

Figure 39: Maze, $\widehat{q}^{+}_{\mathrm{CBF}}$ at $\alpha_0 = 0.30$.

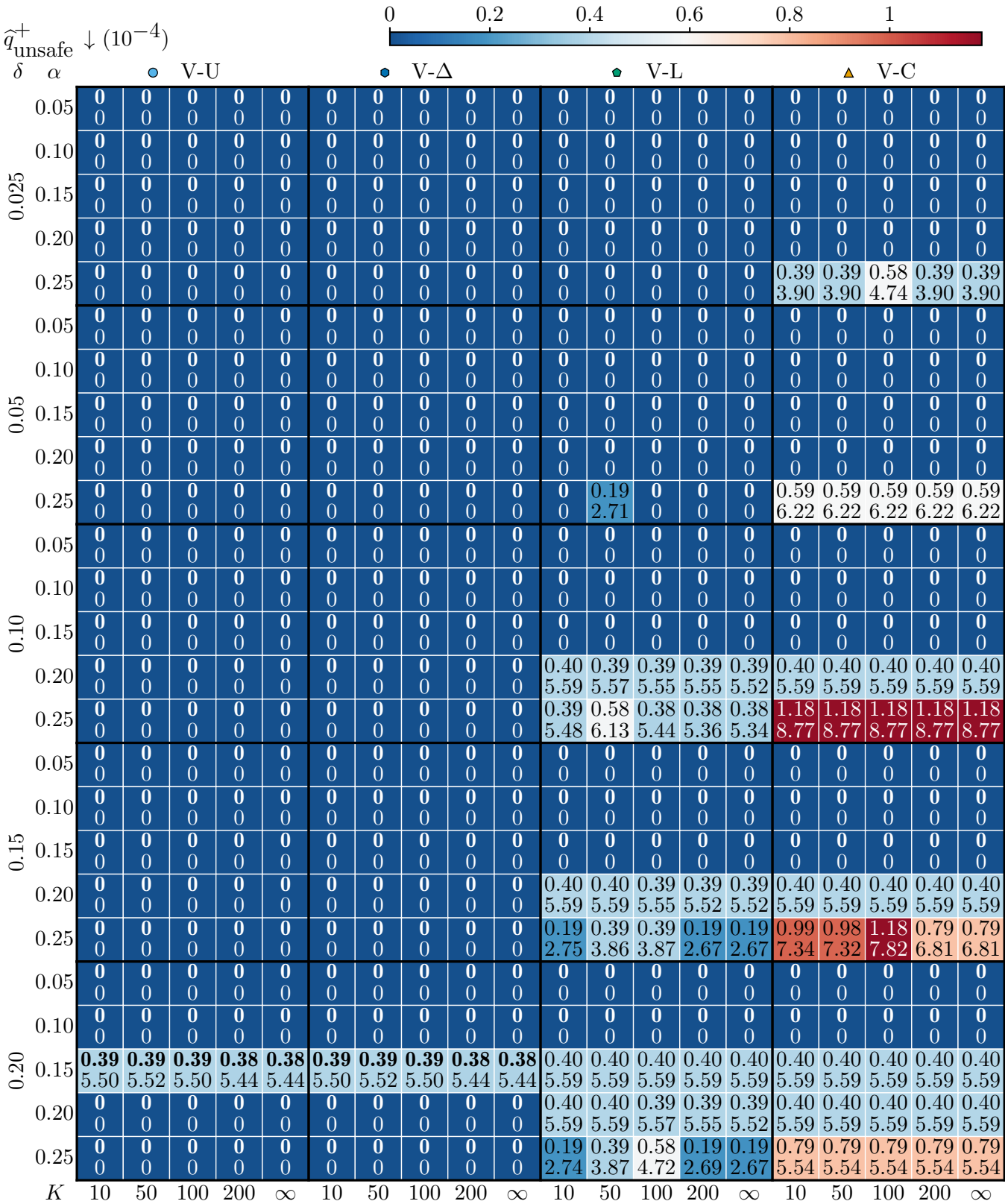


Figure 40: Maze, $\widehat{q}^{+}_{\text{unsafe}}$ at $\alpha_0 = 0.30$.

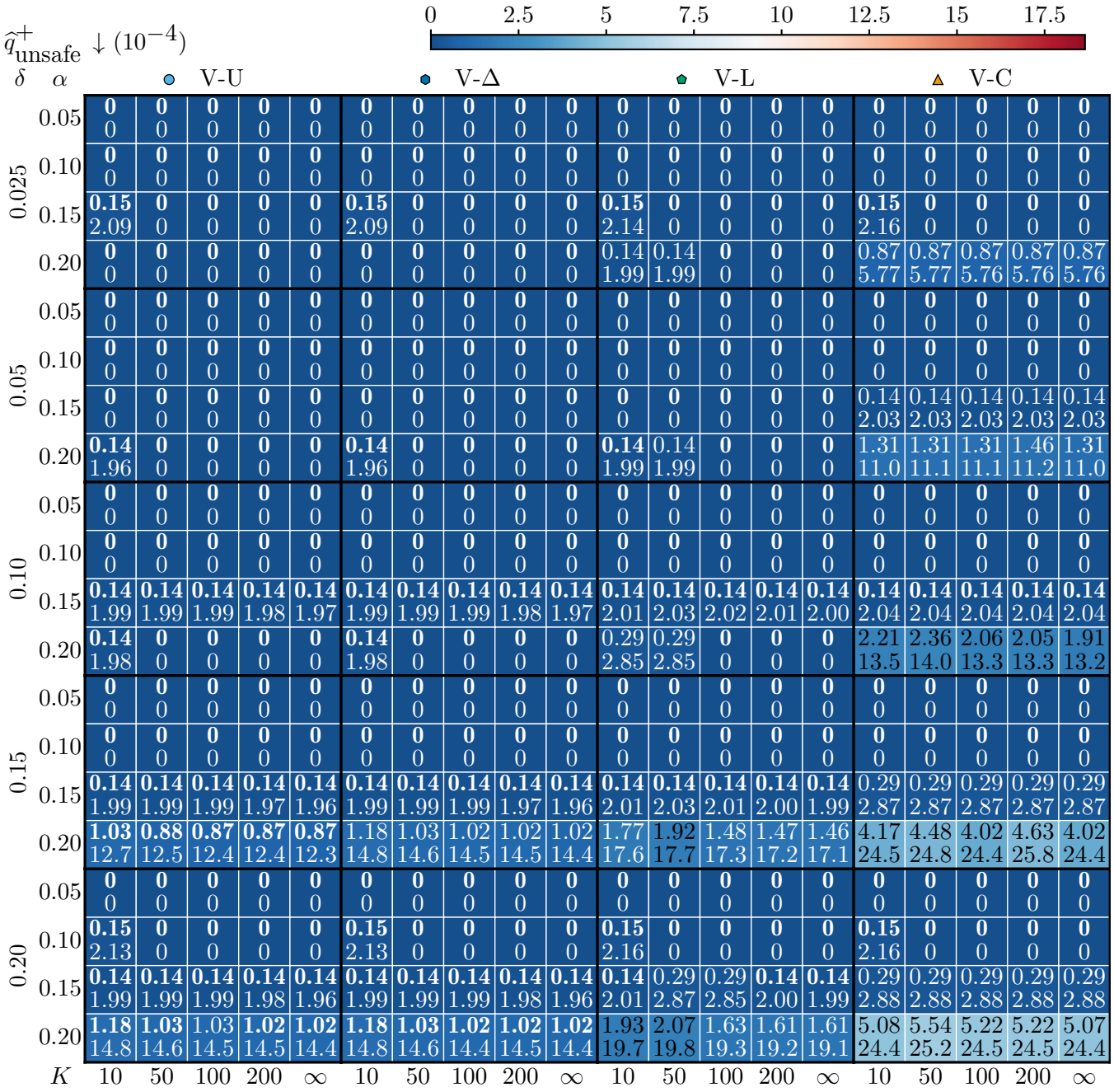


Figure 41: Corridor, $\widehat{q}^{+}_{\text{unsafe}}$ at $\alpha_0 = 0.25$.

Section 2.4 Error Confidence

Table 9: Corridor, error confidence δ, default setting and S1–S5.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C (α_0, α, K) = (0.15, 0.10, 50)					
0.025	0.84±0.05 0.17±0.01 0.93±0.09	1.88±1.07 2.99±1.44 0.00±0.00	76.5±6.9 7.52±2.58 5.3±2.0	21.3±7.1 2.2±0.7 8.66±2.12	0.03±0.01 0.16±0.01 35.21±0.91
0.05	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.10	0.81±0.06 0.17±0.01 0.92±0.09	2.37±1.24 3.82±1.69 0.00±0.00	82.0±5.8 5.38±2.25 6.2±2.3	15.3±6.3 2.7±0.7 10.18±2.40	0.03±0.01 0.16±0.01 35.06±0.91
0.15	0.79±0.06 0.17±0.01 0.92±0.09	2.62±1.39 4.15±1.86 0.00±0.00	83.3±5.5 4.87±2.14 6.6±2.4	13.9±5.9 2.8±0.7 10.64±2.41	0.03±0.01 0.16±0.01 35.01±0.91
0.20	0.78±0.06 0.17±0.01 0.91±0.09	2.86±1.52 4.56±2.07 0.00±0.00	84.7±5.2 4.39±2.02 7.1±2.6	12.5±5.6 2.9±0.7 11.04±2.39	0.03±0.01 0.16±0.01 34.96±0.92

Table 9 (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1 and **S2**: V-C $(\alpha_0, \alpha, K) = (0.30, 0.25, 50)$					
0.025	0.57±0.05 0.17±0.02 0.83±0.09	7.83±1.74 12.14±2.30 0.05±0.19	88.9±1.4 2.74±0.47 18.3±2.9	8.0±1.3 3.1±0.7 26.99±3.13	0.01±0.01 0.14±0.01 34.13±0.89
0.05	0.55±0.05 0.17±0.02 0.82±0.09	8.41±1.80 13.02±2.40 0.08±0.27	90.1±1.5 2.31±0.46 19.1±3.0	6.8±1.3 3.1±0.7 27.95±3.16	0.01±0.01 0.14±0.01 34.03±0.90
0.10	0.53±0.06 0.17±0.02 0.81±0.09	9.00±1.93 13.98±2.76 0.13±0.37	91.0±1.4 1.99±0.40 19.9±3.3	5.8±1.2 3.2±0.8 29.05±3.19	0.01±0.01 0.13±0.01 33.96±0.88
0.15	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
0.20	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
S3: V-L $(\alpha_0, \alpha, K) = (0.30, 0.20, 50)$					
0.025	0.68±0.05 0.17±0.02 0.87±0.09	4.95±1.33 7.74±1.76 0.01±0.04	87.1±1.9 3.39±0.70 12.9±2.9	9.8±2.0 3.1±0.7 21.51±2.91	0.02±0.01 0.15±0.01 34.53±0.87
0.05	0.67±0.05 0.17±0.02 0.86±0.09	5.31±1.34 8.29±1.74 0.01±0.11	88.3±1.9 2.95±0.67 13.4±3.1	8.5±1.9 3.2±0.7 22.33±3.14	0.02±0.01 0.15±0.01 34.47±0.88
0.10	0.64±0.05 0.17±0.02 0.85±0.09	5.71±1.45 9.01±1.95 0.01±0.12	89.7±1.8 2.48±0.60 14.3±3.2	7.2±1.7 3.2±0.7 23.36±3.13	0.02±0.01 0.14±0.01 34.38±0.89

Table 9 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L (α_0, α, K) = (0.30, 0.20, 50)					
0.15	0.63±0.06 0.17±0.02 0.85±0.09	6.06±1.56 9.54±2.09 0.03±0.21	90.5±1.8 2.19±0.61 14.9±3.4	6.3±1.7 3.2±0.8 23.89±3.08	0.01±0.01 0.14±0.01 34.33±0.91
0.20	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
S4: V-Δ (α_0, α, K) = (0.30, 0.25, 100)					
0.025	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
0.05	0.91±0.02 0.17±0.01 0.97±0.08	0.98±0.61 1.94±0.86 0.00±0.00	33.7±3.1 2.31±0.46 12.3±3.5	6.5±1.3 59.8±2.8 17.01±1.90	0.03±0.01 0.16±0.01 35.57±0.88
0.10	0.89±0.02 0.17±0.01 0.96±0.08	1.24±0.71 2.40±0.94 0.01±0.15	37.6±3.2 1.99±0.40 13.6±3.9	5.6±1.1 56.8±3.0 17.82±1.97	0.03±0.01 0.16±0.01 35.49±0.86
0.15	0.88±0.02 0.17±0.01 0.96±0.08	1.44±0.78 2.82±1.11 0.01±0.15	40.0±3.0 1.74±0.32 14.8±4.1	4.9±0.9 55.1±2.8 18.26±2.06	0.03±0.01 0.16±0.01 35.44±0.86
0.20	0.86±0.02 0.17±0.01 0.95±0.08	1.62±0.82 3.15±1.21 0.01±0.15	41.7±2.9 1.53±0.31 15.6±4.4	4.3±0.9 54.0±2.7 18.69±1.99	0.03±0.01 0.16±0.01 35.40±0.85
S5: V-C (α_0, α, K) = (0.30, 0.25, 10)					
0.025	0.57±0.05 0.17±0.02 0.83±0.09	7.83±1.77 12.15±2.36 0.05±0.19	88.9±1.4 2.74±0.47 18.2±2.9	8.0±1.3 3.1±0.7 112.51±13.86	0.01±0.01 0.14±0.01 34.12±0.89
0.05	0.55±0.06 0.17±0.02 0.82±0.09	8.33±1.84 12.89±2.48 0.08±0.26	90.1±1.5 2.31±0.46 18.8±3.0	6.8±1.3 3.1±0.7 115.70±14.07	0.01±0.01 0.14±0.01 34.04±0.90
0.10	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
0.15	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.97 14.40±2.71 0.14±0.36	91.7±1.2 1.74±0.32 20.1±3.2	5.1±0.9 3.2±0.8 122.03±13.45	0.01±0.01 0.13±0.01 33.93±0.88
0.20	0.50±0.06 0.17±0.01 0.80±0.09	9.67±2.09 14.97±2.94 0.18±0.41	92.3±1.1 1.53±0.31 20.8±3.2	4.5±0.9 3.2±0.8 123.83±13.91	0.01±0.01 0.13±0.01 33.88±0.89

Table 10: Maze, as in Table 9.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C (α_0, α, K) = (0.15, 0.10, 50)					
0.025	0.88±0.04 0.17±0.02 0.71±0.04	0.65±0.63 2.05±1.19 0.00±0.00	68.3±10.3 15.71±5.77 4.4±1.7	29.3±10.6 2.4±0.8 5.93±1.92	0.02±0.00 0.21±0.01 53.52±0.47
0.05	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.10	0.84±0.05 0.17±0.02 0.69±0.04	1.07±0.92 2.92±1.63 0.00±0.00	76.7±7.8 11.01±4.29 5.3±2.1	20.6±7.9 2.7±0.8 7.03±1.88	0.02±0.00 0.21±0.01 53.26±0.50
0.15	0.82±0.06 0.17±0.02 0.68±0.04	1.21±1.03 3.30±1.77 0.00±0.00	79.0±7.2 9.78±3.89 5.7±2.3	18.4±7.2 2.6±0.8 7.70±1.95	0.02±0.00 0.21±0.01 53.13±0.53
0.20	0.81±0.07 0.17±0.02 0.66±0.05	1.51±1.31 3.80±2.20 0.00±0.00	81.1±6.6 8.63±3.59 6.2±2.5	16.3±6.7 2.7±0.8 8.09±2.12	0.02±0.00 0.21±0.01 53.03±0.57

Table 10 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1 and **S2**: V-C (α_0, α, K) = (0.30, 0.25, 50)					
0.025	0.60±0.06 0.17±0.02 0.54±0.04	5.77±1.86 11.20±2.48 0.00±0.04	86.6±1.8 5.45±0.85 17.2±3.2	10.5±1.6 2.9±0.8 20.19±2.36	0.01±0.00 0.21±0.01 51.67±0.48
0.05	0.58±0.06 0.17±0.02 0.53±0.04	6.26±1.92 12.11±2.68 0.01±0.06	88.2±1.7 4.56±0.79 17.8±3.3	8.8±1.5 3.0±0.7 20.93±2.32	0.01±0.00 0.21±0.01 51.57±0.49
0.10	0.55±0.07 0.17±0.02 0.52±0.04	6.78±2.18 13.05±3.18 0.01±0.09	89.3±1.6 3.93±0.77 18.7±3.5	7.6±1.5 3.1±0.8 21.78±2.47	0.01±0.00 0.21±0.01 51.44±0.54
0.15	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.20	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52

Table 10 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L (α_0, α, K) = (0.30, 0.20, 50)					
0.025	0.70±0.05 0.17±0.02 0.59±0.04	3.32±1.33 7.18±2.03 0.00±0.00	84.1±2.6 6.75±1.36 11.8±2.8	12.9±2.6 3.0±0.8 15.76±2.14	0.02±0.00 0.21±0.01 52.28±0.46
0.05	0.68±0.06 0.17±0.02 0.58±0.04	3.75±1.51 7.90±2.23 0.00±0.00	85.7±2.4 5.87±1.27 12.6±3.0	11.3±2.4 3.0±0.8 16.57±2.27	0.01±0.00 0.21±0.01 52.16±0.48
0.10	0.65±0.06 0.17±0.02 0.57±0.04	4.26±1.75 8.91±2.56 0.00±0.06	87.6±2.1 4.93±1.09 13.4±3.4	9.4±2.0 3.0±0.8 17.47±2.15	0.01±0.00 0.21±0.01 52.01±0.49
0.15	0.63±0.07 0.17±0.02 0.56±0.04	4.68±1.90 9.54±2.83 0.00±0.06	88.6±2.2 4.31±1.08 14.2±3.7	8.3±2.1 3.1±0.8 17.84±2.27	0.01±0.00 0.21±0.01 51.91±0.53
0.20	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
S4: V-Δ (α_0, α, K) = (0.30, 0.25, 100)					
0.025	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
0.05	0.99±0.00 0.17±0.02 0.81±0.02	0.01±0.07 0.16±0.25 0.00±0.00	3.7±1.3 5.29±4.69 6.4±4.1	8.4±1.5 87.9±0.9 13.23±1.61	0.03±0.00 0.22±0.01 54.42±0.33
0.10	0.98±0.01 0.17±0.02 0.80±0.02	0.03±0.10 0.30±0.34 0.00±0.00	6.5±1.7 4.65±4.66 7.7±4.5	7.2±1.4 86.2±1.4 13.84±1.77	0.03±0.00 0.22±0.01 54.41±0.34
0.15	0.98±0.01 0.17±0.02 0.81±0.02	0.05±0.16 0.44±0.44 0.00±0.00	8.6±1.8 3.55±1.60 9.4±4.9	6.4±1.1 85.1±1.6 14.31±1.72	0.03±0.00 0.22±0.01 54.38±0.35
0.20	0.97±0.01 0.17±0.02 0.80±0.02	0.09±0.18 0.59±0.53 0.00±0.00	10.0±1.9 3.04±0.63 11.0±5.1	5.6±1.2 84.4±1.5 14.57±1.69	0.03±0.00 0.22±0.01 54.37±0.34

Table 10 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
δ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C (α_0, α, K) = (0.30, 0.25, 10)					
0.025	0.60±0.06 0.17±0.02 0.54±0.04	5.73±1.88 11.21±2.53 0.00±0.04	86.6±1.8 5.45±0.85 17.1±3.2	10.5±1.6 2.9±0.8 77.50±9.16	0.01±0.00 0.21±0.01 51.67±0.47
0.05	0.58±0.06 0.17±0.02 0.53±0.04	6.27±1.93 12.09±2.71 0.01±0.06	88.2±1.6 4.56±0.79 17.7±3.2	8.8±1.5 3.0±0.7 80.70±9.27	0.01±0.00 0.21±0.01 51.58±0.50
0.10	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.15	0.53±0.07 0.17±0.02 0.51±0.04	7.22±2.34 13.83±3.42 0.01±0.07	90.3±1.4 3.45±0.62 19.1±3.6	6.7±1.2 2.9±0.8 86.17±10.05	0.01±0.00 0.21±0.01 51.34±0.54
0.20	0.52±0.07 0.17±0.02 0.51±0.04	7.38±2.26 14.39±3.53 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 3.0±0.8 87.29±10.22	0.01±0.00 0.21±0.01 51.26±0.52

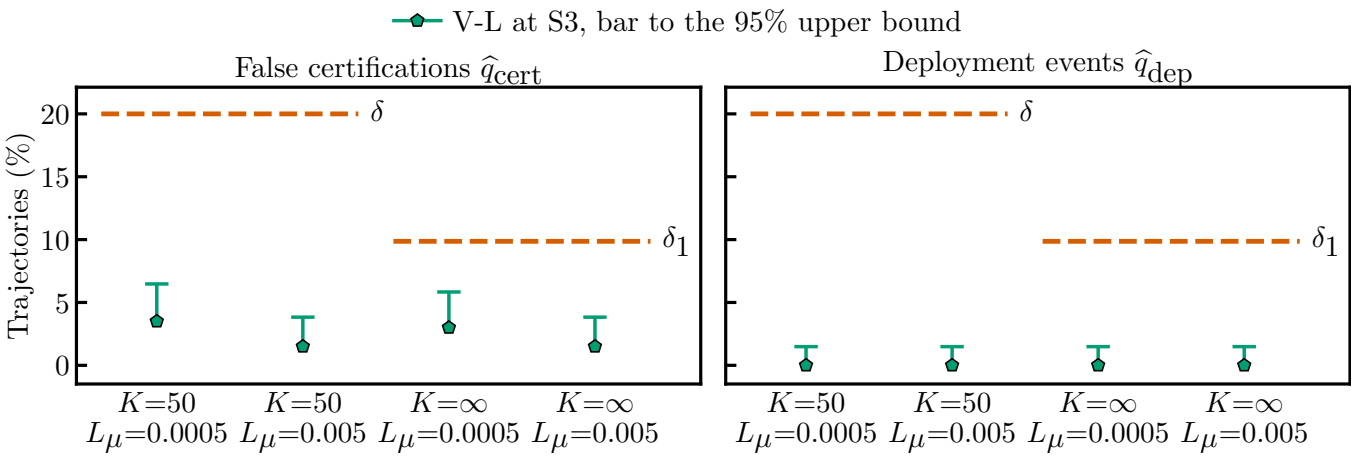


Figure 42: Corridor, S3 with V-L: $\widehat{q}_{\text{cert}}$ and $\widehat{q}_{\text{dep}}$ (28) with upper confidence bounds, with the bounds δ and δ_1 of Theorems 1–3.

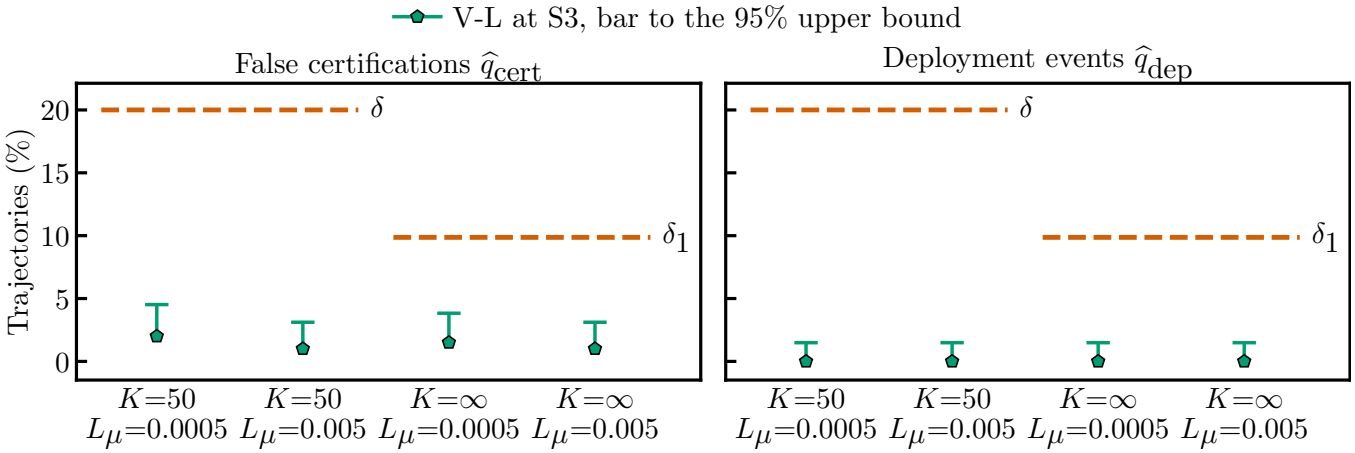


Figure 43: Maze, as in Fig. 42.

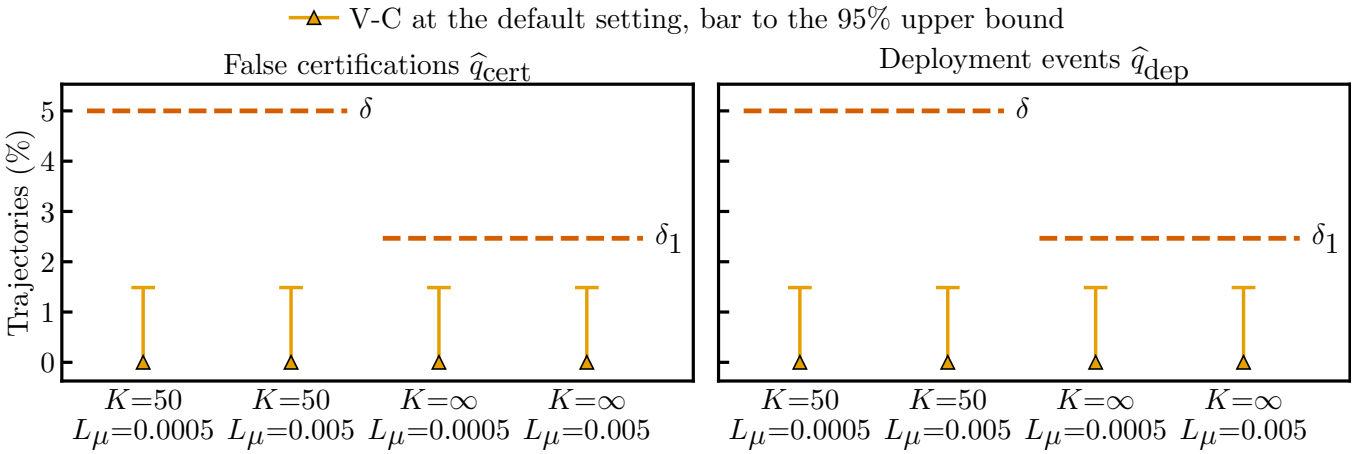


Figure 44: Default setting with V-C, identical in the Corridor and the Maze (no event in either scenario), as in Fig. 42.

Table 11: Corridor, test period K, default setting and S1–S5.

Corridor Baselines $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89

Table 11 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)					
10	0.83±0.05 0.17±0.01 0.93±0.09	2.09±1.14 3.33±1.59 0.00±0.00	79.2±6.2 6.45±2.36 5.6±2.1	18.3±6.6 2.5±0.7 31.73±8.83	0.03±0.01 0.16±0.01 35.14±0.91
50	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
100	0.83±0.05 0.17±0.01 0.93±0.09	2.10±1.17 3.36±1.64 0.00±0.00	79.2±6.2 6.46±2.38 5.7±2.1	18.3±6.6 2.5±0.7 5.73±1.36	0.03±0.01 0.16±0.01 35.14±0.91
200	0.83±0.05 0.17±0.01 0.93±0.09	2.09±1.14 3.33±1.58 0.00±0.00	79.2±6.2 6.46±2.38 5.6±2.1	18.3±6.6 2.5±0.7 3.95±0.97	0.03±0.01 0.16±0.01 35.14±0.91
∞	0.83±0.05 0.17±0.01 0.93±0.09	2.08±1.14 3.32±1.59 0.00±0.00	79.2±6.2 6.46±2.38 5.6±2.1	18.3±6.6 2.5±0.7 2.54±0.59	0.03±0.01 0.16±0.01 35.14±0.91

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)					
10	0.50±0.06 0.17±0.01 0.80±0.09	9.67±2.09 14.97±2.94 0.18±0.41	92.3±1.1 1.53±0.31 20.8±3.2	4.5±0.9 3.2±0.8 123.83±13.91	0.01±0.01 0.13±0.01 33.88±0.89

Table 11 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)					
50	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
100	0.50±0.06 0.17±0.01 0.80±0.09	9.61±2.08 14.92±2.83 0.19±0.41	92.3±1.1 1.53±0.31 21.0±3.3	4.5±0.9 3.2±0.8 17.72±1.96	0.00±0.01 0.13±0.01 33.88±0.87
200	0.52±0.06 0.17±0.01 0.81±0.09	9.32±2.05 14.39±2.80 0.17±0.40	92.3±1.1 1.53±0.31 20.6±3.3	4.5±0.9 3.2±0.8 10.63±1.06	0.01±0.01 0.13±0.01 33.91±0.89
∞	0.52±0.06 0.17±0.01 0.81±0.09	9.15±2.05 13.94±2.92 0.15±0.39	92.3±1.1 1.53±0.31 20.3±3.3	4.5±0.9 3.2±0.8 5.98±0.66	0.01±0.01 0.13±0.01 33.93±0.90
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)					
10	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.97 14.40±2.71 0.14±0.36	91.7±1.2 1.74±0.32 20.1±3.2	5.1±0.9 3.2±0.8 122.03±13.45	0.01±0.01 0.13±0.01 33.93±0.88
50	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
100	0.52±0.06 0.17±0.02 0.81±0.09	9.23±1.94 14.33±2.65 0.14±0.37	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 17.24±1.96	0.01±0.01 0.13±0.01 33.94±0.87
200	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.93 13.83±2.57 0.12±0.35	91.7±1.2 1.74±0.32 20.0±3.2	5.1±0.9 3.2±0.8 10.41±1.09	0.01±0.01 0.13±0.01 33.96±0.88
∞	0.54±0.06 0.17±0.02 0.81±0.09	8.80±1.93 13.45±2.67 0.11±0.33	91.7±1.2 1.74±0.32 19.6±3.2	5.1±0.9 3.2±0.8 5.86±0.65	0.01±0.01 0.13±0.01 33.98±0.89
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)					
10	0.62±0.06 0.17±0.02 0.85±0.09	6.14±1.63 9.66±2.22 0.03±0.21	90.9±1.7 2.04±0.54 15.0±3.7	5.9±1.6 3.2±0.8 97.10±13.54	0.01±0.01 0.14±0.01 34.32±0.91
50	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
100	0.63±0.06 0.17±0.02 0.85±0.09	6.18±1.60 9.72±2.17 0.03±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 14.21±1.91	0.01±0.01 0.14±0.01 34.33±0.91

Table 11 (continued).

K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Corridor $\phi=1,\ L_\mu=0.0005,\ c_{\max}=0.02\,\mathrm{m},\ T_s=0.1\,\mathrm{s},\ N_{\mathrm{MC}}=200$					
S3: V-L (α_0,α,δ) = (0.30, 0.20, 0.20)					
200	0.68±0.05 0.17±0.02 0.87±0.09	4.94±1.58 7.95±2.15 0.03±0.21	82.8±2.5 2.04±0.54 15.1±3.7	5.9±1.5 11.3±2.1 8.83±1.14	0.02±0.01 0.15±0.01 34.56±0.95
∞	0.79±0.05 0.17±0.02 0.91±0.09	3.40±1.39 5.29±1.96 0.03±0.20	49.6±2.3 2.04±0.54 14.2±4.0	5.8±1.5 44.6±1.7 5.02±0.73	0.02±0.01 0.15±0.01 34.92±0.98
S4: V-Δ (α_0,α,δ) = (0.30, 0.25, 0.025)					
10	0.83±0.03 0.17±0.01 0.93±0.08	1.90±0.67 3.23±0.74 0.00±0.02	81.2±7.1 2.74±0.47 11.1±3.4	7.8±1.3 11.0±7.0 115.62±12.63	0.03±0.01 0.16±0.01 35.15±0.90
50	0.89±0.02 0.17±0.01 0.96±0.08	1.18±0.59 2.27±0.77 0.00±0.00	47.8±4.0 2.74±0.47 11.7±3.3	7.7±1.3 44.5±3.9 28.37±2.96	0.03±0.01 0.16±0.01 35.43±0.85
100	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
200	0.95±0.01 0.17±0.01 0.98±0.08	0.43±0.40 0.93±0.62 0.00±0.00	17.7±2.1 2.74±0.47 10.4±3.5	7.6±1.3 74.6±1.8 9.75±0.93	0.04±0.01 0.17±0.01 35.72±0.88
∞	0.97±0.01 0.17±0.01 1.00±0.08	0.13±0.20 0.47±0.42 0.00±0.00	8.5±1.4 2.74±0.47 8.8±4.0	7.6±1.3 83.9±0.8 5.48±0.61	0.05±0.01 0.17±0.01 35.85±0.88
S5: V-C (α_0,α,δ) = (0.30, 0.25, 0.10)					
10	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
50	0.53±0.06 0.17±0.02 0.81±0.09	9.00±1.93 13.98±2.76 0.13±0.37	91.0±1.4 1.99±0.40 19.9±3.3	5.8±1.2 3.2±0.8 29.05±3.19	0.01±0.01 0.13±0.01 33.96±0.88
100	0.53±0.06 0.17±0.02 0.81±0.09	8.90±1.88 13.83±2.58 0.12±0.36	91.0±1.4 1.99±0.40 19.8±3.3	5.8±1.2 3.2±0.8 16.77±1.98	0.01±0.01 0.13±0.01 33.98±0.87
200	0.54±0.06 0.17±0.02 0.81±0.09	8.60±1.87 13.31±2.59 0.11±0.35	91.0±1.4 1.99±0.40 19.5±3.2	5.8±1.2 3.2±0.8 10.25±1.01	0.01±0.01 0.13±0.01 34.00±0.88
∞	0.55±0.06 0.17±0.02 0.82±0.09	8.49±1.87 12.98±2.65 0.10±0.34	91.0±1.4 1.99±0.40 19.2±3.2	5.8±1.2 3.2±0.8 5.78±0.65	0.01±0.01 0.14±0.01 34.01±0.89

Table 12: Maze, as in Table 11.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)					
10	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.73 2.34±1.24 0.00±0.00	72.6±8.9 13.27±4.88 4.7±1.9	24.8±9.0 2.6±0.8 20.37±7.43	0.02±0.00 0.21±0.01 53.41±0.45
50	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
100	0.86±0.05 0.17±0.02 0.70±0.04	0.76±0.71 2.35±1.22 0.00±0.00	72.5±9.0 13.33±4.97 4.7±1.9	24.9±9.2 2.6±0.8 3.95±1.02	0.02±0.00 0.21±0.01 53.41±0.44
200	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.72 2.33±1.24 0.00±0.00	72.5±9.0 13.33±4.97 4.6±1.9	24.9±9.2 2.6±0.7 2.90±0.98	0.02±0.00 0.21±0.01 53.41±0.44
∞	0.86±0.05 0.17±0.02 0.70±0.04	0.75±0.72 2.33±1.24 0.00±0.00	72.5±9.0 13.33±4.97 4.6±1.8	24.9±9.2 2.6±0.7 2.32±0.59	0.02±0.00 0.21±0.01 53.41±0.44

Table 12 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)					
10	0.52±0.07 0.17±0.02 0.51±0.04	7.38±2.26 14.39±3.53 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 3.0±0.8 87.29±10.22	0.01±0.00 0.21±0.01 51.26±0.52
50	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
100	0.52±0.07 0.17±0.02 0.51±0.04	7.42±2.27 14.45±3.56 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 3.0±0.8 14.03±1.74	0.01±0.00 0.21±0.01 51.25±0.52
200	0.52±0.08 0.17±0.02 0.51±0.04	7.32±2.30 14.31±3.61 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 2.9±0.8 9.35±1.29	0.01±0.00 0.21±0.01 51.27±0.52
∞	0.52±0.08 0.17±0.02 0.51±0.04	7.32±2.30 14.31±3.62 0.01±0.06	91.1±1.4 3.04±0.63 19.4±3.5	5.9±1.2 2.9±0.8 5.77±0.69	0.01±0.00 0.21±0.01 51.27±0.52

Table 12 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)					
10	0.53±0.07 0.17±0.02 0.51±0.04	7.22±2.34 13.83±3.42 0.01±0.07	90.3±1.4 3.45±0.62 19.1±3.6	6.7±1.2 2.9±0.8 86.17±10.05	0.01±0.00 0.21±0.01 51.34±0.54
50	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
100	0.53±0.08 0.17±0.02 0.51±0.04	7.21±2.33 13.77±3.45 0.01±0.08	90.3±1.5 3.45±0.62 19.3±3.7	6.7±1.2 2.9±0.8 13.73±1.73	0.01±0.00 0.21±0.01 51.33±0.54
200	0.53±0.08 0.17±0.02 0.51±0.04	7.15±2.36 13.69±3.46 0.01±0.07	90.4±1.5 3.45±0.62 19.0±3.6	6.7±1.2 2.9±0.8 9.22±1.25	0.01±0.00 0.21±0.01 51.35±0.53
∞	0.53±0.08 0.17±0.02 0.51±0.04	7.15±2.36 13.68±3.47 0.01±0.07	90.4±1.5 3.45±0.62 18.9±3.6	6.7±1.2 2.9±0.8 5.68±0.72	0.01±0.00 0.21±0.01 51.35±0.53
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)					
10	0.62±0.07 0.17±0.02 0.56±0.04	4.88±1.86 9.85±2.79 0.00±0.06	89.2±2.0 4.01±0.97 14.2±3.6	7.7±1.9 3.1±0.9 67.48±8.99	0.01±0.00 0.21±0.01 51.87±0.54
50	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
100	0.63±0.07 0.17±0.02 0.56±0.04	4.83±1.81 9.71±2.76 0.00±0.06	89.1±2.0 4.01±0.97 14.4±3.8	7.7±1.8 3.2±0.9 11.26±1.62	0.01±0.00 0.21±0.01 51.88±0.53
200	0.69±0.07 0.17±0.02 0.58±0.04	4.01±1.75 7.98±2.75 0.00±0.06	78.8±2.6 4.01±0.97 14.0±3.7	7.7±1.8 13.5±2.1 7.68±1.19	0.01±0.00 0.21±0.01 52.20±0.52
∞	0.72±0.07 0.17±0.02 0.60±0.04	3.78±1.73 7.36±2.65 0.00±0.06	67.1±2.1 4.01±0.97 13.8±3.7	7.7±1.8 25.2±1.0 4.86±0.77	0.01±0.00 0.21±0.01 52.35±0.53

Table 12 (continued).

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)					
10	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.0±1.6 88.7±1.0 83.68±11.09	0.03±0.00 0.22±0.01 54.44±0.32
50	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.04 0.05±0.15 0.00±0.00	1.7±0.9 7.09±4.38 3.9±3.1	10.0±1.6 88.3±1.0 21.00±2.61	0.03±0.00 0.21±0.01 54.46±0.31
100	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
200	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.03 0.04±0.13 0.00±0.00	1.5±0.8 10.33±13.87 3.9±3.3	10.0±1.6 88.5±1.0 8.55±1.22	0.03±0.00 0.22±0.01 54.44±0.32
∞	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.0±1.6 88.7±1.0 5.29±0.59	0.03±0.00 0.22±0.01 54.44±0.32
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)					
10	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
50	0.55±0.07 0.17±0.02 0.52±0.04	6.78±2.18 13.05±3.18 0.01±0.09	89.3±1.6 3.93±0.77 18.7±3.5	7.6±1.5 3.1±0.8 21.78±2.47	0.01±0.00 0.21±0.01 51.44±0.54
100	0.55±0.07 0.17±0.02 0.52±0.04	6.74±2.17 12.94±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.6±3.4	7.6±1.5 3.0±0.8 13.27±1.57	0.01±0.00 0.21±0.01 51.46±0.54
200	0.56±0.07 0.17±0.02 0.52±0.04	6.71±2.18 12.88±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.4±3.4	7.6±1.5 3.0±0.8 8.82±1.15	0.01±0.00 0.21±0.01 51.47±0.54
∞	0.56±0.07 0.17±0.02 0.52±0.04	6.71±2.18 12.88±3.20 0.01±0.09	89.3±1.6 3.93±0.77 18.4±3.4	7.6±1.5 3.0±0.8 5.58±0.67	0.01±0.00 0.21±0.01 51.47±0.54

Section 2.6 Budget Normalizer

Table 13: Corridor, budget normalizer b, default setting and S1–S5.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
κ	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
2.5	0.83±0.05 0.17±0.01 0.93±0.09	2.08±1.16 3.28±1.58 0.00±0.00	78.6±6.3 6.68±2.38 5.6±2.2	18.9±6.6 2.4±0.7 9.17±2.27	0.03±0.01 0.16±0.01 35.16±0.92
3	0.83±0.05 0.17±0.01 0.93±0.09	2.00±1.11 3.16±1.52 0.00±0.00	77.8±6.5 6.99±2.48 5.5±2.1	19.8±6.9 2.4±0.7 9.03±2.20	0.03±0.01 0.16±0.01 35.17±0.91
4	0.84±0.05 0.17±0.01 0.93±0.08	1.88±1.07 2.98±1.42 0.00±0.00	76.7±6.7 7.44±2.50 5.3±2.0	21.0±6.9 2.3±0.7 8.67±2.13	0.03±0.01 0.16±0.01 35.20±0.91

Table 13 (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
κ	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
2.5	0.51±0.06 0.17±0.02 0.80±0.09	9.54±2.07 14.79±2.89 0.14±0.35	92.0±1.2 1.64±0.33 20.7±3.3	4.8±0.9 3.2±0.8 30.18±3.34	0.01±0.01 0.13±0.01 33.89±0.88
3	0.52±0.06 0.17±0.02 0.81±0.09	9.31±1.93 14.46±2.73 0.14±0.36	91.7±1.2 1.75±0.35 20.3±3.2	5.1±1.0 3.2±0.8 29.67±3.21	0.01±0.01 0.13±0.01 33.93±0.89
4	0.53±0.06 0.17±0.02 0.81±0.09	9.02±1.91 14.01±2.73 0.13±0.37	91.0±1.3 1.98±0.40 19.9±3.2	5.8±1.1 3.2±0.8 29.12±3.23	0.01±0.01 0.13±0.01 33.95±0.88
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
κ	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88
2.5	0.52±0.06 0.17±0.02 0.81±0.09	9.22±1.92 14.27±2.72 0.13±0.35	91.4±1.3 1.85±0.37 20.2±3.2	5.4±1.1 3.1±0.8 29.44±3.17	0.01±0.01 0.13±0.01 33.94±0.88
3	0.53±0.06 0.17±0.02 0.81±0.09	9.02±1.91 14.01±2.73 0.13±0.37	91.0±1.3 1.98±0.40 19.9±3.2	5.8±1.1 3.2±0.8 29.12±3.23	0.01±0.01 0.13±0.01 33.95±0.88
4	0.54±0.06 0.17±0.02 0.81±0.09	8.72±1.83 13.47±2.63 0.11±0.35	90.7±1.5 2.11±0.45 19.6±3.1	6.2±1.3 3.1±0.7 28.67±3.28	0.01±0.01 0.13±0.01 34.00±0.88

Table 13 (continued).

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
κ	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
2.5	0.62±0.06 0.17±0.02 0.85±0.09	6.15±1.58 9.70±2.09 0.03±0.21	90.6±1.7 2.16±0.56 15.1±3.5	6.3±1.6 3.2±0.8 24.02±3.18	0.01±0.01 0.14±0.01 34.32±0.91
3	0.63±0.06 0.17±0.02 0.85±0.09	5.98±1.57 9.42±2.06 0.03±0.19	90.2±1.7 2.27±0.58 14.8±3.5	6.6±1.7 3.2±0.8 23.74±3.07	0.01±0.01 0.14±0.01 34.34±0.91
4	0.64±0.05 0.17±0.02 0.85±0.09	5.76±1.47 9.06±1.94 0.02±0.12	89.7±1.8 2.48±0.60 14.4±3.2	7.2±1.7 3.2±0.7 23.44±3.13	0.02±0.01 0.14±0.01 34.37±0.90
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
κ	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
2.5	0.93±0.02 0.17±0.01 0.98±0.08	0.69±0.49 1.35±0.69 0.00±0.00	29.2±3.3 2.77±0.51 10.7±3.3	7.7±1.4 63.1±3.0 16.26±1.74	0.04±0.01 0.17±0.01 35.63±0.89
3	0.93±0.01 0.17±0.01 0.98±0.08	0.62±0.47 1.23±0.65 0.00±0.00	28.3±3.3 2.88±0.52 10.3±3.2	8.0±1.4 63.7±3.0 16.08±1.81	0.04±0.01 0.17±0.01 35.64±0.89
4	0.94±0.01 0.17±0.01 0.98±0.08	0.56±0.42 1.10±0.57 0.00±0.00	26.7±3.3 3.06±0.49 9.9±3.1	8.5±1.3 64.7±3.0 15.82±1.72	0.04±0.01 0.17±0.01 35.67±0.89
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
κ	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
2.5	0.54±0.06 0.17±0.02 0.81±0.09	8.78±1.88 13.59±2.63 0.11±0.34	90.7±1.4 2.11±0.43 19.5±3.0	6.1±1.2 3.1±0.8 118.62±13.67	0.01±0.01 0.13±0.01 33.99±0.88
3	0.54±0.06 0.17±0.02 0.82±0.09	8.61±1.83 13.33±2.53 0.10±0.32	90.4±1.5 2.22±0.45 19.3±3.0	6.5±1.3 3.1±0.7 117.37±13.53	0.01±0.01 0.13±0.01 34.01±0.89
4	0.55±0.05 0.17±0.02 0.82±0.09	8.35±1.83 12.92±2.47 0.08±0.26	90.1±1.5 2.31±0.46 18.9±3.0	6.8±1.3 3.1±0.7 115.90±14.00	0.01±0.01 0.14±0.01 34.04±0.91

Table 14: Maze, as in Table 13.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
κ	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
2.5	0.86±0.05 0.17±0.02 0.70±0.04	0.72±0.65 2.29±1.26 0.00±0.00	71.6±9.3 13.90±5.13 4.5±1.8	26.0±9.5 2.5±0.7 6.36±1.99	0.02±0.00 0.21±0.01 53.44±0.47
3	0.87±0.05 0.17±0.02 0.71±0.04	0.72±0.67 2.20±1.27 0.00±0.00	70.3±9.8 14.57±5.42 4.5±1.8	27.2±10.0 2.4±0.7 6.18±1.93	0.02±0.00 0.21±0.01 53.45±0.45
4	0.87±0.04 0.17±0.02 0.71±0.04	0.70±0.64 2.09±1.17 0.00±0.00	68.4±10.1 15.60±5.68 4.3±1.8	29.1±10.5 2.5±0.8 5.95±1.93	0.02±0.00 0.21±0.01 53.53±0.47

Table 14 (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
κ	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
2.5	0.53±0.07 0.17±0.02 0.51±0.04	7.32±2.32 14.12±3.48 0.01±0.08	90.6±1.4 3.25±0.64 19.3±3.6	6.3±1.2 3.0±0.8 22.49±2.47	0.01±0.00 0.21±0.01 51.31±0.50
3	0.53±0.08 0.17±0.02 0.51±0.04	7.18±2.36 13.81±3.46 0.01±0.08	90.3±1.5 3.46±0.64 19.1±3.7	6.7±1.2 3.0±0.8 22.28±2.52	0.01±0.00 0.21±0.01 51.34±0.53
4	0.55±0.07 0.17±0.02 0.52±0.04	6.75±2.20 13.07±3.19 0.01±0.09	89.3±1.6 3.92±0.75 18.5±3.5	7.6±1.4 3.1±0.9 21.86±2.45	0.01±0.00 0.21±0.01 51.44±0.51
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
κ	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
2.5	0.54±0.07 0.17±0.02 0.51±0.04	7.09±2.34 13.57±3.52 0.01±0.08	89.8±1.5 3.67±0.65 18.9±3.6	7.1±1.3 3.0±0.8 22.12±2.39	0.01±0.00 0.21±0.01 51.37±0.51
3	0.55±0.07 0.17±0.02 0.52±0.04	6.75±2.20 13.07±3.19 0.01±0.09	89.3±1.6 3.92±0.75 18.5±3.5	7.6±1.4 3.1±0.9 21.86±2.45	0.01±0.00 0.21±0.01 51.44±0.51
4	0.56±0.07 0.17±0.02 0.52±0.04	6.70±2.05 12.83±2.95 0.01±0.07	89.0±1.7 4.16±0.77 18.3±3.4	8.1±1.5 3.0±0.8 21.39±2.45	0.01±0.00 0.21±0.01 51.44±0.52

Table 14 (continued).

b	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Maze $\phi=1,\ L_\mu=0.0005,\ c_{\max}=0.01\,\mathrm{m},\ T_s=0.2\,\mathrm{s},\ N_{\mathrm{MC}}=200$					
S3: V-L $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.20, 0.20, 50)					
κ	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
2.5	0.63±0.07 0.17±0.02 0.56±0.04	4.85±2.01 9.78±2.80 0.00±0.06	88.7±2.0 4.26±0.99 14.3±3.8	8.2±1.9 3.2±0.8 18.08±2.32	0.01±0.00 0.21±0.01 51.88±0.53
3	0.64±0.07 0.17±0.02 0.56±0.04	4.63±1.89 9.39±2.69 0.01±0.06	88.3±2.1 4.50±1.01 14.1±3.6	8.6±2.0 3.1±0.8 17.83±2.29	0.01±0.00 0.21±0.01 51.91±0.53
4	0.65±0.06 0.17±0.02 0.57±0.04	4.31±1.74 9.00±2.54 0.00±0.06	87.5±2.2 4.93±1.09 13.5±3.5	9.4±2.0 3.0±0.8 17.46±2.15	0.01±0.00 0.21±0.01 52.01±0.50
S4: V-Δ $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.025, 100)					
κ	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
2.5	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	1.3±0.8 5.00±0.00 3.6±3.2	10.1±1.7 88.6±1.1 12.75±1.59	0.03±0.00 0.22±0.01 54.43±0.32
3	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.03±0.11 0.00±0.00	1.0±0.6 5.20±0.00 3.4±3.0	10.5±1.7 88.5±1.2 12.61±1.61	0.03±0.00 0.22±0.01 54.43±0.32
4	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.01±0.07 0.00±0.00	0.5±0.3 5.60±0.00 3.1±2.7	11.2±1.7 88.4±1.5 12.38±1.63	0.03±0.00 0.22±0.01 54.44±0.32
S5: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.10, 10)					
κ	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
2.5	0.56±0.07 0.17±0.02 0.52±0.04	6.66±2.09 12.78±2.95 0.01±0.07	88.9±1.7 4.16±0.77 18.3±3.4	8.0±1.5 3.0±0.8 82.98±9.69	0.01±0.00 0.21±0.01 51.47±0.51
3	0.57±0.06 0.17±0.02 0.53±0.04	6.40±1.95 12.53±2.83 0.01±0.07	88.5±1.6 4.39±0.77 18.0±3.1	8.4±1.5 3.0±0.8 81.93±9.55	0.01±0.00 0.21±0.01 51.49±0.50
4	0.58±0.06 0.17±0.02 0.53±0.04	6.27±1.92 12.11±2.75 0.01±0.06	88.2±1.7 4.56±0.79 17.6±3.3	8.8±1.5 3.0±0.7 80.99±9.01	0.01±0.00 0.21±0.01 51.57±0.49

Section 2.7 Bound Factor and Distribution-Drift Bound

Table 15: Corridor, bound factor ϕ and distribution-drift bound L_μ, default setting and S1–S5.

Corridor Baselines $\alpha_0 = 0.15,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.49±0.04 0.17±0.02 0.82±0.09	9.02±1.17 15.26±0.53 0.05±0.17	98.2±0.0 0.60±0.00 33.4±8.6	0.01±0.01 0.14±0.01 34.00±0.87
QT	0.52±0.04 0.17±0.02 0.86±0.09	8.34±0.97 14.38±0.32 0.02±0.10	99.0±0.8 0.10±0.00 49.7±8.0	0.01±0.01 0.14±0.01 34.17±0.89
$L_\mu = 0.005$				
Unfiltered	N-A 0.18±0.05 0.00±0.00	30.50±13.38 N-A 22.41±9.99	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.21±0.02 0.67±0.08	28.69±2.97 49.74±2.87 16.43±4.06	100.0±0.0 0.00±0.00 82.7±8.6	−0.03±0.01 0.09±0.01 32.51±0.82
$c_{\max}$	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.95±0.98
ACI	0.59±0.04 0.20±0.02 0.87±0.09	8.44±1.11 14.21±0.59 0.02±0.13	98.3±0.0 0.60±0.00 33.9±8.6	0.01±0.01 0.14±0.01 34.47±0.91
QT	0.59±0.03 0.20±0.02 0.90±0.09	8.36±0.96 14.38±0.24 0.01±0.05	98.2±1.0 0.10±0.00 48.1±7.5	0.01±0.01 0.14±0.01 34.57±0.92

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.0005$					
0	0.82±0.06 0.17±0.01 0.93±0.09	2.12±1.18 3.46±1.68 0.00±0.00	81.7±6.6 6.43±2.38 6.7±3.1	18.3±6.6 0.0±0.0 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.25	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.5	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
0.75	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.38±1.64 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
1	0.83±0.05 0.17±0.01 0.93±0.09	2.12±1.18 3.37±1.63 0.00±0.00	79.2±6.2 6.45±2.37 5.7±2.2	18.3±6.6 2.5±0.7 9.39±2.38	0.03±0.01 0.16±0.01 35.14±0.91
default setting: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50), $L_\mu = 0.005$					
0	0.87±0.05 0.20±0.02 0.95±0.09	1.72±1.17 2.76±1.63 0.00±0.00	79.9±8.6 7.12±3.12 5.6±3.0	20.1±8.6 0.0±0.0 7.53±2.02	0.03±0.01 0.16±0.01 35.38±0.98
0.25	0.90±0.03 0.20±0.02 0.97±0.09	1.19±0.72 1.86±0.98 0.00±0.00	77.4±8.1 7.14±3.11 4.5±2.5	20.0±8.6 2.6±0.9 7.58±2.07	0.04±0.01 0.16±0.01 35.51±0.98
0.5	0.95±0.02 0.20±0.01 0.99±0.09	0.54±0.36 0.86±0.43 0.00±0.00	47.8±17.2 7.34±3.79 3.3±2.0	19.9±8.6 32.3±15.5 7.64±2.02	0.04±0.01 0.16±0.01 35.72±0.98
0.75	0.99±0.01 0.20±0.01 1.00±0.09	0.12±0.18 0.22±0.23 0.00±0.00	14.2±8.0 7.87±5.52 2.4±1.4	19.8±8.5 66.0±7.1 7.64±2.02	0.04±0.01 0.17±0.01 35.88±0.99
1	1.00±0.00 0.20±0.01 1.01±0.09	0.03±0.08 0.05±0.12 0.00±0.00	3.4±2.8 6.42±2.74 1.7±0.1	19.8±8.5 76.9±6.6 7.70±2.09	0.05±0.01 0.17±0.01 35.94±0.98

Table 15 (continued).

Corridor Baselines $\alpha_0 = 0.30,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.04 0.00±0.00	30.69±13.51 N-A 22.32±10.04	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02 0.66±0.07	28.00±2.99 49.65±2.89 15.27±4.17	100.0±0.0 0.00±0.00 78.8±8.6	−0.03±0.01 0.10±0.01 32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01 1.00±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.91±0.92
ACI	0.22±0.04 0.17±0.02 0.73±0.08	17.68±1.59 30.21±0.72 2.74±1.43	99.1±0.0 0.30±0.00 57.4±10.2	−0.01±0.01 0.12±0.01 33.13±0.81
QT	0.30±0.03 0.17±0.02 0.78±0.09	15.19±1.31 26.54±1.03 0.61±0.52	99.7±0.1 0.10±0.00 60.1±11.5	−0.00±0.00 0.13±0.01 33.44±0.83
$L_\mu = 0.005$				
Unfiltered	N-A 0.18±0.05 0.00±0.00	30.50±13.38 N-A 22.41±9.99	N-A N-A N-A	−0.27±0.05 −0.02±0.01 70.00±0.00
$c = 0$	0.00±0.00 0.21±0.02 0.67±0.08	28.69±2.97 49.74±2.87 16.43±4.06	100.0±0.0 0.00±0.00 82.7±8.6	−0.03±0.01 0.09±0.01 32.51±0.82
$c_{\max}$	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.05±0.01 0.17±0.01 35.95±0.98
ACI	0.33±0.04 0.20±0.02 0.79±0.09	17.07±1.68 29.20±0.64 1.13±0.99	99.1±0.0 0.30±0.00 58.2±11.3	−0.01±0.01 0.12±0.01 33.62±0.87
QT	0.36±0.03 0.20±0.02 0.83±0.09	15.96±1.32 27.34±0.84 0.41±0.45	99.6±0.2 0.10±0.00 63.5±12.0	−0.00±0.00 0.13±0.01 33.78±0.90

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.47±0.07 0.17±0.02 0.80±0.09	10.02±2.38 16.14±3.49 0.22±0.50	95.5±0.9 1.51±0.30 29.7±8.8	4.5±0.9 0.0±0.0 30.40±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.25	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.61±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.6±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.5	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.61±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.6±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
0.75	0.49±0.07 0.17±0.02 0.80±0.09	10.02±2.38 15.60±3.40 0.21±0.45	92.3±1.1 1.53±0.31 21.5±3.5	4.5±0.9 3.2±0.8 30.38±3.39	0.00±0.01 0.13±0.01 33.84±0.89
1	0.50±0.06 0.17±0.02 0.80±0.09	9.74±2.12 15.10±2.94 0.18±0.41	92.3±1.1 1.53±0.31 21.1±3.3	4.5±0.9 3.2±0.8 30.46±3.36	0.01±0.01 0.13±0.01 33.86±0.89
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.005$					
0	0.55±0.09 0.20±0.02 0.84±0.10	10.74±3.52 17.11±5.23 0.21±0.53	95.5±1.0 1.53±0.33 32.2±12.7	4.5±1.0 0.0±0.0 26.61±3.41	0.01±0.01 0.13±0.01 34.19±0.98
0.25	0.66±0.04 0.20±0.02 0.88±0.09	7.23±1.50 11.45±2.03 0.04±0.24	92.1±1.2 1.54±0.34 22.5±5.9	4.4±1.0 3.4±0.9 27.23±3.36	0.01±0.01 0.14±0.01 34.59±0.93
0.5	0.75±0.03 0.20±0.02 0.92±0.09	4.10±0.96 6.78±1.29 0.01±0.15	92.1±1.2 1.54±0.34 18.2±5.0	4.4±1.0 3.5±0.9 27.83±3.42	0.02±0.01 0.15±0.01 34.99±0.93
0.75	0.83±0.02 0.20±0.02 0.95±0.08	2.28±0.85 4.02±1.06 0.01±0.15	91.9±1.4 1.54±0.34 15.4±4.8	4.3±1.0 3.8±1.2 28.33±3.29	0.03±0.01 0.15±0.01 35.28±0.92
1	0.89±0.01 0.20±0.02 0.97±0.08	1.35±0.68 2.36±0.84 0.00±0.02	74.4±3.7 1.54±0.34 12.9±4.7	4.3±0.9 21.3±3.6 28.60±3.39	0.03±0.01 0.16±0.01 35.50±0.96
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.0005$					
0	0.49±0.07 0.17±0.02 0.80±0.09	9.58±2.17 15.50±3.18 0.17±0.44	94.9±0.9 1.72±0.32 28.0±8.4	5.1±0.9 0.0±0.0 29.82±3.24	0.01±0.01 0.13±0.01 33.90±0.88
0.25	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.98±3.09 0.16±0.38	91.7±1.2 1.74±0.32 21.0±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
0.5	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.98±3.09 0.16±0.38	91.7±1.2 1.74±0.32 21.0±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
0.75	0.51±0.07 0.17±0.02 0.80±0.09	9.59±2.17 14.97±3.09 0.16±0.38	91.7±1.2 1.74±0.32 20.9±3.4	5.1±0.9 3.2±0.8 29.80±3.25	0.01±0.01 0.13±0.01 33.90±0.88
1	0.52±0.06 0.17±0.02 0.81±0.09	9.35±1.96 14.52±2.74 0.14±0.36	91.7±1.2 1.74±0.32 20.3±3.3	5.1±0.9 3.2±0.8 29.80±3.27	0.01±0.01 0.13±0.01 33.92±0.88

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.005$					
0	0.57±0.09 0.20±0.02 0.84±0.10	10.07±3.37 16.03±5.05 0.16±0.47	94.9±1.0 1.74±0.35 29.7±11.8	5.1±1.0 0.0±0.0 26.05±3.40	0.01±0.01 0.13±0.01 34.25±0.97
0.25	0.67±0.04 0.20±0.02 0.88±0.09	6.86±1.45 10.83±1.92 0.03±0.22	91.7±1.3 1.75±0.35 21.2±5.9	5.0±1.0 3.3±0.9 26.68±3.27	0.02±0.01 0.14±0.01 34.62±0.93
0.5	0.76±0.03 0.20±0.02 0.92±0.09	3.86±0.92 6.37±1.27 0.01±0.15	91.6±1.3 1.75±0.35 17.2±4.9	5.0±1.0 3.4±0.9 27.34±3.41	0.02±0.01 0.15±0.01 35.03±0.93
0.75	0.84±0.02 0.20±0.02 0.95±0.08	2.07±0.73 3.70±0.95 0.01±0.15	91.1±1.5 1.75±0.35 14.5±4.7	4.9±1.0 3.9±1.3 27.66±3.34	0.03±0.01 0.16±0.01 35.30±0.93
1	0.90±0.01 0.20±0.02 0.97±0.08	1.20±0.62 2.14±0.80 0.00±0.02	71.5±3.9 1.75±0.35 12.2±4.6	4.9±1.0 23.6±3.8 28.06±3.45	0.03±0.01 0.16±0.01 35.54±0.95
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.57±0.08 0.17±0.02 0.83±0.09	7.44±2.15 12.14±3.27 0.07±0.32	94.1±1.6 2.03±0.55 21.9±7.9	5.9±1.6 0.0±0.0 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.25	0.58±0.07 0.17±0.02 0.83±0.09	7.44±2.16 11.73±3.19 0.06±0.27	90.9±1.7 2.04±0.54 16.6±3.8	5.9±1.6 3.2±0.8 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.5	0.58±0.07 0.17±0.02 0.83±0.09	7.44±2.16 11.73±3.19 0.06±0.27	90.9±1.7 2.04±0.54 16.6±3.8	5.9±1.6 3.2±0.8 24.29±3.09	0.01±0.01 0.14±0.01 34.18±0.91
0.75	0.59±0.07 0.17±0.02 0.83±0.09	7.15±1.96 11.18±2.78 0.06±0.26	90.9±1.7 2.04±0.54 16.1±3.6	5.9±1.6 3.2±0.8 24.31±3.05	0.01±0.01 0.14±0.01 34.20±0.91
1	0.62±0.06 0.17±0.02 0.84±0.09	6.38±1.57 10.03±2.14 0.04±0.21	90.9±1.7 2.04±0.54 15.3±3.6	5.9±1.6 3.2±0.8 24.33±3.09	0.01±0.01 0.14±0.01 34.28±0.90
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.005$					
0	0.64±0.09 0.20±0.02 0.87±0.10	7.79±3.15 12.33±4.75 0.06±0.31	94.1±1.6 2.05±0.57 22.3±10.0	5.9±1.6 0.0±0.0 20.96±2.99	0.01±0.01 0.14±0.01 34.52±0.99
0.25	0.83±0.03 0.20±0.02 0.95±0.09	2.61±0.88 4.30±1.32 0.01±0.17	76.5±5.6 2.06±0.57 14.4±5.5	5.8±1.6 17.7±5.4 22.21±3.01	0.03±0.01 0.15±0.01 35.31±0.97
0.5	0.97±0.01 0.20±0.01 1.00±0.09	0.36±0.43 0.68±0.61 0.01±0.12	13.0±3.2 2.16±1.16 10.2±5.2	5.7±1.6 81.3±2.7 22.74±3.02	0.04±0.01 0.17±0.01 35.83±0.97
0.75	1.00±0.00 0.20±0.01 1.01±0.08	0.08±0.18 0.11±0.21 0.00±0.00	2.1±1.0 2.49±1.86 5.8±4.9	5.7±1.6 92.2±0.9 22.74±2.99	0.04±0.01 0.17±0.01 35.94±0.99
1	1.00±0.00 0.20±0.01 1.01±0.09	0.02±0.07 0.03±0.09 0.00±0.00	0.6±0.3 1.81±0.04 5.1±4.9	5.7±1.6 93.7±1.3 22.75±2.94	0.05±0.01 0.17±0.01 35.95±0.99

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
0	0.55±0.06 0.17±0.02 0.83±0.09	8.03±1.85 12.97±2.61 0.06±0.21	92.0±1.3 2.73±0.47 24.5±7.7	8.0±1.3 0.0±0.0 15.14±1.83	0.01±0.01 0.14±0.01 34.11±0.89
0.25	0.58±0.05 0.17±0.02 0.83±0.09	7.51±1.62 11.74±2.11 0.05±0.19	88.9±1.4 2.74±0.47 18.1±3.1	8.0±1.3 3.1±0.7 15.15±1.86	0.01±0.01 0.14±0.01 34.16±0.87
0.5	0.70±0.04 0.17±0.02 0.88±0.09	4.66±0.98 7.10±1.27 0.01±0.05	84.5±2.7 2.74±0.47 15.2±3.2	7.9±1.3 7.7±2.4 15.57±1.90	0.02±0.01 0.15±0.01 34.65±0.90
0.75	0.86±0.02 0.17±0.01 0.94±0.08	1.97±0.82 3.10±1.04 0.00±0.00	48.6±3.0 2.74±0.47 13.1±3.3	7.7±1.3 43.6±2.8 16.09±1.97	0.03±0.01 0.16±0.01 35.30±0.92
1	0.92±0.02 0.17±0.01 0.97±0.08	0.75±0.52 1.44±0.71 0.00±0.00	29.8±3.3 2.74±0.47 11.2±3.4	7.7±1.3 62.5±3.0 16.38±1.82	0.03±0.01 0.17±0.01 35.62±0.88
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
0	0.64±0.06 0.20±0.02 0.87±0.10	7.78±2.43 12.29±3.57 0.02±0.11	92.0±1.4 2.76±0.50 22.8±9.2	8.0±1.4 0.0±0.0 13.49±2.03	0.01±0.01 0.14±0.01 34.51±0.94
0.25	0.99±0.00 0.20±0.01 1.00±0.09	0.10±0.18 0.18±0.25 0.00±0.00	4.6±1.6 4.02±5.26 6.4±3.8	7.7±1.4 87.7±1.3 15.02±2.07	0.04±0.01 0.17±0.01 35.91±0.97
0.5	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98
0.75	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98
1	1.00±0.00 0.20±0.01 1.01±0.08	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	7.7±1.4 92.3±1.4 15.03±2.07	0.05±0.01 0.17±0.01 35.95±0.98

Table 15 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.10, 10)$, $L_\mu = 0.0005$					
0	0.51±0.07 0.17±0.02 0.81±0.09	9.20±2.07 14.89±3.11 0.15±0.44	94.2±1.2 1.97±0.40 27.5±8.6	5.8±1.2 0.0±0.0 119.56±13.82	0.01±0.01 0.13±0.01 33.95±0.88
0.25	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.40±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.4±3.4	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
0.5	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.40±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.4±3.4	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
0.75	0.52±0.07 0.17±0.02 0.81±0.09	9.20±2.08 14.39±3.06 0.14±0.37	91.0±1.4 1.99±0.40 20.3±3.3	5.8±1.2 3.2±0.8 119.63±13.83	0.01±0.01 0.13±0.01 33.95±0.88
1	0.53±0.06 0.17±0.02 0.81±0.09	8.94±1.88 13.84±2.67 0.12±0.36	91.0±1.4 1.99±0.40 19.6±3.2	5.8±1.2 3.2±0.8 119.71±13.69	0.01±0.01 0.13±0.01 33.97±0.88
S5: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.10, 10)$, $L_\mu = 0.005$					
0	0.59±0.08 0.20±0.02 0.85±0.10	9.46±3.08 15.01±4.61 0.11±0.39	94.2±1.2 1.99±0.43 28.4±11.0	5.8±1.2 0.0±0.0 103.12±14.06	0.01±0.01 0.14±0.01 34.33±0.96
0.25	0.69±0.04 0.20±0.02 0.89±0.09	6.34±1.28 9.95±1.77 0.02±0.21	91.0±1.4 2.00±0.43 19.8±5.7	5.7±1.2 3.3±0.9 105.33±13.81	0.02±0.01 0.14±0.01 34.69±0.93
0.5	0.77±0.03 0.20±0.02 0.92±0.09	3.66±0.82 6.07±1.05 0.01±0.15	90.9±1.5 2.00±0.43 15.7±4.8	5.7±1.2 3.4±0.9 107.94±13.88	0.02±0.01 0.15±0.01 35.04±0.94
0.75	0.83±0.02 0.20±0.02 0.94±0.09	2.24±0.71 3.78±0.87 0.01±0.15	90.9±1.4 2.00±0.43 13.0±4.5	5.6±1.2 3.4±0.9 109.88±13.10	0.03±0.01 0.15±0.01 35.28±0.95
1	0.88±0.01 0.20±0.02 0.96±0.09	1.32±0.58 2.27±0.69 0.00±0.02	89.2±3.6 2.00±0.43 10.4±3.9	5.6±1.2 5.2±3.4 111.21±14.56	0.03±0.01 0.16±0.01 35.48±0.96

Table 16: Maze, as in Table 15.

Maze **Baselines** $\alpha_0 = 0.15,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.49±0.05 0.17±0.02 0.50±0.03	7.45±1.47 14.94±1.34 0.01±0.06	97.7±0.0 1.20±0.00 27.6±7.3	0.01±0.00 0.21±0.01 51.16±0.40
QT	0.72±0.02 0.17±0.02 0.61±0.02	2.78±0.88 6.40±0.81 0.00±0.00	99.6±0.1 0.20±0.00 15.3±3.7	0.02±0.00 0.21±0.01 52.52±0.33
$L_\mu = 0.005$				
Unfiltered	N-A 0.22±0.02 0.00±0.00	54.91±1.61 N-A 53.15±1.09	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.17±0.16
$c = 0$	0.00±0.00 0.22±0.02 0.32±0.02	28.79±2.72 49.92±3.15 5.43±2.54	100.0±0.0 0.00±0.00 72.2±14.1	−0.01±0.00 0.20±0.01 48.69±0.28
$c_{\max}$	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.21±0.01 54.50±0.34
ACI	0.59±0.04 0.22±0.02 0.56±0.03	7.22±1.42 15.57±1.34 0.00±0.06	97.7±0.0 1.20±0.00 28.8±7.7	0.01±0.00 0.21±0.01 51.85±0.38
QT	0.75±0.02 0.22±0.02 0.64±0.02	3.11±0.91 7.89±0.74 0.00±0.00	99.6±0.1 0.20±0.00 15.7±3.0	0.02±0.00 0.21±0.01 52.74±0.33

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
default setting: V-C $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$, $L_\mu = 0.0005$					
0	0.86±0.05 0.17±0.02 0.70±0.04	0.81±0.72 2.48±1.31 0.00±0.00	75.2±9.0 13.27±4.84 5.2±2.3	24.8±9.0 0.0±0.0 6.49±1.94	0.02±0.00 0.21±0.01 53.40±0.45
0.25	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.5	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
0.75	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
1	0.86±0.05 0.17±0.02 0.70±0.04	0.77±0.73 2.36±1.25 0.00±0.00	72.6±8.8 13.28±4.84 4.7±1.9	24.8±9.0 2.6±0.7 6.49±1.88	0.02±0.00 0.21±0.01 53.40±0.44
default setting: V-C $(\alpha_0, \alpha, \delta, K) = (0.15, 0.10, 0.05, 50)$, $L_\mu = 0.005$					
0	0.89±0.05 0.22±0.02 0.72±0.04	0.71±0.76 2.64±1.77 0.00±0.00	73.1±11.8 14.47±6.43 5.2±3.0	26.9±11.8 0.0±0.0 4.44±1.32	0.02±0.00 0.21±0.01 53.63±0.50
0.25	0.90±0.04 0.21±0.02 0.73±0.04	0.54±0.58 2.08±1.22 0.00±0.00	70.7±11.6 14.48±6.42 4.3±2.4	26.9±11.8 2.4±0.7 4.43±1.32	0.02±0.00 0.21±0.01 53.76±0.44
0.5	0.95±0.02 0.22±0.02 0.76±0.03	0.19±0.26 1.04±0.61 0.00±0.00	49.6±19.2 14.18±6.40 3.4±2.1	26.8±11.8 23.6±14.3 4.47±1.37	0.02±0.00 0.21±0.01 54.05±0.39
0.75	0.98±0.01 0.22±0.02 0.79±0.02	0.05±0.14 0.31±0.33 0.00±0.00	17.0±10.5 14.79±8.71 2.6±1.5	26.6±11.8 56.4±8.1 4.40±1.31	0.03±0.00 0.21±0.01 54.34±0.35
1	1.00±0.00 0.21±0.02 0.80±0.02	0.01±0.06 0.07±0.17 0.00±0.00	4.8±4.0 12.29±5.13 1.7±0.1	26.6±11.8 68.6±8.8 4.48±1.27	0.03±0.00 0.21±0.01 54.41±0.34

Table 16 (continued).

Maze Baselines $\alpha_0 = 0.30,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$L_\mu = 0.0005$				
Unfiltered	N-A 0.17±0.02 0.00±0.00	55.68±1.58 N-A 53.27±0.95	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02 0.32±0.02	24.93±2.57 49.79±2.97 3.86±2.16	100.0±0.0 0.00±0.00 66.8±12.7	−0.01±0.00 0.21±0.01 48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.22±0.01 54.44±0.32
ACI	0.22±0.04 0.17±0.02 0.39±0.02	15.84±1.93 29.86±1.85 0.48±0.65	98.8±0.0 0.60±0.00 47.1±8.6	−0.00±0.00 0.21±0.01 49.71±0.32
QT	0.50±0.03 0.17±0.02 0.50±0.02	8.04±1.31 15.68±1.10 0.05±0.17	99.6±0.0 0.20±0.00 33.8±7.6	0.01±0.00 0.21±0.01 51.18±0.32
$L_\mu = 0.005$				
Unfiltered	N-A 0.22±0.02 0.00±0.00	54.91±1.61 N-A 53.15±1.09	N-A N-A N-A	−0.08±0.00 0.21±0.01 42.17±0.16
$c = 0$	0.00±0.00 0.22±0.02 0.32±0.02	28.79±2.72 49.92±3.15 5.43±2.54	100.0±0.0 0.00±0.00 72.2±14.1	−0.01±0.00 0.20±0.01 48.69±0.28
$c_{\max}$	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	0.03±0.00 0.21±0.01 54.50±0.34
ACI	0.33±0.05 0.22±0.02 0.44±0.03	17.20±1.96 31.05±1.81 0.25±0.45	98.8±0.0 0.60±0.00 47.6±8.1	0.00±0.00 0.21±0.01 50.35±0.39
QT	0.57±0.03 0.22±0.02 0.54±0.02	9.29±1.32 18.33±0.96 0.01±0.09	99.6±0.0 0.20±0.00 34.9±8.7	0.01±0.00 0.21±0.01 51.63±0.30

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.50±0.08 0.17±0.02 0.50±0.04	7.53±2.28 15.01±3.62 0.01±0.09	94.1±1.2 3.04±0.63 24.1±7.3	5.9±1.2 0.0±0.0 22.68±2.53	0.01±0.00 0.21±0.01 51.21±0.52
0.25	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
0.5	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
0.75	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.53±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.8±3.6	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
1	0.52±0.07 0.17±0.02 0.51±0.04	7.40±2.27 14.52±3.57 0.01±0.06	91.1±1.4 3.04±0.63 19.7±3.5	5.9±1.2 2.9±0.8 22.83±2.54	0.01±0.00 0.21±0.01 51.25±0.52
S1: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.20, 50)$, $L_\mu = 0.005$					
0	0.56±0.10 0.22±0.02 0.54±0.05	9.24±3.91 18.31±6.00 0.03±0.21	94.1±1.2 3.05±0.63 27.6±9.7	5.9±1.2 0.0±0.0 17.42±1.89	0.01±0.00 0.21±0.01 51.63±0.67
0.25	0.65±0.05 0.22±0.02 0.58±0.03	6.13±1.79 12.92±2.86 0.00±0.03	91.1±1.3 3.05±0.63 22.5±6.1	5.8±1.2 3.0±0.8 17.61±1.98	0.01±0.00 0.21±0.01 52.16±0.43
0.5	0.75±0.03 0.22±0.02 0.64±0.03	2.98±1.02 7.77±1.68 0.00±0.00	91.1±1.4 3.05±0.63 18.3±5.2	5.8±1.2 3.1±0.8 17.79±2.03	0.02±0.00 0.21±0.01 52.82±0.38
0.75	0.82±0.02 0.22±0.02 0.68±0.03	1.52±0.71 4.57±1.32 0.00±0.00	90.9±1.7 3.05±0.63 15.3±5.0	5.7±1.2 3.4±1.3 17.96±1.94	0.02±0.00 0.21±0.01 53.30±0.35
1	0.88±0.01 0.21±0.02 0.72±0.02	0.71±0.57 2.76±0.99 0.00±0.00	77.1±4.4 3.05±0.63 12.9±5.0	5.7±1.2 17.2±4.3 18.03±1.93	0.02±0.00 0.21±0.01 53.68±0.37
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.0005$					
0	0.52±0.08 0.17±0.02 0.51±0.04	7.30±2.38 14.44±3.51 0.01±0.09	93.3±1.2 3.45±0.62 23.3±7.1	6.7±1.2 0.0±0.0 22.33±2.57	0.01±0.00 0.21±0.01 51.31±0.53
0.25	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.5	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
0.75	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.90±3.47 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53
1	0.53±0.08 0.17±0.02 0.51±0.04	7.28±2.37 13.89±3.46 0.01±0.07	90.3±1.5 3.45±0.62 19.2±3.7	6.7±1.2 2.9±0.8 22.37±2.50	0.01±0.00 0.21±0.01 51.33±0.53

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S2: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.15, 50)$, $L_\mu = 0.005$					
0	0.58±0.09 0.22±0.02 0.54±0.05	8.72±3.75 17.40±5.70 0.02±0.13	93.3±1.2 3.46±0.63 27.0±10.1	6.7±1.2 0.0±0.0 16.89±1.92	0.01±0.00 0.21±0.01 51.69±0.63
0.25	0.67±0.05 0.22±0.02 0.58±0.03	5.75±1.81 12.28±2.73 0.00±0.00	90.2±1.3 3.46±0.63 21.4±6.0	6.6±1.2 3.2±0.8 17.24±2.04	0.01±0.00 0.21±0.01 52.24±0.41
0.5	0.76±0.03 0.22±0.02 0.64±0.03	2.80±1.08 7.33±1.51 0.00±0.00	90.4±1.4 3.46±0.63 17.5±5.0	6.5±1.2 3.1±0.8 17.45±1.99	0.02±0.00 0.21±0.01 52.88±0.36
0.75	0.83±0.02 0.22±0.02 0.69±0.02	1.36±0.72 4.20±1.14 0.00±0.00	89.9±1.9 3.46±0.63 14.6±4.8	6.5±1.2 3.6±1.6 17.64±1.92	0.02±0.00 0.21±0.01 53.32±0.36
1	0.89±0.01 0.21±0.02 0.72±0.02	0.65±0.53 2.47±0.90 0.00±0.00	74.4±4.8 3.46±0.63 11.7±4.6	6.4±1.2 19.2±4.7 17.67±1.92	0.02±0.00 0.21±0.01 53.73±0.37
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.0005$					
0	0.59±0.08 0.17±0.02 0.55±0.04	5.42±2.13 11.07±3.41 0.01±0.08	92.2±1.8 4.01±0.97 18.0±6.3	7.8±1.8 0.0±0.0 18.22±2.34	0.01±0.00 0.21±0.01 51.76±0.57
0.25	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
0.5	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
0.75	0.61±0.08 0.17±0.02 0.55±0.04	5.36±2.18 10.62±3.32 0.00±0.06	89.0±2.0 4.01±0.97 15.2±3.8	7.7±1.8 3.2±0.9 18.23±2.33	0.01±0.00 0.21±0.01 51.79±0.58
1	0.62±0.07 0.17±0.02 0.55±0.04	5.04±1.93 9.98±2.87 0.00±0.06	89.1±2.0 4.01±0.97 14.5±3.8	7.7±1.9 3.2±0.9 18.19±2.35	0.01±0.00 0.21±0.01 51.86±0.54
S3: V-L $(\alpha_0, \alpha, \delta, K) = (0.30, 0.20, 0.20, 50)$, $L_\mu = 0.005$					
0	0.66±0.09 0.22±0.02 0.58±0.05	6.07±3.41 12.95±5.24 0.01±0.12	92.3±1.9 4.04±1.01 20.1±8.6	7.7±1.9 0.0±0.0 13.80±1.88	0.01±0.00 0.21±0.01 52.14±0.60
0.25	0.83±0.04 0.22±0.02 0.70±0.03	1.84±1.05 4.93±1.71 0.00±0.00	77.9±6.8 4.04±1.01 14.2±5.6	7.6±1.9 14.5±6.8 14.11±1.81	0.02±0.00 0.21±0.01 53.30±0.40
0.5	0.97±0.01 0.22±0.02 0.79±0.02	0.26±0.31 0.82±0.64 0.00±0.00	15.7±3.5 4.10±1.40 9.7±5.1	7.4±1.9 76.8±3.2 14.34±2.02	0.03±0.00 0.21±0.01 54.26±0.36
0.75	0.99±0.01 0.21±0.02 0.81±0.02	0.00±0.03 0.16±0.27 0.00±0.00	2.7±1.3 5.05±3.85 6.4±5.0	7.4±1.9 89.8±1.0 14.34±2.03	0.03±0.00 0.21±0.01 54.49±0.34
1	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.04±0.13 0.00±0.00	0.6±0.3 3.60±0.00 5.4±4.8	7.4±1.9 92.0±1.6 14.36±2.07	0.03±0.00 0.21±0.01 54.50±0.34

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S4: V-Δ $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.025, 100)$, $L_\mu = 0.0005$					
0	0.59±0.06 0.17±0.02 0.54±0.04	5.82±1.84 11.47±2.58 0.00±0.04	89.5±1.6 5.45±0.84 19.9±6.0	10.5±1.6 0.0±0.0 12.22±1.49	0.01±0.00 0.21±0.01 51.64±0.50
0.25	0.61±0.06 0.17±0.02 0.55±0.04	5.43±1.74 10.61±2.25 0.01±0.05	86.7±1.8 5.45±0.85 16.6±3.3	10.5±1.6 2.8±0.7 12.25±1.43	0.01±0.00 0.21±0.01 51.73±0.45
0.5	0.91±0.02 0.17±0.02 0.73±0.03	0.92±0.69 1.96±0.99 0.00±0.00	35.5±2.8 5.45±0.85 11.1±3.6	10.1±1.6 54.4±2.4 12.73±1.70	0.02±0.00 0.21±0.01 53.70±0.34
0.75	0.98±0.01 0.17±0.02 0.80±0.02	0.14±0.24 0.38±0.37 0.00±0.00	9.6±2.2 5.53±1.66 7.5±3.7	10.0±1.6 80.4±1.9 12.84±1.66	0.03±0.00 0.21±0.01 54.34±0.31
1	1.00±0.00 0.17±0.01 0.81±0.02	0.00±0.00 0.05±0.13 0.00±0.00	1.4±0.8 7.85±7.25 3.9±3.1	10.0±1.6 88.5±1.0 12.86±1.58	0.03±0.00 0.22±0.01 54.43±0.32
S4: V-Δ $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.025, 100)$, $L_\mu = 0.005$					
0	0.66±0.07 0.22±0.02 0.58±0.04	6.04±2.71 12.95±4.13 0.00±0.06	89.5±1.7 5.50±0.91 21.7±7.6	10.5±1.7 0.0±0.0 9.31±1.17	0.01±0.00 0.21±0.01 52.14±0.53
0.25	1.00±0.00 0.21±0.02 0.81±0.02	0.01±0.04 0.08±0.19 0.00±0.00	2.5±1.2 6.48±4.88 6.1±4.3	10.1±1.7 87.4±0.9 9.83±1.35	0.03±0.00 0.21±0.01 54.50±0.32
0.5	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34
0.75	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34
1	1.00±0.00 0.21±0.02 0.81±0.02	0.00±0.00 0.00±0.00 0.00±0.00	0.0±0.0 N-A N-A	10.1±1.7 89.9±1.7 9.86±1.33	0.03±0.00 0.21±0.01 54.50±0.34

Table 16 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$ $\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$ $\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
S5: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.10, 10)$, $L_\mu = 0.0005$					
0	0.54±0.07 0.17±0.02 0.52±0.04	6.88±2.10 13.56±3.19 0.02±0.11	92.4±1.5 3.93±0.77 22.3±6.6	7.6±1.5 0.0±0.0 83.81±9.69	0.01±0.00 0.21±0.01 51.40±0.49
0.25	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.5	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
0.75	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.04±3.14 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.4	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
1	0.55±0.07 0.17±0.02 0.52±0.04	6.76±2.17 13.03±3.13 0.01±0.09	89.3±1.7 3.93±0.77 18.5±3.3	7.6±1.5 3.0±0.8 83.58±9.71	0.01±0.00 0.21±0.01 51.43±0.52
S5: V-C $(\alpha_0, \alpha, \delta, K) = (0.30, 0.25, 0.10, 10)$, $L_\mu = 0.005$					
0	0.60±0.08 0.22±0.02 0.55±0.05	7.99±3.50 16.17±5.05 0.02±0.13	92.4±1.5 3.94±0.78 25.1±8.8	7.6±1.5 0.0±0.0 61.57±7.92	0.01±0.00 0.21±0.01 51.80±0.60
0.25	0.69±0.05 0.22±0.02 0.59±0.03	5.25±1.85 11.34±2.51 0.00±0.00	89.3±1.7 3.94±0.78 20.1±5.8	7.5±1.5 3.1±0.8 62.80±7.58	0.01±0.00 0.21±0.01 52.29±0.42
0.5	0.77±0.03 0.21±0.02 0.64±0.03	2.76±1.00 6.96±1.38 0.00±0.00	89.4±1.6 3.94±0.78 15.8±4.7	7.5±1.5 3.2±0.8 64.77±7.77	0.02±0.00 0.21±0.01 52.85±0.36
0.75	0.83±0.02 0.21±0.02 0.69±0.03	1.46±0.62 4.38±1.03 0.00±0.00	89.4±1.7 3.94±0.78 12.9±4.3	7.4±1.5 3.2±1.0 65.62±7.79	0.02±0.00 0.21±0.01 53.26±0.37
1	0.88±0.02 0.22±0.02 0.72±0.02	0.71±0.49 2.76±0.80 0.00±0.00	87.1±5.6 3.94±0.78 10.3±3.9	7.3±1.5 5.5±5.5 65.88±8.11	0.02±0.00 0.21±0.01 53.64±0.31

Table 17: Corridor, S1–S5 with their variants, bound factor ϕ and distribution-drift bound L_μ: events and bound excesses.

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^V(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.47±0.07	10.02±2.38	91 (51.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.49±0.07	10.02±2.38	0 (1.5) ✓	32 (20.9)	123 (67.3)
1	0.50±0.06	9.74±2.12	0 (1.5) ✓	0 (1.5)	0 (1.5)
S1: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.005$					
$0^\dagger$	0.55±0.09	10.74±3.52	106 (59.0)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.66±0.04	7.23±1.50	16 (11.9) ✓	199 (100.0)	200 (100.0)
$0.5^\dagger$	0.75±0.03	4.10±0.96	1 (2.4) ✓	168 (88.2)	168 (88.2)
$0.75^\dagger$	0.83±0.02	2.28±0.85	1 (2.4) ✓	5 (5.2)	5 (5.2)
1	0.89±0.01	1.35±0.68	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.49±0.07	9.58±2.17	75 (43.5)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^\dagger$	0.51±0.07	9.59±2.17	0 (1.5) ✓	24 (16.5)	116 (63.9)
1	0.52±0.06	9.35±1.96	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.005$					
$0^\dagger$	0.57±0.09	10.07±3.37	85 (48.6)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.67±0.04	6.86±1.45	10 (8.4) ✓	199 (100.0)	200 (100.0)
$0.5^\dagger$	0.76±0.03	3.86±0.92	1 (2.4) ✓	161 (85.0)	161 (85.0)
$0.75^\dagger$	0.84±0.02	2.07±0.73	0 (1.5) ✓	5 (5.2)	5 (5.2)
1	0.90±0.01	1.20±0.62	0 (1.5) ✓	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0,\alpha,\delta,K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.0005$					
$0^\dagger$	0.57±0.08	7.44±2.15	32 (20.9)	200 (100.0)	200 (100.0)
$0.25^\dagger$	0.58±0.07	7.44±2.16	0 (1.5) ✓	45 (27.9)	54 (32.7)
$0.5^\dagger$	0.58±0.07	7.44±2.16	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^\dagger$	0.59±0.07	7.15±1.96	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.62±0.06	6.38±1.57	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 17 (continued).

Corridor $c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.64±0.09	7.79±3.15	40 (25.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.83±0.03	2.61±0.88	3 (3.9) ✓	11 (9.0)	12 (9.6)
$0.5^{\dagger}$	0.97±0.01	0.36±0.43	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	1.00±0.00	0.08±0.18	0 (1.5) ✓	0/181 (1.5)	0/181 (1.5)
1	1.00±0.00	0.02±0.07	0 (1.5) ✓	0/158 (1.5)	0/158 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
$0^{\dagger}$	0.55±0.06	8.03±1.85	43 (26.9)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.58±0.05	7.51±1.62	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.5^{\dagger}$	0.70±0.04	4.66±0.98	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.86±0.02	1.97±0.82	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.92±0.02	0.75±0.52	0 (1.5) ✓	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
$0^{\dagger}$	0.64±0.06	7.78±2.43	34 (22.0)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.99±0.00	0.10±0.18	0 (1.5) ✓	0/199 (1.5)	0/199 (1.5)
$0.5^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
$0.75^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
$0^{\dagger}$	0.51±0.07	9.20±2.07	69 (40.5)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.52±0.07	9.20±2.08	0 (1.5) ✓	36 (23.1)	108 (60.0)
1	0.53±0.06	8.94±1.88	0 (1.5) ✓	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
$0^{\dagger}$	0.59±0.08	9.46±3.08	80 (46.1)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.69±0.04	6.34±1.28	8 (7.2) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.77±0.03	3.66±0.82	0 (1.5) ✓	146 (78.2)	146 (78.2)
$0.75^{\dagger}$	0.83±0.02	2.24±0.71	0 (1.5) ✓	1 (2.4)	1 (2.4)
1	0.88±0.01	1.32±0.58	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 18: Maze, as in Table 17.

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.0005$					
$0^{\dagger}$	0.50±0.08	7.53±2.28	40 (25.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.52±0.07	7.40±2.27	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.52±0.07	7.40±2.27	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.52±0.07	7.40±2.27	0 (1.5) ✓	106 (59.0)	121 (66.3)
1	0.52±0.07	7.40±2.27	0 (1.5) ✓	0 (1.5)	0 (1.5)
S1: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.56±0.10	9.24±3.91	77 (44.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.65±0.05	6.13±1.79	16 (11.9) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.75±0.03	2.98±1.02	1 (2.4) ✓	166 (87.3)	166 (87.3)
$0.75^{\dagger}$	0.82±0.02	1.52±0.71	0 (1.5) ✓	10 (8.4)	12 (9.6)
1	0.88±0.01	0.71±0.57	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.0005$					
$0^{\dagger}$	0.52±0.08	7.30±2.38	35 (22.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.53±0.08	7.28±2.37	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.53±0.08	7.28±2.37	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.53±0.08	7.28±2.37	0 (1.5) ✓	99 (55.6)	112 (62.0)
1	0.53±0.08	7.28±2.37	0 (1.5) ✓	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.58±0.09	8.72±3.75	63 (37.4)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.67±0.05	5.75±1.81	9 (7.8) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.76±0.03	2.80±1.08	1 (2.4) ✓	157 (83.2)	157 (83.2)
$0.75^{\dagger}$	0.83±0.02	1.36±0.72	0 (1.5) ✓	6 (5.9)	8 (7.2)
1	0.89±0.01	0.65±0.53	0 (1.5) ✓	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.0005$					
$0^{\dagger}$	0.59±0.08	5.42±2.13	11 (9.0) ✓	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.61±0.08	5.36±2.18	0 (1.5) ✓	44 (27.4)	45 (27.9)
$0.5^{\dagger}$	0.61±0.08	5.36±2.18	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.61±0.08	5.36±2.18	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	0.62±0.07	5.04±1.93	0 (1.5) ✓	0 (1.5)	0 (1.5)

Table 18 (continued).

Maze $c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$ †: $d_t^{\mathrm{V}}(c;\sigma,\tau)$ scaled by $\phi < 1$					
ϕ	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S3: V-L $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50), $L_\mu = 0.005$					
$0^{\dagger}$	0.66±0.09	6.07±3.41	24 (16.5) ✓	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.83±0.04	1.84±1.05	3 (3.9) ✓	10 (8.4)	11 (9.0)
$0.5^{\dagger}$	0.97±0.01	0.26±0.31	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.99±0.01	0.00±0.03	0 (1.5) ✓	0/184 (1.5)	0/184 (1.5)
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	0/161 (1.5)	0/161 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.0005$					
$0^{\dagger}$	0.59±0.06	5.82±1.84	12 (9.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.61±0.06	5.43±1.74	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.5^{\dagger}$	0.91±0.02	0.92±0.69	0 (1.5) ✓	0 (1.5)	0 (1.5)
$0.75^{\dagger}$	0.98±0.01	0.14±0.24	0 (1.5) ✓	0 (1.5)	0 (1.5)
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	0/171 (1.5)	0/171 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100), $L_\mu = 0.005$					
$0^{\dagger}$	0.66±0.07	6.04±2.71	29 (19.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	1.00±0.00	0.01±0.04	0 (1.5) ✓	0/194 (1.5)	0/194 (1.5)
$0.5^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
$0.75^{\dagger}$	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
1	1.00±0.00	0.00±0.00	0 (1.5) ✓	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.0005$					
$0^{\dagger}$	0.54±0.07	6.88±2.10	29 (19.3)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	200 (100.0)	200 (100.0)
$0.75^{\dagger}$	0.55±0.07	6.76±2.17	0 (1.5) ✓	88 (50.1)	96 (54.1)
1	0.55±0.07	6.76±2.17	0 (1.5) ✓	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10), $L_\mu = 0.005$					
$0^{\dagger}$	0.60±0.08	7.99±3.50	50 (30.6)	200 (100.0)	200 (100.0)
$0.25^{\dagger}$	0.69±0.05	5.25±1.85	6 (5.9) ✓	200 (100.0)	200 (100.0)
$0.5^{\dagger}$	0.77±0.03	2.76±1.00	0 (1.5) ✓	145 (77.7)	145 (77.7)
$0.75^{\dagger}$	0.83±0.02	1.46±0.62	0 (1.5) ✓	0 (1.5)	2 (3.2)
1	0.88±0.02	0.71±0.49	0 (1.5) ✓	0 (1.5)	0 (1.5)

Section 2.8 Certification, Deployment, and Assumption Checks

Table 19: Corridor, detailed checks at $L_\mu = 0.0005$.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	26
QT	200 (100.0)	200	12

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	200	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	0	0 (1.5)	0 (1.5)

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	198
QT	200 (100.0)	200	156

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	3 (3.9) ✓	0 (1.5) ✓	200	51	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	45	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	1 (2.4) ✓	0 (1.5) ✓	200	43	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	36	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	7 (6.5) ✓	0 (1.5) ✓	200	13	0 (1.5)	0 (1.5)
∞	6 (5.9) ✓	0 (1.5) ✓	200	10	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	178	0	0 (1.5)	0 (1.5)
∞	0 (1.5)	0 (1.5)	74	0	0 (1.5)	0 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	200	36	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	30	0 (1.5)	0 (1.5)

Table 20: Corridor, detailed checks at $L_\mu = 0.005$.

Corridor **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$			
Method	$\widehat{q}_{\text{dep}}\downarrow$	$\widehat{q}_{\text{CBF}}\downarrow$	$\widehat{q}_{\text{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	8
QT	200 (100.0)	200	3

Corridor $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$						
K	$\widehat{q}_{\text{cert}}\downarrow$	$\widehat{q}_{\text{dep}}\downarrow$	$\widehat{q}_{\text{CBF}}\downarrow$	$\widehat{q}_{\text{unsafe}}\downarrow$	$\widehat{q}_{\text{A2}}$	$\widehat{q}_{\text{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	19	0	0/139 (1.5)	0/139 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	13	0	0/107 (1.5)	0/107 (1.5)

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$			
Method	$\widehat{q}_{\text{dep}}\downarrow$	$\widehat{q}_{\text{CBF}}\downarrow$	$\widehat{q}_{\text{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	171
QT	200 (100.0)	200	128

Corridor $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$						
K	$\widehat{q}_{\text{cert}}\downarrow$	$\widehat{q}_{\text{dep}}\downarrow$	$\widehat{q}_{\text{CBF}}\downarrow$	$\widehat{q}_{\text{unsafe}}\downarrow$	$\widehat{q}_{\text{A2}}$	$\widehat{q}_{\text{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	0 (1.5) ✓	0 (1.5) ✓	198	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	119	1	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	0 (1.5) ✓	0 (1.5) ✓	199	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	114	1	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	3 (3.9) ✓	0 (1.5) ✓	14	0	0/158 (1.5)	0/158 (1.5)
∞	3 (3.9) ✓	0 (1.5) ✓	14	0	0/158 (1.5)	0/158 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	N-A	N-A
∞	0 (1.5)	0 (1.5)	0	0	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	1 (2.4) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	108	1	0 (1.5)	0 (1.5)

Table 21: Maze, detailed checks at $L_\mu = 0.0005$.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	195
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	2
QT	85–87 (49.6)	200	0

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	155	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	154	0	0 (1.5)	0 (1.5)

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	195
$c_{\max}$	0 (1.5)	0	0
ACI	199 (100.0)	200	102
QT	131 (71.1)	200	17

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	1 (2.4) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	3	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	4 (4.6) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
∞	3 (3.9) ✓	0 (1.5) ✓	200	1	0 (1.5)	0 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	0/171 (1.5)	0/171 (1.5)
∞	0 (1.5)	0 (1.5)	0	0	0/148 (1.5)	0/148 (1.5)
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	200	4	0 (1.5)	0 (1.5)

Table 22: Maze, detailed checks at $L_\mu = 0.005$.

Maze **Baselines** $\alpha_0 = 0.15,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	199 (100.0)	200	1
QT	111 (61.5)	200	0

Maze $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
default setting: V-C $(\alpha_0, \alpha, \delta)$ = (0.15, 0.10, 0.05)						
50	0 (1.5) ✓	0 (1.5) ✓	6	0	0/136 (1.5)	0/136 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	5	0	0/110 (1.5)	0/110 (1.5)

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$
Unfiltered	N-A	200	200
$c = 0$	200 (100.0)	200	200
$c_{\max}$	0 (1.5)	0	0
ACI	200 (100.0)	200	65
QT	133 (72.1)	200	4

Maze $\phi = 1,\ L_\mu = 0.005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
K	$\widehat{q}_{\mathrm{cert}}\downarrow$	$\widehat{q}_{\mathrm{dep}}\downarrow$	$\widehat{q}_{\mathrm{CBF}}\downarrow$	$\widehat{q}_{\mathrm{unsafe}}\downarrow$	$\widehat{q}_{\mathrm{A2}}$	$\widehat{q}_{\mathrm{pw}}$
S1: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.20)						
50	1 (2.4) ✓	0 (1.5) ✓	166	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	66	0	0 (1.5)	0 (1.5)
S2: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.15)						
50	1 (2.4) ✓	0 (1.5) ✓	166	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	58	0	0 (1.5)	0 (1.5)
S3: V-L $(\alpha_0, \alpha, \delta)$ = (0.30, 0.20, 0.20)						
50	2 (3.2) ✓	0 (1.5) ✓	0	0	0/161 (1.5)	0/161 (1.5)
∞	2 (3.2) ✓	0 (1.5) ✓	0	0	0/161 (1.5)	0/161 (1.5)
S4: V-Δ $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.025)						
100	0 (1.5) ✓	0 (1.5) ✓	0	0	N-A	N-A
∞	0 (1.5)	0 (1.5)	0	0	N-A	N-A
S5: V-C $(\alpha_0, \alpha, \delta)$ = (0.30, 0.25, 0.10)						
10	0 (1.5) ✓	0 (1.5) ✓	167	0	0 (1.5)	0 (1.5)
∞	0 (1.5) ✓	0 (1.5) ✓	57	0	0 (1.5)	0 (1.5)

Section 2.9 Variation in Robustness Margin and Estimated Exceedance Risk

Table 23: Corridor, $\widehat{V}_v$ and $\widehat{J}_v$, default setting and S1–S5.

Corridor **Baselines** $L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
$\alpha_0 = 0.15$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.633±0.582	3.108±1.339	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	21.158±7.176	5.841±1.733	3.553±1.710	2.694±0.958	200, 200
QT	41.791±7.940	10.394±0.125	8.745±2.405	5.152±0.606	200, 200
$\alpha_0 = 0.30$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.633±0.582	3.108±1.339	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	19.851±6.237	6.803±1.683	6.433±2.338	4.360±1.196	200, 200
QT	41.165±6.218	13.125±0.171	15.176±1.820	9.347±0.644	200, 200

Corridor $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
V-U	0.883±1.430	1.404±1.255	0.023±0.059	0.194±0.236	200, 200
V-Δ	0.883±1.430	1.404±1.255	0.023±0.059	0.194±0.236	200, 200
V-L	5.719±5.367	2.950±1.337	0.168±0.262	0.450±0.306	200, 200
V-C	9.966±6.831	3.754±1.522	0.373±0.393	0.650±0.383	200, 200
S1: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
V-U	38.489±13.559	10.377±2.082	3.039±1.783	2.687±0.805	200, 200
V-Δ	38.486±13.556	10.364±2.066	3.039±1.782	2.683±0.798	200, 200
V-L	31.330±10.482	10.693±2.072	3.182±1.717	2.975±0.841	200, 200
V-C	35.985±11.153	11.789±2.294	3.936±1.849	3.444±0.976	200, 200
S2: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
V-U	36.427±13.399	9.859±2.081	2.718±1.673	2.491±0.779	200, 200
V-Δ	36.424±13.395	9.846±2.077	2.718±1.672	2.487±0.777	200, 200
V-L	30.562±10.395	10.304±1.967	2.908±1.635	2.806±0.788	200, 200
V-C	36.287±10.834	11.407±2.181	3.801±1.740	3.276±0.907	200, 200
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
V-U	26.713±11.116	9.101±2.042	2.252±1.615	2.311±0.782	200, 200
V-Δ	26.675±11.017	9.064±1.965	2.249±1.606	2.301±0.754	200, 200
V-L	29.294±10.894	9.777±2.240	2.584±1.555	2.574±0.883	200, 200
V-C	29.239±10.806	9.733±2.145	2.579±1.546	2.560±0.848	200, 200
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
V-U	21.183±9.515	6.521±1.713	1.132±0.798	1.487±0.557	200, 200
V-Δ	21.188±9.513	6.523±1.713	1.133±0.798	1.488±0.558	200, 200
V-L	35.672±11.899	8.240±1.720	2.505±1.346	2.013±0.612	200, 200
V-C	36.754±11.519	9.761±1.907	3.367±1.675	2.622±0.737	200, 200
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
V-U	22.998±10.232	8.394±1.770	1.790±1.306	2.084±0.679	200, 200
V-Δ	22.991±10.221	8.385±1.755	1.790±1.304	2.081±0.673	200, 200
V-L	29.026±10.255	9.674±1.906	2.558±1.534	2.539±0.740	200, 200
V-C	36.889±10.943	11.005±2.135	3.755±1.746	3.119±0.869	200, 200

Table 24: Maze, as in Table 23.

Maze **Baselines** $L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
$\alpha_0 = 0.15$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.189±0.277	1.416±1.322	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	16.729±5.454	4.931±1.900	2.431±1.098	1.996±0.933	200, 200
QT	14.664±3.332	0.472±0.032	1.147±0.393	0.203±0.119	200, 200
$\alpha_0 = 0.30$					
Unfiltered	N-A	N-A	N-A	N-A	0, 0
$c = 0$	0.000±0.000	0.000±0.000	0.189±0.277	1.416±1.322	200, 200
$c_{\max}$	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
ACI	17.309±6.033	6.280±1.812	4.700±2.066	3.382±1.162	200, 200
QT	42.553±7.890	0.749±0.015	6.713±1.861	0.482±0.278	200, 200

Maze $\phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$					
Method	$10^3\widehat{V}_{c/c_{\max}}\downarrow$	$100\widehat{J}_{c/c_{\max}}\downarrow$	$10^3\widehat{V}_{\widehat{p}}\downarrow$	$100\widehat{J}_{\widehat{p}}\downarrow$	$\lvert\mathcal{K}_{c/c_{\max}}\rvert, \lvert\mathcal{K}_{\widehat{p}}\rvert$
default setting: $(\alpha_0, \alpha, \delta, K)$ = (0.15, 0.10, 0.05, 50)					
V-U	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
V-Δ	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	200, 200
V-L	4.908±4.252	2.717±1.385	0.131±0.165	0.404±0.278	200, 200
V-C	8.343±5.253	3.627±1.401	0.284±0.268	0.607±0.322	200, 200
S1: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.20, 50)					
V-U	12.962±8.699	6.149±1.909	0.774±0.797	1.390±0.653	200, 200
V-Δ	13.241±8.720	5.833±1.868	0.782±0.798	1.335±0.644	200, 200
V-L	28.807±10.025	10.428±2.437	3.003±1.674	2.908±0.896	200, 200
V-C	32.238±8.987	11.641±2.981	3.538±1.465	3.432±1.132	200, 200
S2: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.15, 50)					
V-U	9.916±7.184	5.593±1.762	0.561±0.646	1.209±0.590	200, 200
V-Δ	10.208±7.168	5.318±1.698	0.572±0.645	1.164±0.574	200, 200
V-L	29.416±10.534	10.069±2.565	2.969±1.730	2.771±0.915	200, 200
V-C	33.493±9.698	11.186±2.772	3.590±1.569	3.246±1.044	200, 200
S3: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.20, 0.20, 50)					
V-U	26.345±12.501	7.669±2.054	1.681±1.388	1.741±0.696	200, 200
V-Δ	26.011±12.344	7.296±1.974	1.666±1.376	1.666±0.674	200, 200
V-L	26.642±9.971	9.821±2.738	2.358±1.401	2.593±0.986	200, 200
V-C	26.537±9.947	9.635±2.782	2.353±1.398	2.552±0.995	200, 200
S4: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.025, 100)					
V-U	0.956±1.324	2.009±1.405	0.039±0.084	0.348±0.334	200, 200
V-Δ	1.010±1.364	1.959±1.355	0.040±0.085	0.340±0.325	200, 200
V-L	34.866±12.512	8.010±1.804	2.396±1.406	1.978±0.599	200, 200
V-C	34.436±10.441	9.665±2.205	3.198±1.522	2.635±0.807	200, 200
S5: $(\alpha_0, \alpha, \delta, K)$ = (0.30, 0.25, 0.10, 10)					
V-U	5.706±5.234	4.568±1.920	0.290±0.430	0.913±0.564	200, 200
V-Δ	5.791±5.249	4.486±1.877	0.293±0.430	0.898±0.555	200, 200
V-L	26.952±9.363	9.377±2.368	2.401±1.375	2.453±0.806	200, 200
V-C	33.277±9.583	10.767±2.728	3.356±1.457	3.051±0.996	200, 200